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ABBREVIATED FIRING TABLES

THE 75-mm. GUN,
M1897, M1897A1, M1897A2, M1897A3, M1897A4, and M2

THE 105-mm. HOWITZER,
M2 and M2A1

and

THE 155-mm. HOWITZER,
M1917, M1917A1, M1918, and M1918A1

1942 EDITION

THE DISTRIBUTION OF THIS BOOK WILL BE LIMITED TO MILITARY PERSONNEL AND TO OTHER PERSONS COMING WITHIN THE MEANING OF THE CLASSIFICATION "RESTRICTED."



PREPARED AT THE FIELD ARTILLERY SCHOOL
FORT SILL, OKLAHOMA
1942

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SECTION I

GENERAL

1. **General.**—Firing tables contain data, pertinent to the application of fire, based on *standard trajectories* for each type of weapon and combination of projectile, fuze, and propelling charge. A standard trajectory is the one theoretically existing under known, arbitrarily selected conditions of weather, position, and materiel. The amount of important variance that may be expected in a trajectory fired under conditions differing from standard may be computed from the values listed in the tables.

2. **Scope.**—The tables contained herein are extracts from Firing Tables 75-B-4 for the 75-mm. gun, M1897 (and modifications), Firing Tables 105-H-3 for the 105-mm. howitzer M2 and M2A1, and Firing Tables 155-D-2 for charges 3, 4, 5, and 6, and 155-D-3 for charge 7, for the 155-mm. howitzer, M1917, M1917A1, M1918, and M1918A1. Of the various combinations of projectile, fuze, and charge available for these weapons, there have been included herein three combinations for the 75-mm. gun, three combinations for the 105-mm. howitzer, and five combinations for the 155-mm. howitzer.

3. **Purpose.**—The purpose of this manual is to present information and instruction regarding firing tables. For the 75-mm. gun the manual includes sufficient data for the solution of problems involving selection and application of the appropriate combination of projectile and fuze for various missions; for the 105-mm. howitzer and the 155-mm. howitzer, sufficient data to enable the student to select and apply the appropriate combination of fuze and charge for HE shell, within the usual ranges for division artillery.

4. Arrangement.

a. Section I contains an explanation of the tables and instructions for their use.

b. Section II contains data applicable to any combination of projectile, charge, and fuze. In it are found:

- (1) Conversion factors.
- (2) Natural trigometric functions of angles in mils.
- (3) *s* and *d* tables.

(4) Wind-components table.

(5) Tables and formulas for reducing temperature and density at the altitude of the meteorological datum plane to those at the altitude of the battery.

(6) Conversion table, degrees and minutes to mils.

c. Section III contains tables for the 75-mm. gun. It is divided into three parts. One part contains data for shrapnel, Mark I; the other two, data for HE shell, Mark I, point detonating fuzes M46 and M47, reduced charge and normal charge, respectively. Each part contains three tables: A, B, and C.

(1) Table A contains information as to certain elements of a standard trajectory, range and deflection effects of differences from standard, and correction for the angle of site.

(2) Table B gives the change in muzzle velocity due to variations in the temperature of the powder.

(3) Table C gives the deflection effect for cant of axle. (The correction for this effect is not necessary for weapons equipped with sights having cross-leveling bubbles.)

d. Section IV contains tables for the 105-mm. howitzer. It is divided into three parts containing data for HE shell, Mark I, fuzes M48 and M54, charges 3, 5, and 7, respectively. Each part consists of three tables: A, B, and C, containing data as explained in c (1), (2), and (3) above.

e. Section V contains tables for the 155-mm. howitzer. It is divided into five parts containing data for HE shell, Mark I, point detonating fuzes M46 and M47, charges 3, 4, 5, 6, and 7, respectively. Each part consists of three tables: A, B, and C, containing data as explained in c (1), (2), and (3) above.

5. **Explanation of tables.**—The tables in Section II are self-explanatory. The tables of Sections III, IV and V are explained below.

a. *Table A.*—This explanation is based upon the Table A for the 75-mm. gun firing HE shell, Mark I, normal charge, fuzes M46 and M47. An explanation of the variations between this table and the table for other combinations of projectile, fuze, and charge is given in d below.

Column

- (1) The *range in yards* for each hundred yards. The range given in these tables is regarded as measured along the surface of a sphere concentric with the earth and passing through the muzzle of the piece. For practical purposes, the

Column

range may be considered as measured along a horizontal plane passing through the piece. All *effects*, throughout a computation, are based on this range (to the nearest hundred yards). The determination of one effect is not based upon a range corrected for another effect. For example, the map range being 4123, a range of 4100 would be used throughout the computation as an argument in entering the tables, although the elevation for the map range, 4123, would be determined to the nearest mil, by interpolation. The algebraic sum of all range *corrections* is converted to mils of elevation and applied to this map-range elevation, as is the site.

- (2) The *elevation* in mils for the range shown in Column 1.
- (3) The *fork* in mils of elevation. (The fork is the change in elevation for four-probable-errors change in range.)
- (4) The *change in elevation* for a 100-yard change in range. (This is *c*. It applies to ranges within 50 yards of the tabulated range.)
- (5) The *change in range* for a 1-mil change in elevation. (Often referred to as "yards per mil". Applies to ranges within 50 yards of tabulated range.)
- (6) The *time of flight* to the level point.
- (7) The *probable error in range*. (The symbol for range probable error is e_{pr} .)
- (8) The *probable error in deflection*. (The symbol for deflection probable error is e_{pd} .)
- (9) The *slope of fall* (the tangent of the angle of fall), expressed as $1/a$, where 1 is the vertical distance and a is the horizontal distance.
- (10) The *line number of metro message*. This is the initial digit of that seven-figure group which is to be used for the particular range. The seven-figure group on this line contains those data appropriate for the maximum ordinate of the range considered. The maximum ordinate is not given in Table A. Its value in feet may be approximated by squaring the time of flight and multiplying by four. A condensed table of maximum ordinates is given in Appendix I.
- (11) The *deflection effect of drift*. The effect given in these tables includes side jump. Drift proper is always to the right, but the included left side-jump effect exceeds the drift effect in the shorter ranges. (Although a standard

Column

trajectory has drift, for convenience drift is considered a deflection effect.)

- (12) The *deflection effect of a cross wind* of one mile per hour. The direction of the effect is given in the wind-components table. A wind from the right causes a *left effect*; a wind from the left, a *right effect*. (See wind-components table and description of metro message.)

(12 and

- 15-19) In Columns 12 and 15 to 19 are given the *effects of a unit increase* above standard in materiel and weather conditions. Range effects of a *unit decrease* below standard in materiel and weather conditions are equal in value but opposite in sign to the effects shown in the table. A condition causing the trajectory to have greater range than standard has a *plus* range effect; one causing the trajectory to fall shorter than standard has a *minus* range effect. A condition causing the trajectory to fall to the right of a standard trajectory has a *right* deflection effect; one causing it to fall to the left, a *left* deflection effect. The range, usually measured from a map or firing chart, is one of the known values or arguments used in determining these effects. It is taken to the nearest 100 yards.

(13 and

- 14) The *CORRECTION* which must be applied to each mil of computed angle of site, to compensate for the nonrigidity of the trajectory. It is given to the nearest hundredth of a mil.
- (15) The range effect of an *increase* of one cross (or square) above standard, in weight of projectile. (See table of projectile weights.) For example, a 75-mm. HE shell of weight "4 crosses" is two crosses *above* standard (2), giving an effect twice as great as the unit effect shown in the table and of the *same sign*; a 155-mm. HE shell of weight "1 square" is three squares *below* standard (4), giving an effect three times as great as the unit effect shown in the table and of *opposite sign*.
- (16) The range effect of an *increase* of muzzle velocity of one foot per second above standard. The sign of the effect is the same as the sign of the variation. (See Table B.)
- (17) The range effect of an *increase* in air temperature of one degree above standard (59° F.).

Column

- (18) The *range effect* of a rear wind of one mile per hour. The sign of the *effect* is the same as that of the range component.
- (19) The *range effect* of an *increase* in air density of one percent above standard (100%).
- (20) The *range setting* to be placed on the range scale of a piece firing from the M1897 carriage (and the modifications thereof) in order to reach the range given in column 1.
- (21) The *range setting* similarly required when firing from the M2 carriage.
- (1) *Range*. Same as in Column 1 on left side; repeated for convenience.

b. *Table B*.—This table gives the change in muzzle velocity due to variation in the powder temperature. Whenever possible the actual temperature of the powder should be taken. With fixed ammunition the temperature of the place where the ammunition is stored at the battery is used. If the temperature cannot be taken at the battery, the air temperature given in the metro message, corrected for the altitude of the battery position, is used as a guide. With the effect obtained from this table, the column headed "Range Effect of Increase of—One foot per sec. in MV" in Table A is entered to obtain the final effect in range.

c. *Table C*.—This table gives the deflection effect of 10-mils cant of the carriage axle. The effect of any other amount is proportional. The correction is applied to individual pieces by the executive.

d. *Variations in tables*.—The Tables B and the Tables C are similar in form for all combinations of projectile, fuze, and charge. Variations in the Tables A from the arrangement described in a above are described below.

(1) *75-mm. gun firing shrapnel, Mark I, 21-second combination fuze, and 105-mm. howitzer*.—Table A contains four additional columns showing:

- (9) The *probable error in height of burst*, expressed in mils.
- (21) The *fuze setting*, in seconds, to produce a graze burst.
- (22 and
- 23) The *displacement of the burst* resulting from a change of five points in the corrector.

In the table for the 75-mm. gun firing shrapnel, Mark I, 21-second combination fuze, no column of data is given for effect of weight of projectile.

(2) *155-mm. howitzer, firing HE shell, Mark I, fuzes M46 and M47 and 105-mm. howitzer.*—Separate tables are required for each of the different charges of propelling powder used with this materiel. The tables are the same as those described in *a* above except that columns 20 and 21 (range settings) do not apply and are therefore omitted or otherwise used.

6. The metro message.

a. The metro message is a coded tabulation of meteorological conditions, on the surface of the earth and at varying zones above it, for use by the field artillery in applying these meteorological effects to its firing data. It consists of a group of three letters to designate the sending station, followed by one five-figure group and a series of seven-figure groups, as explained by the table which follows. If the first figure of the *five-figure group* is 2 the message applies to antiaircraft or other high-angle fire; if the first figure is 3 the message applies to field artillery.

b.

(1)	(2)
Example of Metro Message	Explanation
MIFMIF	The letter M indicates a metro message; the letters IF are the code designation of the sending station. (The group is usually repeated as indicated.)
31256	The figure 3 indicates a metro message for the field artillery; 12 indicates that the MDP (meteorological datum plane) is 1,200 feet above sea level; 56 indicates that the temperature at the MDP is 56° Fahrenheit.

0820802
1351101
2371499
3401598
4441697
5481896
6502195
7532295
*562495
622795

(3)	(4)	(5)	(6)	(7)
Zones in height (feet)	Line of metro message	Azimuth of wind, mils	Velocity of wind, m/h	Density %
Surface	0	3200	8	102
Surface to 600	1	3500	11	101
600 to 1500	2	3700	14	99
1500 to 3000	3	4000	15	98
3000 to 4500	4	4400	16	97
4500 to 6000	5	4800	18	96
6000 to 9000	6	5000	21	95
9000 to 12,000	7	5300	22	95
12,000 to 15,000	8	5600	24	95
15,000 to 18,000	9	6200	27	95

The tabulation above explains the seven-figure groups.

The first figure of each group gives the line number of the metro message, Column (4). (Also see Column 10, Table A).

The second and third figures, explained in Column (5), give the azimuth of the direction from which the ballistic wind blows, measured clockwise from the true North, in hundreds of mils. In computations this is usually assumed to be the same as Y-North.

The fourth and fifth figures give the velocity of the ballistic wind, Column (6).

The sixth and seventh figures, explained in Column (7), give the ballistic density in percentage of standard. If greater than standard the first digit does not appear in the metro message; for example, 02 would mean 102, or an increase of 2% above normal, while 98 means a decrease of 2% below normal. This density must be corrected for the difference in altitude between the battery and the MDP by means of the table for correcting density (page 17). The density is the mean effective density for the whole layer of atmosphere having a thickness equal to the height given.

*Each additional line refers to an additional 6000-foot zone in height.

c. Further explanation of the use of the metro message appears in paragraph 7 below.

7. Computation of metro corrections; type problem.

a. *General procedure.*—To compute metro corrections quickly and reliably it is necessary to follow a procedure that is systematic and orderly. Much of the work (such as determining the uncorrected direction and quadrant elevation, ascertaining from the battery executive the weight of projectile and temperature of powder, and listing all effects due to materiel) can usually be done *in advance*, leaving only a few simple, planned operations to be performed upon receipt of the metro message. The tables are so arranged as to facilitate this. Note that for a given target nearly all matter to be extracted from the tables appears on a single line, namely that line of Table A which is identified by the map range (to nearest 100 yards).

b. *Example.*—For the 75-mm. gun M2 firing HE shell Mark I, normal charge, fuze M46, to determine the corrections in *direction* and *range* for a single check point:

(1) Record map data from firing chart (map, wide angle photo, or grid sheet):

Map range (to nearest 10 yards)	= 4780 yards
Altitude of check point (target)	= 1665 feet
Altitude of battery	= 1505 feet
Map shift (from base point)	= BDR 50
Direction of fire (Y-azimuth taken to nearest 100 mils)	= 4900

(2) Determine the elevation for range. From Table A (column 2) note the elevations corresponding to 4700 (168.2) and 4800 (174.4). For range 4780 interpolate as follows:

$$\begin{aligned} \text{Elevation corresponding to 4700} &= 168.2 \\ 80/100 \times (174.4 - 168.2)** &= 5.0 \end{aligned}$$

$$\text{Elevation corresponding to 4780} = 173.2 \text{ (use 173)}$$

(3) Compute the site. The target is 160 feet above the battery, giving a difference in altitude of +53 yards (minus when target is below guns). For range 4800 (nearest 100 yards) the site is +53/4.8 or +11 mils. To correct the site (nonrigidity of trajectory) enter Table A, line 4800 and column 13, and note that for each +1 mil angle of site the correction is +.05 mil. The com-

**Instead of making this subtraction it is equally good practice to use the value c, found in column 4.

plementary angle of site is, then, $11 \times (+.05)$ or $+.55$ mil. This is taken as 1 mil (nearest mil), making the corrected site $+12$ mils ($11 + 1$). (Alternative method: Compute the corrected site directly by multiplying $+11$ by 1.05 . This method adapts itself to slide-rule computation. The factor 1.05 is the numerical, not the algebraic, sum of unity plus the site correction for one mil.)

(4) The quadrant elevation is the algebraic sum of the corrected site and the elevation for range:

Elevation for range 4780 (subpar (2) above)	173 mils
Corrected site (subpar (3) above)	+12
Quadrant elevation	185 mils

(5) The battery executive reports that: The weight marking on the projectiles is $+++$, or one cross above standard (par. 5 a (15)). The powder temperature is 40° F. Now enter Table B and note that a powder temperature of 40° changes the muzzle velocity from standard by -22 f/s.

(6) Enter Table A, line 4800 and column 10, and note that the metro message line-number that will apply is 2.

(7) The following metro message is received:

MIF	MIF
3	12 35
0	57 25 97
1	59 27 96
2	59 29 96
3	60 32 95
4	60 33 95

The pertinent parts of the metro message are:

3	12 35
* * * * *	
2	59 29 96
* * * * *	

- 3 (Message applies to field artillery.)
- 12 (Altitude of MDP is 1200 feet.)
- 35 (Air temperature at MDP is 35° F. This must be corrected to level of battery. See (8) below.)
- 2 (Line number that applies. See (6) above.)
- 59 (Ballistic wind blows from Y-azimuth 5900.)
- 29 (Velocity of ballistic wind is 29 mph.)
- 96 (Ballistic density at MDP is 96% of standard. This must be corrected to level of battery. See (8) below.)

(8) Correct the air temperature and density to the level of the battery, using the corrected-density-and-temperature table and notes pertaining thereto on page 17.

Battery is 300 feet above MDP (1505 - 1200).

Corrected temperature $= 35 - (.2 \times 3) = 34.4^\circ$. (Use 34° .)

Corrected density $= 96 - (.3 \times 3) = 95.1\%$. (Use 95% .)

(9) The variations from standard are now found to be:
Variation in weight of projectile = 3 (actually +++) -2
 (++ is standard) = +1. (Par. 5 a (15).)

Temperature *variation* = 34° (at battery) - 59° (standard) =
 - 25° . (Par. 5 a (17).)

Density *variation* = 95% (at battery) - 100% (standard) =
 - 5% . (Par. 5 a (19).)

(10) Determine *chart* direction of wind:

Wind direction (subpar (7)) (add 6400 if
 necessary = 5900

Minus direction of fire (subpar (1)) = 4900

Chart direction 1000

(11) Determine the wind components: Enter the wind-component table, page 16, with chart direction 1000 (subpar (10)) and note that for a wind of 1 mph the cross wind is *left .83 mph* and the range wind is *-.56 mph*. Multiplying these factors by 29, the reported wind velocity (subpar (7)), the cross wind is found to be *left 24 mph* and the range wind *-16 mph*.

NOTE.—The foregoing paragraphs contain the preliminary ground work necessary for determining the corrections for direction and range. Those that follow show in detail how to determine and apply these corrections.

(12) *Direction*.—Compute the weather (cross wind) correction, combine it with the drift correction and apply the net correction thus determined to the map shift. To compute the weather effect multiply the cross wind of left 24 mph (subpar (11)) by the effect in mils of deflection (.4 mil in this case—Table A, line 4800, column 12) of a cross wind of *one* mph. The result (left 9.6) taken to the nearest mil is *left 10*.

FACTOR	EFFECT	CORRECTION
Drift (Table A, line 4800, column 11)....	Right 5	Left 5
Weather (cross wind).....	Left 10	Right 10
Net correction.....		Right 5
Map shift.....	BD Right 50	
Net correction.....		Right 5
Corrected direction =	BD Right 55	

(13) *Range*.—Calculate the net combined effect (in yards) of the various factors affecting range. The effect of any single factor is the product of its variation from standard by its unit effect in yards. The sign of the effect is that of the algebraic product. (Plus $\times$ plus = *plus*; minus $\times$ minus = *plus*; plus $\times$ minus = *minus*; zero $\times$ either plus or minus = *zero*.) The unit effects appear in Table A, line 4800, columns 15 to 19 inclusive, in an order which should be followed in tabulating the factors. Convert the net combined effect to a correction (in yards) by changing its sign, then convert the latter (which for most purposes is not in a usable form) into a correction in mils of quadrant elevation by dividing it by “yards per mil” (Table A, line 4800, column 5).

Column No.	Factor	Variation from Std.	Unit Effect	Range Effect (yds)	
				+	-
15	Wt. of proj. (subpar (9))	+ 1 (cross)	+13	13	
16	Pow. temp. (subpar (5))	-22 (f/s)	+ 2.1		46
17	Air temp. (subpar (9))	-25 (degrees)	+ 1.8		45
18	Range wind (subpar (11))	-16 (mph)	+ 5.7		91
19	Density (subpar (9))	- 5 (per cent)	-18	90	
Totals of effects (yards).....				+103	-182
Net effect (yards) (+103 - 182).....				=	- 79
Net <i>correction</i> (yards).....				=	+ 79
Yards per mil at 4800 (Table A, column 5) = 16					
Elevation correction (mils) (+79/16).....				=	+ 5
Uncorrected quadrant elevation (subpar (4))...				185	
Elevation correction.....				+5	
Corrected quadrant elevation.....				190	

8. Selection of appropriate charge.

a. With the 105-mm. howitzer and the 155-mm. howitzer the propeling charge appropriate for use depends upon the range. A given range-line can usually be reached by more than one charge. The heavier the charge the greater the erosion of the bore. The lighter the charge the steeper the angle of fall and, in general, the greater the dispersion. Selecting too light a charge may necessitate stopping an adjustment and starting anew with a heavier charge. A registration made near the long limit of a charge will be useless for transfers to targets at greater ranges.

b. *Procedure*.—Mark each Table A to indicate 75% and 85%, respectively, of the maximum range of its charge. For observed fires the charge selected should be such that the computed range

to the target is not greater than 75% of the maximum range for that charge. Subject to these restrictions, select the lightest charge possible. Example: Adjust on a check point at range 6600 with 155-mm. howitzer, shell Mark I, fuze M46. The maximum initial ranges for adjustment with charges 4, 5, and 6 (indicated in each case by the 75% mark in the appropriate Table A) are, respectively, 5250, 6800, and 8250. Charge 4 is manifestly too light. Charge is needlessly heavy. Charge 5, then, is the proper selection.

c. For unobserved fires that are prepared from an accurate firing chart a charge can be used up to 85% of its maximum range without serious impairment of accuracy by dispersion.

SECTION II

TABLES COMMON TO ALL WEAPONS

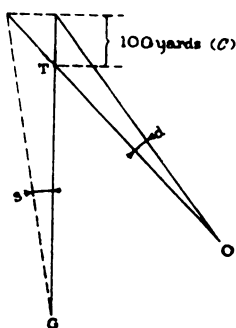
CONVERSION FACTORS

1 Yard...	=	0.9144 Meters
1 Meter...	=	1.0936 Yards
1°	=	17.7778 Mils
1'	=	0.2963 Mils
1 Mil.....	=	0.0563°
1 Mil.....	=	3.375'

NATURAL FUNCTIONS OF ANGLES IN MILS

Mils	sin	cos	tan	cot		Mils	sin	cos	tan	cot	
0	.0000	1.0000	.0000		1600	400	.3827	.9239	.4142	2.414	1200
10	.0098	1.0000	.0098	101.9	90	10	.3917	.9201	.4258	2.349	90
20	.0196	.9998	.0196	50.92	80	20	.4008	.9162	.4374	2.286	80
30	.0295	.9996	.0295	33.94	70	30	.4097	.9122	.4492	2.226	70
40	.0393	.9992	.0393	25.45	60	40	.4187	.9081	.4610	2.169	60
50	.0491	.9988	.0491	20.36	50	50	.4276	.9040	.4730	2.114	50
60	.0589	.9983	.0590	16.96	40	60	.4364	.8998	.4850	2.062	40
70	.0687	.9976	.0688	14.53	30	70	.4452	.8954	.4972	2.011	30
80	.0785	.9969	.0787	12.71	20	80	.4540	.8910	.5095	1.963	20
90	.0882	.9961	.0886	11.29	10	90	.4627	.8865	.5220	1.916	10
100	.0980	.9952	.0985	10.15	1500	500	.4714	.8819	.5345	1.871	1100
10	.1078	.9942	.1084	9.224	90	10	.4800	.8773	.5472	1.827	90
20	.1175	.9931	.1184	8.449	80	20	.4886	.8725	.5600	1.786	80
30	.1273	.9919	.1283	7.793	70	30	.4972	.8677	.5730	1.745	70
40	.1370	.9906	.1383	7.230	60	40	.5057	.8627	.5861	1.706	60
50	.1467	.9892	.1483	6.741	50	50	.5141	.8577	.5994	1.668	50
60	.1564	.9877	.1584	6.314	40	60	.5225	.8526	.6128	1.632	40
70	.1661	.9861	.1685	5.936	30	70	.5308	.8475	.6264	1.596	30
80	.1758	.9844	.1786	5.600	20	80	.5391	.8422	.6401	1.562	20
90	.1855	.9827	.1887	5.299	10	90	.5474	.8369	.6541	1.529	10
200	.1951	.9808	.1989	5.027	1400	600	.5556	.8315	.6682	1.497	1000
10	.2047	.9788	.2091	4.781	90	10	.5637	.8260	.6825	1.465	90
20	.2143	.9768	.2194	4.558	80	20	.5718	.8204	.6970	1.435	80
30	.2239	.9746	.2297	4.353	70	30	.5798	.8148	.7117	1.405	70
40	.2335	.9724	.2401	4.165	60	40	.5878	.8090	.7265	1.376	60
50	.2430	.9700	.2505	3.992	50	50	.5957	.8032	.7417	1.348	50
60	.2525	.9676	.2610	3.832	40	60	.6036	.7973	.7570	1.321	40
70	.2620	.9651	.2715	3.684	30	70	.6114	.7914	.7725	1.294	30
80	.2714	.9625	.2820	3.546	20	80	.6191	.7853	.7883	1.268	20
90	.2809	.9597	.2927	3.417	10	90	.6268	.7792	.8044	1.243	10
300	.2903	.9569	.3034	3.297	1300	700	.6344	.7730	.8207	1.219	900
10	.2997	.9540	.3141	3.184	90	10	.6420	.7667	.8372	1.194	90
20	.3090	.9511	.3249	3.078	80	20	.6495	.7604	.8541	1.171	80
30	.3183	.9480	.3358	2.978	70	30	.6569	.7540	.8712	1.148	70
40	.3276	.9448	.3468	2.884	60	40	.6643	.7475	.8886	1.125	60
50	.3369	.9415	.3578	2.795	50	50	.6716	.7410	.9064	1.103	50
60	.3461	.9382	.3689	2.711	40	60	.6788	.7343	.9244	1.082	40
70	.3553	.9348	.3801	2.631	30	70	.6860	.7276	.9428	1.061	30
80	.3645	.9312	.3914	2.555	20	80	.6931	.7209	.9615	1.040	20
90	.3736	.9276	.4028	2.483	10	90	.7001	.7140	.9806	1.020	10
400	.3827	.9239	.4142	2.414	1200	800	.7071	.7071	1.0000	1.000	800
	cos	sin	cot	tan	Mils		cos	sin	cot	tan	Mils

EXPLANATION OF s AND d TABLES



G is the gun.

O is the observer.

T is the target. It is also the *angle T* (OTG) (often referred to as the *observer displacement* and as the *target offset in mils*).

R is range *GT* in thousands of yards.
r is the distance *OT* in thousands of yards.

d is the deviation, as seen from O, caused by a range change of 100 yards (an elevation change of one c). Its value depends on the values of T and r.

s is the shift in deflection necessary to keep a shot on the OT line when making a range change of 100 yards (an elevation change of one c).

s TABLE

Range GT in yards	T in mils														
	100	200	300	400	500	600	700	800	900	1000	1100	1150	1200	1250	1300
2000	5	10	15	21	27	34	42	51	62	76	95	108	123	142	168
2100	5	10	15	20	26	32	40	49	59	73	91	103	117	136	160
2200	5	9	14	19	25	31	38	46	56	69	87	98	112	129	153
2300	4	9	13	18	24	30	36	44	54	66	83	94	107	124	146
2400	4	8	13	18	23	28	35	42	52	64	79	90	102	119	140
2500	4	8	12	17	22	27	33	41	50	61	76	86	98	114	134
2600	4	8	12	16	21	26	32	39	48	59	73	83	95	109	129
2700	4	8	11	16	20	25	31	38	46	56	71	80	91	105	124
2800	4	7	11	15	19	24	30	36	44	54	68	77	88	102	120
2900	3	7	11	15	19	23	29	35	43	53	66	74	85	98	116
3000	3	7	10	14	18	23	28	34	41	51	64	72	82	95	112
3200	3	6	10	13	17	21	26	32	39	48	60	67	77	89	105
3400	3	6	9	12	16	20	25	30	37	45	56	63	72	84	99
3600	3	6	9	12	15	19	23	28	34	42	53	60	68	79	93
3800	3	5	8	11	14	18	22	27	33	40	50	57	65	75	88
4000	3	5	8	11	14	17	21	25	31	38	48	54	61	71	84
4500	2	5	7	9	12	15	19	23	28	34	42	48	55	63	75
5000	2	4	6	8	11	14	17	20	25	31	38	43	49	57	67
5500	2	4	6	8	10	12	15	19	23	28	35	39	45	52	61
6000	2	3	5	7	9	11	14	17	21	25	32	36	41	47	56
6500	2	3	5	6	8	10	13	16	19	23	29	33	38	44	52
7000	1	3	4	6	8	10	12	15	18	22	27	31	35	41	48
7500	1	3	4	6	7	9	11	14	17	20	25	29	33	38	45
8000	1	3	4	5	7	9	10	13	16	19	24	27	31	36	42
8500	1	2	4	5	6	8	10	12	15	18	22	25	29	33	40
9000	1	2	3	5	6	8	9	11	14	17	21	24	27	32	37
9500	1	2	3	4	6	7	9	11	13	16	20	23	26	30	35
10000	1	2	3	4	5	7	8	10	12	15	19	22	25	28	34

d TABLE

Distance OT in yards	T in mils												
	100	200	300	400	500	600	700	800	900	1000	1100	1200	1300
1000	10	20	30	39	48	57	65	72	79	85	90	94	97
1100	9	18	27	35	44	51	59	65	72	77	82	86	89
1200	8	17	25	32	40	47	54	60	66	71	75	78	81
1300	8	15	23	30	37	44	50	55	61	65	69	72	75
1400	7	14	21	28	34	40	46	51	56	60	64	67	70
1500	7	13	20	26	32	38	43	48	52	56	60	63	65
1600	6	12	18	24	30	35	40	45	49	53	56	59	61
1700	6	12	17	23	28	33	38	42	46	50	53	55	57
1800	6	11	16	22	27	31	36	40	44	47	50	52	54
1900	5	10	16	21	25	30	34	38	41	45	47	50	51
2000	5	10	15	20	24	28	32	36	39	42	45	47	49
2100	5	9	14	19	23	27	31	34	37	40	43	45	46
2200	5	9	13	18	22	26	29	33	36	38	41	43	44
2300	4	9	13	17	21	25	28	31	34	37	39	41	42
2400	4	8	12	16	20	24	27	30	33	35	37	39	41
2500	4	8	12	16	19	23	26	29	31	34	36	38	39
2600	4	8	11	15	18	22	25	28	30	33	35	36	37
2700	4	7	11	14	18	21	24	27	29	31	33	35	36
2800	4	7	11	14	17	20	23	26	28	30	32	34	35
2900	3	7	10	13	17	20	22	25	27	29	31	32	34
3000	3	7	10	13	16	19	22	24	26	28	30	31	32
3200	3	6	9	12	15	18	20	23	25	26	28	29	30
3400	3	6	9	11	14	17	19	21	23	25	26	28	29
3600	3	6	8	11	13	16	18	20	22	24	25	26	27
3800	3	5	8	10	13	15	17	19	21	22	24	25	26
4000	2	5	7	10	12	14	16	18	20	21	22	24	24
4500	2	4	7	9	11	13	14	16	18	19	20	21	22
5000	2	4	6	8	10	11	13	14	16	17	18	19	19
5500	2	4	5	7	9	10	12	13	14	15	16	17	18
6000	2	3	5	6	8	9	11	12	13	14	15	16	16
6500	2	3	5	6	7	9	10	11	12	13	14	14	15
7000	1	3	4	6	7	8	9	10	11	12	13	13	14
7500	1	3	4	5	6	8	9	10	11	11	12	13	13
8000	1	2	4	5	6	7	8	9	10	11	11	12	12

WIND COMPONENTS FOR ONE-MILE-PER-HOUR WIND

Chart Direction of Wind	Cross Wind MPH	Range Wind MPH	Chart Direction of Wind	Cross Wind MPH	Range Wind MPH
0	0	-1.00	3200	0	+1.00
100	L .10	- .99	3300	R .10	+ .99
200	L .20	- .98	3400	R .20	+ .98
300	L .29	- .96	3500	R .29	+ .96
400	L .38	- .92	3600	R .38	+ .92
500	L .47	- .88	3700	R .47	+ .88
600	L .56	- .83	3800	R .56	+ .88
700	L .63	- .77	3900	R .63	+ .77
800	L .71	- .71	4000	R .71	+ .71
900	L .77	- .63	4100	R .77	+ .63
1000	L .83	- .56	4200	R .83	+ .56
1100	L .88	- .47	4300	R .88	+ .47
1200	L .92	- .38	4400	R .92	+ .38
1300	L .96	- .29	4500	R .96	+ .29
1400	L .98	- .20	4600	R .98	+ .20
1500	L .99	- .10	4700	R .99	+ .10
1600	L 1.00	.00	4800	R 1.00	.00
1700	L .99	+ .10	4900	R .99	- .10
1800	L .98	+ .20	5000	R .98	- .20
1900	L .96	+ .29	5100	R .96	- .29
2000	L .92	+ .38	5200	R .92	- .38
2100	L .88	+ .47	5300	R .88	- .47
2200	L .83	+ .56	5400	R .83	- .56
2300	L .77	+ .63	5500	R .77	- .63
2400	L .71	+ .71	5600	R .71	- .71
2500	L .63	+ .77	5700	R .63	- .77
2600	L .56	+ .83	5800	R .56	- .88
2700	L .47	+ .88	5900	R .47	- .88
2800	L .38	+ .92	6000	R .38	- .92
2900	L .29	+ .96	6100	R .29	- .96
3000	L .20	+ .98	6200	R .20	- .98
3100	L .10	+ .99	6300	R .10	- .99
3200	0	+1.00	6400	0	-1.00

This table divides a wind of one mile per hour, blowing from the *chart direction*, into two components: the cross wind, perpendicular to the plane of fire, and the range wind, parallel to the plane of fire. The chart direction is the Y-azimuth of the wind direction as given in the metro message (increased by 6400 when necessary) minus the Y-azimuth of the direction of fire.

CORRECTED DENSITY AND TEMPERATURE

Height of battery with reference to the M. D. P., ft.	Change in	
	Density in percent	Temperature in °F.
+600	-1.8	-1.2
+500	-1.5	-1.0
+400	-1.2	-0.8
+300	-0.9	-0.6
+200	-0.6	-0.4
+100	-0.3	-0.2
Same	0	0
-100	+0.3	+0.2
-200	+0.6	+0.4
-300	+0.9	+0.6
-400	+1.2	+0.8
-500	+1.5	+1.0
-600	+1.8	+1.2

Density decreases 0.3 percent for each 100 feet battery is above the M.D.P.

Temperature decreases 0.2° F. for each 100 feet battery is above the M.D.P.

CONVERSION TABLE

Degrees and minutes to mils

Degrees by 10's	Mils	De- grees	Mils	Minutes by 10's	Mils	Minutes	Mils	Minutes in tenths	Mils
10°	177.78	1°	17.78	10'	2.96	1'	.30	.1'	.03
20°	355.56	2°	35.56	20'	5.93	2'	.59	.2'	.06
30°	533.33	3°	53.33	30'	8.89	3'	.89	.3'	.09
40°	711.11	4°	71.11	40'	11.85	4'	1.19	.4'	.12
50°	888.89	5°	88.89	50'	14.82	5'	1.48	.5'	.15
60°	1066.67	6°	106.67			6'	1.78	.6'	.18
70°	1244.44	7°	124.44			7'	2.07	.7'	.21
80°	1422.22	8°	142.22			8'	2.37	.8'	.24
90°	1600.00	9°	160.00			9'	2.67	.9'	.27
100°	1777.78								
110°	1955.56								
120°	2133.33								
130°	2311.11								
140°	2488.89								
150°	2666.67								
160°	2844.44								
170°	3022.22								

For example, to convert 78°43.6' to mils:			
70°	=	1244.44	mils
8°	=	142.22	"
40'	=	11.85	"
3'	=	.89	"
.6'	=	.18	"
78°43.6'	=	1399.58	mils

SECTION III

FIRING TABLES, 75-mm. GUN

CHARACTERISTICS

75-mm. Gun, M1897, M1897A1, M1897A2, M1897A3, and M1897A4

75-mm. Gun

Firing HE Shell Mark 1 and Shrapnel Mark 1

Diameter of the bore between lands	inches	2.953
Diameter of the bore between grooves	inches	2.992
Total length	inches	107.126
Length of rifled portion	inches	87.772
Travel of projectile	inches	89.9
Capacity of powder chamber	cubic inches	83
Number of grooves		24
Character of rifling	{ uniform twist 1 in 25.6 calibers	
Maximum pressure for which gun is designed	lb./sq. in.	36,000
Weight of gun and breech mechanism	pounds	1,035

75-mm. Gun Carriage, M-2

		On wheels	On firing jack
Maximum traverse, right	mils	800	800
Maximum traverse, left	mils	711	711
Least possible elevation	mils	-178	-178
Greatest possible elevation	mils	818	821
Traverse for one turn of the traversing hand- wheel	mils	19.0	19.0
Change in elevation for one turn of elevating handwheel	mils	10	10
Maximum range scale setting	yards	9760	9760

75-mm. Gun Carriage, M1897 (and modifications)

Total traverse (one-half on each side)	mils	106
Least possible elevation	mils	-178
Greatest possible elevation	mils	338
Traverse for one turn of traversing handwheel	mils	1.8
Change in elevation for one turn of elevating handwheel	mils	8
Maximum range scale setting	meters	5500

Projectile—Mean Weight of Fuzed Projectile in Pounds

Shrapnel.—Standardized at 15.96 pounds.

HE Shell, Mark I.—P. D. Fuzes M46 and M47.

Variations in weight are indicated by markings stenciled on the projectile as follows:

Marking	Weight
L	11.58
+	11.91
+ + (standard)	12.24
+ + +	12.57
+ + + +	12.90

Fuzes

21-second combination time and percussion.

Point detonating fuzes:

M46 (nose painted white)..... superquick.

M47 (nose painted black)..... delay.

75-mm. GUN

Characteristics of combinations of Projectile, Charge, and Fuze for which tables are not included herein.

HE SHELL MK I, FUZES (SHORT) MK IV AND MK V, REDUCED CHARGE: Muzzle velocity, 1130 f/s; maximum range, 6965 yards.

HE SHELL MK I, FUZES (SHORT) MK IV AND MK V, NORMAL CHARGE: Muzzle velocity, 1805 f/s; maximum range, 8915 yards.

HE SHELL MK I, FUZE (LONG) MK III, NORMAL CHARGE: Muzzle velocity, 1778 f/s; maximum range, 8175 yards.

CHEMICAL SHELL MK II, FUZES M46 AND M47, REDUCED CHARGE: Muzzle velocity, 1115 f/s; maximum range, 6425 yards.

CHEMICAL SHELL MK II, FUZES M46 AND M47, NORMAL CHARGE: Muzzle velocity, 1758 f/s; maximum range, 8810 yards.

CHEMICAL SHELL MK II, FUZES (SHORT) MK IV AND MK V, REDUCED CHARGE: Muzzle velocity, 1130 f/s; maximum range, 6400 yards.

CHEMICAL SHELL MK II, FUZES (SHORT) MK IV AND MK V, NORMAL CHARGE: Muzzle velocity, 1805 f/s; maximum range, 8400 yards.

CHEMICAL SHELL MK II, FUZE (LONG) MK III, NORMAL CHARGE: Muzzle velocity, 1778 f/s; maximum range, 7925 yards.

75-mm. GUN

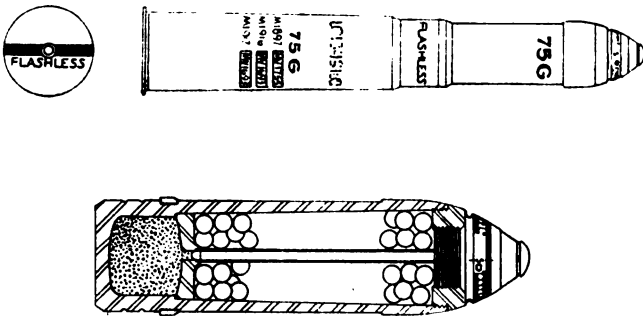
Shrapnel, Mark I

Fuze, 21-second combination

NOTE: Elevations given in this table are for Carriage M2. Add 0.8 mils to these elevations when firing from Carriage M1897 (and modifications).

Maximum Range: 9760 yards.

Data for ranges 0—8000 included herein.



**75-mm. GUN
SHRAPNEL, MK I
FUZE, 21-SEC. COMB.
M.V. 1755 f/s**

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation El	F Fork	Change in elevation for 100-yd. change in range c	Change in range for 1-mil. change in elevation 1 mil	Time of flight Time	Probable Error			Slope of fall Slope	Line number of metro message Line	Deflection Effect	
						Range e _{pr}	Deflection e _{pd}	Height of burst			Drift* Dft.	Lateral wind of 1 mile per hour(+) W-D
yds.	m	m	m	yds.	sec.	yds.	yds.	m	1/-	No.	m	m
0	0.2	1	1.6	67	0.0	11	0	1		1	L 1	.0
100	1.8	1	1.6	64	0.1	11	0	1	1018	1	L 1	.0
200	3.4	1	1.6	61	0.3	11	0	1	407	1	L 1	.0
300	5.0	1	1.8	58	0.5	11	0	1	255	1	0	.1
400	6.8	1	1.8	56	0.7	11	0	1	173	1	0	.1
500	8.6	1	1.8	54	0.9	11	0	1	124	1	0	.1
600	10.6	1	2.0	52	1.1	11	0	1	95	1	0	.1
700	12.6	1	2.0	50	1.3	11	0	1	77	1	0	.1
800	14.6	1	2.0	48	1.5	11	0	1	64	1	0	.1
900	16.8	1	2.2	47	1.7	11	0	1	54	1	0	.1
1000	19.0	1	2.2	45	1.9	11	1	1	46	1	0	.1
1100	21.2	1	2.2	44	2.1	11	1	1	40	1	0	.1
1200	23.6	1	2.4	42	2.3	11	1	1	35	1	0	.1
1300	26.2	1	2.4	41	2.6	11	1	1	31	1	0	.2
1400	28.8	1	2.4	40	2.8	11	1	1	28	1	R 1	.2
1500	31.4	1	2.6	38	3.0	11	1	1	25	1	R 1	.2
1600	34.0	1	2.6	37	3.2	11	1	1	23	1	R 1	.2
1700	36.8	1	2.6	36	3.4	11	1	1	21	1	R 1	.2
1800	39.6	1	2.8	35	3.7	11	1	1	19.7	1	R 1	.2
1900	42.6	1	2.8	34	4.0	11	1	1	18.2	1	R 1	.2
2000	45.6	1	3.0	33	4.3	11	1	1	16.8	1	R 1	.2
2100	48.6	1	3.0	32	4.6	11	1	1	15.6	1	R 1	.2
2200	51.6	1	3.2	32	4.9	11	1	1	14.5	1	R 1	.2
2300	54.8	1	3.2	31	5.1	11	1	1	13.5	1	R 1	.3
2400	58.2	1	3.4	30	5.4	11	1	1	12.6	1	R 1	.3
2500	61.8	1	3.4	29	5.7	11	1	1	11.8	1	R 1	.3
2600	65.4	2	3.4	29	6.0	11	1	1	11.2	1	R 2	.3
2700	69.0	2	3.6	28	6.3	11	1	1	10.6	1	R 2	.3
2800	72.6	2	3.6	27	6.5	11	1	1	10.0	1	R 2	.3
2900	76.2	2	3.6	27	6.8	11	1	1	9.4	1	R 2	.3
3000	80.0	2	3.8	26	7.1	11	1	1	8.9	1	R 2	.3
3100	83.8	2	3.8	26	7.4	11	1	1	8.4	1	R 2	.3
3200	87.6	2	4.0	25	7.7	11	1	1	8.0	1	R 2	.3
3300	91.6	2	4.0	25	8.0	11	1	1	7.6	1	R 3	.4
3400	95.8	2	4.2	24	8.3	11	1	1	7.2	1	R 3	.4
3500	100.0	2	4.2	24	8.6	11	1	1	6.9	1	R 3	.4
3600	104.2	2	4.2	23	8.9	11	1	1	6.6	1	R 3	.4
3700	108.6	2	4.4	23	9.2	11	1	1	6.3	1	R 3	.4
3800	113.0	2	4.4	23	9.5	11	1	1	6.0	1	R 3	.4
3900	117.4	2	4.4	22	9.9	11	1	1	5.8	1	R 3	.4
4000	122.0	2	4.6	22	10.2	11	1	1	5.5	1	R 4	.4

*Drift includes side jump.

75-mm. GUN
SHRAPNEL, MK. I
FUZE, 21-SEC. COMB.
M.V. 1755 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	24	25	1
Comp. angle of site for each		Range Effect of Increase of—						Displ. of burst for change of 5 points in fuze setter corrector in		Range setting for 1897 Carriage (and Mods.)	Range setting for M2 Carriage	Range
+ 1 mil of site	- 1 mil of site	One percent in wt. of projectile Std. wt. 16.96 lbs.	One foot per sec. in M.V.	Air temperature 1 degree Standard is 59° F.	Rear wind 1 mile per hour	One percent in air density	Fuze setting for graze burst					
m	m	Wt. yds.	VE yds.	Temp. yds.	W-R yds.	Den. yds.	F.S. sec.	Height m.	Range yds.	Set. m.	Set. yds.	R yds.
.00	.00	0	0.0	0.0	0.0	0	0.0			160	-50	0
.00	.00	-1	+0.1	0.0	0.0	0	0.2	4.1	230	250	+50	100
.00	.00	-1	+0.2	0.0	+0.1	0	0.4	4.1	226	340	150	200
.00	.00	-2	+0.3	0.0	+0.1	0	0.6	4.2	221	425	250	300
.00	.00	-2	+0.4	0.0	+0.1	0	0.8	4.2	217	510	355	400
.00	.00	-3	+0.5	0.0	+0.2	-1	1.0	4.3	213	595	455	500
.00	.00	-3	+0.6	0.0	+0.2	-1	1.2	4.4	209	680	555	600
.00	.00	-4	+0.7	0.0	+0.2	-1	1.4	4.5	205	765	655	700
.00	.00	-4	+0.8	0.0	+0.3	-1	1.6	4.5	202	850	755	800
.00	.00	-5	+0.9	0.0	+0.3	-2	1.9	4.6	199	935	855	900
.00	.00	-5	+1.0	0.0	+0.4	-2	2.1	4.7	196	1020	955	1000
.00	.00	-5	+1.1	0.0	+0.4	-2	2.3	4.7	193	1110	1060	1100
.00	.00	-5	+1.1	0.0	+0.5	-2	2.6	4.8	190	1200	1165	1200
.00	.00	-5	+1.2	0.0	+0.5	-3	2.8	4.9	188	1290	1270	1300
.00	.00	-5	+1.2	0.0	+0.6	-3	3.1	4.9	185	1380	1375	1400
.00	.00	-5	+1.3	0.0	+0.7	-3	3.3	5.0	183	1470	1480	1500
.00	.00	-5	+1.3	0.0	+0.7	-4	3.6	5.1	180	1560	1580	1600
.00	.00	-5	+1.4	0.0	+0.8	-4	3.8	5.1	178	1650	1680	1700
.00	.00	-5	+1.5	0.0	+0.9	-5	4.1	5.2	176	1745	1780	1800
.00	.00	-5	+1.5	+0.1	+1.0	-5	4.3	5.3	174	1840	1880	1900
+0.1	-0.1	-5	+1.6	+0.1	+1.1	-5	4.6	5.3	172	1935	1980	2000
+0.1	-0.1	-5	+1.6	+0.1	+1.2	-6	4.9	5.4	170	2030	2080	2100
+0.1	-0.1	-5	+1.7	+0.1	+1.3	-6	5.2	5.5	168	2125	2180	2200
+0.1	-0.1	-4	+1.7	+0.2	+1.4	-6	5.5	5.5	166	2220	2280	2300
+0.1	-0.1	-4	+1.8	+0.2	+1.5	-7	5.8	5.6	165	2315	2380	2400
+0.1	-0.1	-4	+1.8	+0.3	+1.6	-7	6.0	5.7	163	2415	2480	2500
+0.1	-0.1	-4	+1.9	+0.3	+1.7	-8	6.3	5.8	162	2515	2580	2600
+0.1	-0.1	-4	+1.9	+0.3	+1.8	-8	6.6	5.9	160	2615	2685	2700
+0.1	-0.1	-3	+2.0	+0.4	+1.9	-9	6.9	5.9	159	2715	2785	2800
+0.1	-0.1	-3	+2.0	+0.4	+2.0	-9	7.2	6.0	157	2810	2885	2900
+0.1	-0.1	-3	+2.0	+0.5	+2.2	-9	7.5	6.0	156	2905	2985	3000
+0.1	-0.1	-3	+2.1	+0.5	+2.3	-10	7.9	6.1	155	3000	3085	3100
+0.1	-0.1	-3	+2.1	+0.6	+2.4	-10	8.2	6.1	153	3095	3180	3200
+0.1	-0.1	-2	+2.1	+0.6	+2.6	-11	8.5	6.2	152	3195	3280	3300
+0.1	-0.1	-2	+2.2	+0.7	+2.7	-11	8.8	6.3	150	3290	3380	3400
+0.1	-0.1	-2	+2.2	+0.8	+2.9	-11	9.1	6.3	149	3385	3480	3500
+0.1	-0.1	-2	+2.2	+0.9	+3.0	-12	9.4	6.4	148	3480	3580	3600
+0.1	-0.1	-2	+2.3	+0.9	+3.2	-12	9.7	6.4	147	3580	3680	3700
+0.1	-0.1	-1	+2.3	+1.0	+3.3	-13	10.0	6.5	146	3675	3780	3800
+0.1	-0.1	-1	+2.3	+1.0	+3.5	-13	10.4	6.6	144	3770	3880	3900
+0.1	-0.1	-1	+2.3	+1.1	+3.7	-14	10.7	6.6	143	3865	3980	4000

75-mm. GUN
SHRAPNEL, MK I
FUZE, 21-SEC. COMB.
M.V. 1755 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range c	Change in range for 1-mil change in elevation 1 mil	Time of flight Time	Probable Error			Slope of fall Slope	Line number of metro message Line	Deflection Effect	
						Range e _{pr}	Deflection e _{pd}	Height of burst			Drift Dft.	Lateral wind of 1 mile per hour (+) W-D
yds.	m	m	m	yds.	sec.	yds.	yds.	m	1/-	No.	m	m
4000	122.0	2	4.6	22	10.2	11	1	1	5.5	1	R 4	.4
4100	126.6	2	4.6	21	10.5	12	2	1	5.3	1	R 4	.4
4200	131.4	2	4.8	21	10.9	12	2	1	5.1	1	R 4	.4
4300	136.2	2	4.8	21	11.2	12	2	1	4.9	1	R 4	.5
4400	141.0	2	4.8	20	11.6	12	2	1	4.7	1	R 4	.5
4500	146.0	2	5.0	20	11.9	12	2	1	4.5	2	R 4	.5
4600	151.0	3	5.0	20	12.3	12	2	1	4.3	2	R 4	.5
4700	156.0	3	5.0	19	12.6	12	2	1	4.2	2	R 4	.5
4800	161.2	3	5.2	19	13.0	12	2	1	4.0	2	R 5	.5
4900	166.6	3	5.2	19	13.3	13	2	1	3.9	2	R 5	.5
5000	172.0	3	5.4	19	13.7	13	3	1	3.7	2	R 5	.5
5100	177.6	3	5.4	18	14.0	13	3	1	3.6	2	R 5	.5
5200	183.2	3	5.6	18	14.4	13	3	1	3.5	2	R 5	.5
5300	188.8	3	5.6	18	14.8	14	3	1	3.4	2	R 6	.6
5400	194.4	3	5.8	17	15.2	14	3	1	3.2	2	R 6	.6
5500	200.2	3	5.8	17	15.6	14	3	1	3.1	2	R 6	.6
5600	206.2	3	6.0	17	16.0	14	3	1	3.0	2	R 6	.6
5700	212.2	3	6.0	16	16.4	15	3	1	2.9	2	R 6	.6
5800	218.4	4	6.2	16	16.9	15	3	1	2.8	2	R 7	.6
5900	224.6	4	6.2	16	17.3	15	3	1	2.8	2	R 7	.6
6000	231.0	4	6.4	16	17.7	16	3	1	2.7	2	R 7	.6
6100	237.6	4	6.4	15	18.1	16	3	1	2.6	2	R 7	.6
6200	244.2	4	6.6	15	18.6	17	3	1	2.5	2	R 7	.6
6300	251.0	5	6.6	15	19.0	17	3	2	2.4	3	R 8	.7
6400	257.8	5	6.8	15	19.4	18	3	2	2.4	3	R 8	.7
6500	264.6	5	7.0	14	19.9	18	3	2	2.3	3	R 8	.7
6600	271.6	5	7.0	14	20.4	19	3	2	2.2	3	R 9	.7
6700	278.8	6	7.2	14	20.8	19	3	2	2.2	3	R 9	.7
6800	286.2	6	7.4	14	21.3	20	3		2.1	3	R 9	.7
6900	293.8	6	7.6	13	21.8	20	3		2.0	3	R 10	.7
7000	301.4	6	7.6	13	22.3	21	3		2.0	3	R 10	.7
7100	309.2	6	7.8	13	22.8	22	4		1.93	3	R 10	.7
7200	317.2	7	8.0	13	23.3	22	4		1.87	3	R 11	.7
7300	325.4	7	8.2	12	23.8	23	4		1.81	3	R 11	.8
7400	333.6	8	8.2	12	24.3	23	4		1.76	3	R 12	.8
7500	342.0	8	8.4	12	24.9	24	4		1.71	3	R 12	.8
7600	350.6	9	8.6	12	25.4	25	4		1.66	3	R 12	.8
7700	359.4	9	8.8	11	26.0	26	4		1.61	3	R 13	.8
7800	368.6	10	9.0	11	26.5	26	4		1.57	3	R 13	.8
7900	378.0	10	9.2	11	27.1	27	4		1.52	4	R 14	.8
8000	387.6	11	9.6	10	27.7	28	4		1.47	4	R 14	.8

*Drift includes side jump.

75-mm. GUN
SHRAPNEL, MK. I
FUZE, 21-SEC. COMB.
M.V. 1755 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	24	25	1
Comp. angle of site for each		Range Effect of Increase of—						Displ. of burst for change of 5 points in fuze setter corrector in		Range setting for 1897 Carriage (and Mods.)	Range setting for M2 Carriage	Range
+1 mil of site	-1 mil of site	One percent in wt. of projectile. Std. wt. 15.96 lbs.	One foot per sec. in M.V.	Air temperature 1 degree. Standard is 59° F.	Rear wind 1 mile per hour	One percent in air density	Fuze setting for graze burst					
Wt.	VE	Temp.	W-R	Den.	F.S.	Height	Range	Set.	Set.	R		
yd.	yd.	yd.	yd.	yd.	yd.	sec.	ft.	m.	yd.	yd.	yd.	yd.
+01	-01	-1	+2.3	+1.1	+3.7	-14	10.7	6.6	143	3865	3980	4000
+01	-01	-1	+2.4	+1.2	+3.8	-14	11.0	6.7	142	3965	4080	4100
+01	-01	-1	+2.4	+1.2	+4.0	-15	11.4	6.7	140	4060	4185	4200
+02	-02	0	+2.4	+1.3	+4.2	-15	11.7	6.8	139	4155	4285	4300
+02	-02	0	+2.5	+1.4	+4.4	-15	12.1	6.9	138	4250	4385	4400
+02	-02	0	+2.5	+1.4	+4.6	-16	12.4	6.9	137	4345	4485	4500
+02	-02	0	+2.5	+1.5	+4.8	-16	12.8	7.0	136	4440	4585	4600
+02	-02	+1	+2.5	+1.6	+5.0	-16	13.1	7.1	135	4535	4685	4700
+02	-02	+1	+2.6	+1.7	+5.2	-17	13.5	7.1	134	4630	4785	4800
+02	-02	+1	+2.6	+1.7	+5.4	-17	13.8	7.2	133	4725	4885	4900
+02	-02	+2	+2.6	+1.8	+5.6	-18	14.2	7.2	132	4820	4985	5000
+02	-02	+2	+2.6	+1.9	+5.8	-18	14.5	7.3	131	4915	5090	5100
+03	-03	+2	+2.6	+1.9	+6.0	-19	14.9	7.3	130	5010	5190	5200
+03	-03	+2	+2.6	+2.0	+6.2	-19	15.2	7.4	129	5100	5290	5300
+03	-03	+3	+2.7	+2.1	+6.4	-20	15.6	7.4	128	5190	5390	5400
+03	-03	+3	+2.7	+2.2	+6.7	-20	16.0	7.5	127	5280	5485	5500
+03	-03	+3	+2.7	+2.2	+6.9	-21	16.3	7.5	126	5370	5585	5600
+03	-03	+4	+2.7	+2.3	+7.1	-21	16.7	7.6	125	5460	5685	5700
+04	-04	+4	+2.7	+2.4	+7.3	-22	17.1	7.7	125		5785	5800
+04	-04	+4	+2.7	+2.5	+7.5	-22	17.5	7.7	124		5885	5900
+04	-04	+5	+2.8	+2.6	+7.8	-23	17.9	7.8	123		5985	6000
+04	-04	+5	+2.8	+2.6	+8.0	-23	18.3	7.8	122		6090	6100
+05	-05	+6	+2.8	+2.7	+8.2	-24	18.7	7.9	121		6190	6200
+05	-05	+6	+2.8	+2.8	+8.4	-24	19.1	8.0	121		6290	6300
+06	-06	+7	+2.8	+2.9	+8.7	-25	19.5	8.0	120		6390	6400
+06	-06	+7	+2.8	+3.0	+8.9	-25	20.0	8.1	119		6490	6500
+07	-07	+7	+2.9	+3.0	+9.2	-26	20.5	8.1	118		6590	6600
+07	-07	+8	+2.9	+3.1	+9.4	-26	21.0	8.2	117		6690	6700
+08	-08	+8	+2.9	+3.2	+9.6	-27					6790	6800
+08	-08	+9	+2.9	+3.3	+9.8	-27					6890	6900
+09	-09	+9	+2.9	+3.4	+10.1	-28					6990	7000
+10	-10	+9	+3.0	+3.5	+10.3	-28					7090	7100
+11	-11	+10	+3.0	+3.5	+10.6	-29					7190	7200
+12	-11	+10	+3.0	+3.6	+10.8	-29					7290	7300
+13	-12	+11	+3.0	+3.7	+11.0	-30					7390	7400
+14	-13	+11	+3.0	+3.8	+11.3	-30					7490	7500
+16	-14	+11	+3.0	+3.9	+11.5	-31					7590	7600
+17	-15	+12	+3.0	+4.0	+11.7	-31					7690	7700
+19	-17	+12	+3.0	+4.0	+12.0	-32					7790	7800
+20	-18	+13	+3.1	+4.1	+12.2	-32					7890	7900
+22	-19	+13	+3.1	+4.2	+12.5	-33					7990	8000

NOTE: Since the shrapnel is armed with a 21-sec. comb. fuze, all information which is based on the behavior of this fuze is not given beyond the point where the fuze setting for graze burst is 21-sec.

Quadrant aiming

75-mm. GUN
SHRAPNEL, MK I
FUZE, 21-SEC. COMB.
M.V. 1755 f/s

TABLE B

Change in velocity due to change in temperature of powder

Temperature of Powder, F.	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second....	-51	-44	-37	-29	-22	-15	-8	0	+8	+16	+25

TABLE C

Cant of carriage axle, deflection effect in mils due to

Range, yards.....	1000	2000	3000	4000	5000	6000	7000
Cant of 10 mils, effect	0.2	0.4	0.8	1.2	1.7	2.3	3.0
Range, yards.....	7000	8000	9000	9500	9600	9700	9760
Cant of 10 mils, effect ..	3.0	4.0	5.4	6.8	7.3	8.0	9.5

NOTE: Right wheel above left causes left deflection effect.

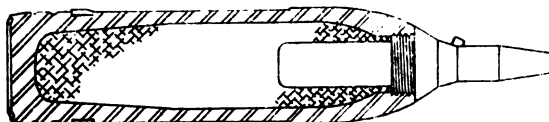
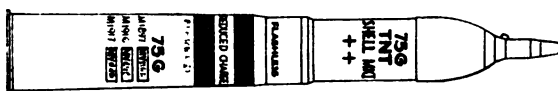
The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be so cross leveled.

75-mm. GUN
HE Shell, Mk. I
Fuze M46 and M47
Reduced Charge

NOTE: Elevations given in this table are for Carriage M2. Add 1.0 mils to these elevations when firing from Carriage M1897 (and modifications).

Maximum Range: 6700 yards

Data for ranges 0—6700 included herein



75-mm. GUN
SHELL, H.E., MK. I
FUZE, M46, M47
REDUCED CHARGE
M.V. 1115 F/S

TABLE A

1	2	3	4	5	6	7 8		9	10	11 12	
Range R yds	Elevation El m	Fork F m	Change in elevation for 100-yd. change in range c m	Change in range for 1-mil change in elevation 1 mil yds	Time of flight Time sec	Probable Error		Slope of fall Slope 1/-	Line number of metro message L No	Deflection Effect	
						Range e _{pr} yds	Deflection e _{pd} yds			Drift* Dft m	Lateral wind of 1 mile per hour (+) W-D m
0	-1.2	1	4.0	25	0.0	8	0		1	L 1	0.0
100	+2.8	1	4.0	25	0.3	8	0	239	1	L 1	0.0
200	6.8	1	4.2	24	0.6	8	0	119	1	L 1	0.0
300	11.0	1	4.2	24	0.8	9	0	78	1	L 1	0.0
400	15.2	1	4.2	24	1.1	9	0	58	1	L 1	0.0
500	19.6	2	4.4	23	1.4	9	0	45	1	L 1	0.0
600	24.0	2	4.4	23	1.7	9	0	37	1	0	0.0
700	28.4	2	4.6	22	2.0	9	0	31	1	0	0.0
800	33.0	2	4.6	22	2.3	10	0	27	1	0	0.0
900	37.8	2	4.8	21	2.6	10	0	24	1	0	0.0
1000	42.6	2	4.8	21	2.9	10	0	21	1	0	0.1
1100	47.4	2	4.8	21	3.2	10	0	18.8	1	0	0.1
1200	52.4	2	5.0	20	3.5	10	0	16.9	1	0	0.1
1300	57.4	2	5.0	20	3.9	11	0	15.3	1	0	0.1
1400	62.6	2	5.2	20	4.2	11	0	14.0	1	R 1	0.1
1500	67.8	2	5.2	19	4.5	11	0	12.9	1	R 1	0.1
1600	73.2	2	5.4	19	4.8	11	0	11.9	1	R 1	0.1
1700	78.6	2	5.4	19	5.2	11	0	11.0	1	R 1	0.1
1800	84.2	3	5.6	18	5.5	12	0	10.3	1	R 1	0.1
1900	89.8	3	5.6	18	5.9	12	0	9.6	1	R 1	0.1
2000	95.4	3	5.8	18	6.2	12	0	9.0	1	R 1	0.1
2100	101.2	3	5.8	17	6.6	12	0	8.4	1	R 1	0.1
2200	107.2	3	6.0	17	6.9	13	0	7.9	1	R 1	0.2
2300	113.2	3	6.0	17	7.3	13	0	7.4	1	R 1	0.2
2400	119.4	3	6.2	16	7.6	13	0	7.0	1	R 1	0.2
2500	125.6	3	6.2	16	8.0	13	0	6.6	1	R 1	0.2
2600	132.0	3	6.4	16	8.4	14	0	6.3	1	R 1	0.2
2700	138.4	3	6.6	15	8.8	14	0	6.0	1	R 1	0.2
2800	145.0	4	6.6	15	9.1	14	0	5.7	1	R 2	0.2
2900	151.8	4	6.8	15	9.5	14	0	5.4	1	R 2	0.2
3000	158.6	4	6.8	15	9.9	15	0	5.1	1	R 2	0.2
3100	165.6	4	7.0	14	10.3	15	0	4.9	1	R 2	0.2
3200	172.6	4	7.2	14	10.7	15	0	4.7	1	R 3	0.2
3300	179.8	5	7.4	14	11.1	15	0	4.4	1	R 3	0.3
3400	187.2	5	7.4	14	11.5	16	0	4.2	1	R 3	0.3
3500	194.8	5	7.6	13	11.9	16	0	4.1	1	R 4	0.3
3600	202.4	5	7.8	13	12.3	16	0	3.9	2	R 4	0.3
3700	210.2	5	8.0	13	12.8	17	0	3.7	2	R 4	0.3
3800	218.2	6	8.2	12	13.2	17	0	3.6	2	R 5	0.3
3900	226.4	6	8.2	12	13.7	17	0	3.4	2	R 5	0.3
4000	234.6	6	8.4	12	14.1	18	0	3.3	2	R 5	0.3

*Drift includes side jump.

75-mm. GUN
SHELL, H.E., MK. I
FUZE, M46, M47
REDUCED CHARGE
M.V. 1115 F/S

TABLE A

13	14	15	16	17	18	19	20	21	1
Complementary angle of site for each		Range effect of increase of							
+ 1 mil of site	- 1 mil of site	One + in wt of proj. Two + + is stan- dard.	One foot per sec in M.V.	Air temperature 1 degree. Standard is 59° F.	Rear wind 1 mile per hour	One percent in air density	Range setting for 1897 carriage (and Mods.)	Range setting for M2 carriage	Range
m	m	Wt yds	VE yds	Temp yds	W-R yds	Den yds	Set m	Set yds	R yds
.00	.00	0	0.0	0.0	0.0	0	85	-145	0
.00	.00	- 2	+0.2	0.0	0.0	0	340	+130	100
.00	.00	- 3	+0.3	+0.1	0.0	0	535	380	200
.00	.00	- 5	+0.5	+0.1	+0.1	0	715	605	300
.00	.00	- 6	+0.6	+0.2	+0.1	0	890	815	400
.00	.00	- 8	+0.8	+0.2	+0.1	-1	1065	1010	500
.00	.00	- 9	+0.9	+0.3	+0.2	-1	1230	1195	600
.00	.00	-10	+1.0	+0.3	+0.2	-1	1390	1375	700
.00	.00	-11	+1.1	+0.4	+0.3	-1	1550	1550	800
.00	.00	-12	+1.2	+0.4	+0.3	-1	1605	1720	900
.00	.00	-13	+1.3	+0.5	+0.4	-1	1860	1890	1000
.00	.00	-14	+1.4	+0.6	+0.5	-1	2010	2050	1100
+.01	.00	-14	+1.5	+0.6	+0.6	-2	2160	2205	1200
+.01	.00	-15	+1.6	+0.7	+0.8	-2	2310	2360	1300
+.01	.00	-15	+1.7	+0.7	+0.9	-2	2455	2515	1400
+.01	.00	-16	+1.8	+0.8	+1.0	-2	2600	2665	1500
+.01	-.01	-16	+1.8	+0.9	+1.2	-2	2740	2810	1600
+.01	-.01	-16	+1.9	+0.9	+1.3	-2	2880	2955	1700
+.02	-.01	-17	+2.0	+1.0	+1.5	-3	3020	3100	1800
+.02	-.01	-17	+2.0	+1.0	+1.6	-3	3155	3240	1900
+.02	-.02	-17	+2.1	+1.1	+1.8	-3	3290	3380	2000
+.02	-.02	-17	+2.2	+1.2	+2.0	-3	3425	3520	2100
+.02	-.02	-17	+2.2	+1.2	+2.2	-4	3560	3660	2200
+.03	-.02	-16	+2.3	+1.3	+2.3	-4	3690	3800	2300
+.03	-.03	-16	+2.3	+1.3	+2.5	-4	3820	3935	2400
+.03	-.03	-16	+2.4	+1.4	+2.7	-4	3950	4070	2500
+.03	-.03	-16	+2.4	+1.5	+2.9	-5	4080	4205	2600
+.04	-.03	-15	+2.5	+1.5	+3.1	-5	4210	4340	2700
+.04	-.04	-15	+2.5	+1.6	+3.3	-5	4335	4470	2800
+.04	-.04	-14	+2.5	+1.6	+3.5	-6	4460	4605	2900
+.04	-.04	-14	+2.6	+1.7	+3.7	-6	4585	4735	3000
+.05	-.05	-13	+2.6	+1.8	+3.9	-6	4710	4870	3100
+.05	-.05	-13	+2.7	+1.9	+4.1	-7	4835	5000	3200
+.05	-.05	-12	+2.7	+1.9	+4.4	-7	4955	5135	3300
+.06	-.06	-12	+2.7	+2.0	+4.6	-7	5075	5265	3400
+.06	-.06	-12	+2.8	+2.1	+4.8	-8	5195	5395	3500
+.06	-.06	-11	+2.8	+2.2	+5.0	-8	5320	5530	3600
+.07	-.07	-11	+2.8	+2.2	+5.3	-8	5440	5660	3700
+.07	-.07	-10	+2.9	+2.3	+5.5	-9	5560	5790	3800
+.08	-.08	-10	+2.9	+2.3	+5.8	-9		5920	3900
+.08	-.08	-10	+3.0	+2.4	+6.0	-9		6050	4000

75-mm. GUN
SHELL, H.E., MK. I
FUZE, M46, M47
REDUCED CHARGE
M.V. 1115 F/S

TABLE A

[illegible]

*Drift includes side jump.

TABLE A

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75-mm. GUN
SHELL, H.E., MK. I
FUZE, M46, M47.
REDUCED CHARGE
M.V. 1115 F/S

TABLE B

Change in velocity due to change in temperature of powder

Temperature of Powder, F.	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second	-23	-20	-17	-14	-11	-7	-4	0	+3	+7	+11

TABLE C

Cant of Carriage axle, deflection effect in mils due to

Range, yards	1000	2000	3000	4000	5000	5500	6000
Cant of 10 mils, effect	0.4	0.9	1.6	2.4	3.4	4.0	4.9
Range, yards	6000	6500	6600	6700	6775	6700	
Cant of 10 mils, effect	4.9	6.3	6.7	7.4	8.7	10.4	

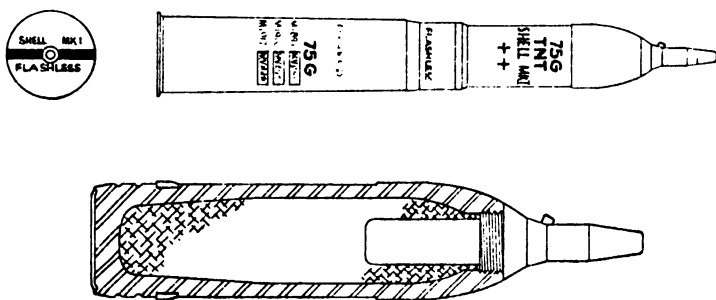
NOTE: Right wheel above left causes left deflection effect. The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be so cross leveled.

75-mm. GUN
HE Shell, Mark I
Fuze M46 and M47
Normal Charge

NOTE: Elevations given in this table are for Carriage M2. Add 0.8 mil to these elevations when firing from Carriage M1897 (and modifications).

Maximum Range: 9000 yards.

Data for ranges 0—8000 included herein.



75-mm. GUN
SHELL, H.E., MK I
FUZE, M46, M47
NORMAL CHARGE
M.V. 1784 f/s

TABLE A

1	2	3	4	5	6	7 8		9	10	11 12	
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range c	Change in range for 1-mil change in elevation 1 mil	Time of flight Time	Probable Error		Slope of fall	Line number of metro message	Deflection Effect	
yds.	m	m	m	yds.	sec.	Range e _{pr}	Deflection e _{pd}	Slope 1/-	Line No.	Drift* m	Lateral wind of 1 mile per hour(+) W-D m
0	-1.0	1	1.4	71	0.0	10	0		1	L 1	.0
100	+0.6	1	1.4	68	0.1	10	0	599	1	L 1	.0
200	2.2	1	1.6	65	0.3	10	0	291	1	L 1	.0
300	3.8	1	1.6	62	0.5	11	0	189	1	0	.0
400	5.6	1	1.8	59	0.7	11	0	140	1	0	.0
500	7.4	1	1.8	56	0.9	11	0	109	1	0	.0
600	9.2	1	2.0	53	1.1	11	0	88	1	0	.0
700	11.2	1	2.0	50	1.3	11	0	73	1	0	.0
800	13.2	1	2.2	48	1.5	11	0	62	1	0	.0
900	15.2	1	2.2	46	1.7	12	0	53	1	0	.0
1000	17.4	1	2.2	44	1.9	12	0	46	1	0	.0
1100	19.6	1	2.4	42	2.2	12	0	41	1	0	.0
1200	22.0	1	2.4	41	2.4	12	0	36	1	0	.0
1300	24.4	1	2.6	39	2.6	12	0	32	1	0	.0
1400	27.0	1	2.6	38	2.9	12	0	29	1	0	.0
1500	29.6	1	2.8	37	3.1	13	0	26	1	R 1	.1
1600	32.4	1	2.8	36	3.4	13	0	23	1	R 1	.1
1700	35.2	2	3.0	34	3.6	13	0	21	1	R 1	.1
1800	38.0	2	3.0	33	3.9	13	0	19.2	1	R 1	.1
1900	41.0	2	3.2	32	4.2	14	0	17.6	1	R 1	.1
2000	44.2	2	3.2	31	4.4	14	0	16.2	1	R 1	.1
2100	47.4	2	3.4	30	4.7	14	0	14.9	1	R 1	.1
2200	50.8	2	3.4	29	5.0	14	0	13.8	1	R 1	.1
2300	54.2	2	3.6	28	5.3	15	0	12.8	1	R 1	.1
2400	57.8	2	3.6	27	5.6	15	0	11.9	1	R 2	.1
2500	61.4	2	3.8	26	5.9	15	0	11.1	1	R 2	.1
2600	65.2	2	3.8	26	6.2	15	0	10.4	1	R 2	.1
2700	69.0	3	4.0	25	6.5	16	0	9.8	1	R 2	.2
2800	73.0	3	4.0	25	6.8	16	0	9.2	1	R 2	.2
2900	77.0	3	4.2	24	7.1	16	0	8.6	1	R 2	.2
3000	81.2	3	4.2	24	7.4	17	0	8.1	1	R 2	.2
3100	85.4	3	4.4	23	7.7	17	0	7.7	1	R 2	.2
3200	89.8	3	4.4	23	8.1	17	0	7.3	1	R 3	.2
3300	94.4	3	4.6	22	8.4	18	0	6.9	1	R 3	.2
3400	99.0	3	4.6	22	8.7	18	0	6.6	1	R 3	.2
3500	103.6	3	4.8	21	9.1	18	0	6.3	1	R 3	.2
3600	108.4	4	4.8	21	9.4	19	0	6.0	1	R 3	.2
3700	113.4	4	5.0	20	9.8	19	0	5.7	1	R 3	.2
3800	118.4	4	5.0	20	10.1	20	0	5.4	1	R 3	.3
3900	123.4	4	5.2	19	10.5	20	0	5.2	1	R 4	.3
4000	128.6	4	5.2	19	10.8	21	0	5.0	1	R 4	.3

*Drift includes side jump.

75-mm. GUN
SHELL, H.E., MK I
FUZE, M46, M47
NORMAL CHARGE
M.V. 1784 f/s

TABLE A

13	14	15	16	17	18	19	20	21	1
Complementary angle of site for each		Range Effect of Increase of—					Range setting for 1897 Carriage (and M46.)	Range setting for M2 Carriage	Range R
+ 1 mil of site	- 1 mil of site	One + in wt. of proj. Two + + is stan- dard. Wt.	One foot per sec. in M.V. VE	Air temperature 1 degree. Standard is 59° F. Temp.	Rear wind 1 mile per hour W-R	One percent in air density Den.			
m	m	yds.	yds.	yds.	yds.	yds.	m.	yds.	yds.
.00	.00	0	0.0	0.0	0.0	0	85	-135	0
.00	.00	- 1	+0.1	0.0	0.0	0	170	- 30	100
.00	.00	- 2	+0.2	0.0	0.0	0	270	+ 75	200
.00	.00	- 4	+0.3	0.0	0.0	- 1	360	170	300
.00	.00	- 5	+0.4	0.0	0.0	- 1	445	270	400
.00	.00	- 6	+0.5	0.0	0.0	- 1	530	380	500
.00	.00	- 7	+0.6	0.0	+0.1	- 2	615	490	600
.00	.00	- 7	+0.7	0.0	+0.1	- 2	700	600	700
.00	.00	- 8	+0.7	+0.1	+0.1	- 2	785	700	800
.00	.00	- 8	+0.8	+0.1	+0.2	- 2	870	800	900
.00	.00	- 8	+0.9	+0.1	+0.2	- 3	960	900	1000
.00	.00	- 9	+0.9	+0.1	+0.3	- 3	1050	1000	1100
.00	.00	- 9	+1.0	+0.1	+0.3	- 3	1140	1100	1200
.00	.00	- 9	+1.1	+0.2	+0.4	- 3	1230	1205	1300
.00	.00	- 9	+1.1	+0.2	+0.4	- 4	1320	1305	1400
.00	.00	-10	+1.2	+0.2	+0.5	- 4	1410	1410	1500
.00	.00	-10	+1.3	+0.2	+0.5	- 4	1510	1520	1600
.00	.00	-10	+1.3	+0.2	+0.6	- 5	1605	1625	1700
+.01	-.01	-10	+1.4	+0.3	+0.7	- 5	1700	1725	1800
+.01	-.01	-10	+1.5	+0.3	+0.8	- 6	1800	1830	1900
+.01	-.01	-10	+1.5	+0.3	+0.9	- 6	1900	1935	2000
+.01	-.01	- 9	+1.6	+0.3	+1.0	- 6	2000	2040	2100
+.01	-.01	- 9	+1.6	+0.4	+1.1	- 7	2105	2145	2200
+.01	-.01	- 9	+1.6	+0.4	+1.2	- 7	2205	2250	2300
+.01	-.01	- 8	+1.7	+0.4	+1.3	- 8	2305	2360	2400
+.01	-.01	- 8	+1.7	+0.5	+1.5	- 8	2410	2470	2500
+.01	-.01	- 7	+1.7	+0.5	+1.6	- 8	2515	2575	2600
+.01	-.01	- 7	+1.7	+0.6	+1.8	- 9	2615	2685	2700
+.02	-.02	- 6	+1.8	+0.6	+1.9	- 9	2720	2795	2800
+.02	-.02	- 5	+1.8	+0.7	+2.1	-10	2825	2905	2900
+.02	-.02	- 4	+1.8	+0.7	+2.3	-10	2930	3015	3000
+.02	-.02	- 3	+1.8	+0.8	+2.5	-10	3035	3120	3100
+.02	-.02	- 2	+1.8	+0.8	+2.7	-11	3140	3230	3200
+.02	-.02	- 1	+1.9	+0.9	+2.8	-11	3250	3345	3300
+.02	-.02	0	+1.9	+0.9	+3.0	-12	3360	3460	3400
+.02	-.02	0	+1.9	+1.0	+3.2	-12	3470	3570	3500
+.02	-.02	+ 1	+1.9	+1.0	+3.4	-13	3580	3680	3600
+.03	-.03	+ 2	+1.9	+1.1	+3.6	-13	3685	3790	3700
+.03	-.03	+ 3	+1.9	+1.1	+3.7	-14	3790	3900	3800
+.03	-.03	+ 4	+1.9	+1.2	+3.9	-14	3895	4015	3900
+.03	-.03	+ 5	+2.0	+1.3	+4.1	-15	4000	4125	4000

75-mm. GUN
SHELL, H.E., MK I
FUZE, M46, M47
NORMAL CHARGE
M.V. 1784 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range c	Change in range for 1-mil. change in elevation 1 mil	Time of flight Time	Probable Error		Slope of fall Slope	Line number of metro message Line	Deflection Effect	
						Range e _{pr}	Deflection e _{pd}			Drift* Dft.	Lateral wind of 1 mile per hour(+) W-D
yds.	m	m	m	yds.	sec.	yds.	yds.	1/-	No.	m	m
4000	128.6	4	5.2	19	10.8	21	.0	5.0	1	R 4	.3
4100	134.0	5	5.4	19	11.2	21	0	4.8	1	R 4	.3
4200	139.4	5	5.4	18	11.6	22	0	4.6	1	R 4	.3
4300	144.8	5	5.6	18	12.0	22	0	4.4	2	R 4	.3
4400	150.4	5	5.8	18	12.3	22	1	4.2	2	R 4	.3
4500	156.2	5	5.8	17	12.7	23	1	4.0	2	R 5	.3
4600	162.2	6	6.0	17	13.1	23	1	3.9	2	R 5	.3
4700	168.2	6	6.0	17	13.5	24	1	3.7	2	R 5	.4
4800	174.4	6	6.2	16	13.9	24	1	3.6	2	R 5	.4
4900	180.6	6	6.4	16	14.3	25	1	3.5	2	R 5	.4
5000	187.0	6	6.4	16	14.7	25	1	3.3	2	R 6	.4
5100	193.4	7	6.6	15	15.1	26	1	3.2	2	R 6	.4
5200	200.0	7	6.6	15	15.6	26	1	3.1	2	R 6	.4
5300	206.8	7	6.8	15	16.0	27	1	3.0	2	R 6	.4
5400	213.6	8	7.0	14	16.4	27	1	2.9	2	R 7	.4
5500	220.6	8	7.0	14	16.9	28	1	2.8	2	R 7	.5
5600	227.8	8	7.2	14	17.3	29	1	2.7	2	R 7	.5
5700	235.0	9	7.4	13	17.7	29	1	2.6	2	R 8	.5
5800	242.4	9	7.6	13	18.2	30	1	2.5	2	R 8	.5
5900	250.0	9	7.6	13	18.7	30	1	2.4	2	R 8	.5
6000	257.6	10	7.8	13	19.1	31	1	2.3	3	R 9	.5
6100	265.6	10	8.0	12	19.6	32	1	2.3	3	R 9	.5
6200	273.6	11	8.2	12	20.1	32	1	2.2	3	R 9	.6
6300	281.8	11	8.4	12	20.5	33	1	2.1	3	R 10	.6
6400	290.2	12	8.6	12	21.0	34	1	2.0	3	R 10	.6
6500	298.8	12	8.8	11	21.5	34	1	1.98	3	R 11	.6
6600	307.6	13	9.0	11	22.0	35	1	1.92	3	R 11	.6
6700	316.6	13	9.2	11	22.6	36	1	1.86	3	R 12	.6
6800	325.8	14	9.4	11	23.1	37	1	1.80	3	R 12	.7
6900	335.4	14	9.6	10	23.6	37	1	1.74	3	R 13	.7
7000	345.2	15	9.8	10	24.2	38	1	1.68	3	R 13	.7
7100	355.2	16	10.0	10	24.7	39	1	1.63	3	R 14	.7
7200	365.4	16	10.2	10	25.2	40	1	1.58	3	R 14	.7
7300	376.0	17	10.6	9	25.8	41	1	1.53	3	R 15	.8
7400	386.8	18	10.8	9	26.4	42	1	1.48	4	R 15	.8
7500	398.0	19	11.2	9	27.1	43	1	1.43	4	R 16	.8
7600	409.4	21	11.6	9	27.7	44	1	1.38	4	R 16	.8
7700	421.2	22	12.0	8	28.3	45	1	1.34	4	R 17	.8
7800	433.4	23	12.4	8	29.0	46	1	1.29	4	R 18	.8
7900	446.0	24	13.0	8	29.7	47	1	1.25	4	R 19	.9
8000	459.2	26	13.6	7	30.4	48	1	1.20	4	R 20	.9

*Drift includes side jump.

75-mm. GUN
SHELL, H.E., MK I
FUZE, M46, M47
NORMAL CHARGE
M.V. 1784 f/s

TABLE A

13	14	15	16	17	18	19	20	21	1
Complementary angle of site for each		Range Effect of Increase of—					Range setting for 1897 Carriage (and Mods.)	Range setting for M2 Carriage	Range
+ 1 mil of site	— 1 mil of site	One + in wt. of proj. Two + + is standard.	One foot per sec. in M.V.	Air temperature 1 degree. Standard is 59° F.	Rear wind 1 mile per hour	One percent in air density			
Wt.	VE	Temp.	W-R	Den.	Set.	Set.	R		
yds.	yds.	yds.	yds.	yds.	yds.	m.	yds.	yds.	
+ .03	— .03	+ 5	+2.0	+1.3	+ 4.1	—15	4000	4125	4000
+ .03	— .03	+ 6	+2.0	+1.3	+ 4.3	—15	4105	4240	4100
+ .03	— .03	+ 7	+2.0	+1.4	+ 4.5	—16	4210	4350	4200
+ .04	— .04	+ 8	+2.0	+1.5	+ 4.7	—16	4320	4455	4300
+ .04	— .04	+ 9	+2.0	+1.5	+ 4.9	—17	4430	4565	4400
+ .04	— .04	+10	+2.0	+1.6	+ 5.1	—17	4540	4680	4500
+ .04	— .04	+11	+2.0	+1.6	+ 5.3	—17	4645	4795	4600
+ .04	— .04	+12	+2.0	+1.7	+ 5.5	—18	4750	4910	4700
+ .05	— .05	+13	+2.1	+1.8	+ 5.7	—18	4855	5025	4800
+ .05	— .05	+14	+2.1	+1.9	+ 5.9	—19	4955	5140	4900
+ .05	— .05	+16	+2.1	+2.0	+ 6.2	—19	5060	5250	5000
+ .05	— .05	+17	+2.1	+2.1	+ 6.4	—20	5165	5360	5100
+ .05	— .05	+18	+2.1	+2.2	+ 6.7	—20	5270	5475	5200
+ .06	— .06	+20	+2.1	+2.2	+ 6.9	—21	5375	5590	5300
+ .06	— .06	+21	+2.1	+2.3	+ 7.1	—21	5480	5700	5400
+ .06	— .06	+22	+2.1	+2.4	+ 7.4	—22	5575	5815	5500
+ .07	— .07	+23	+2.1	+2.5	+ 7.6	—22	Quadrant aiming	5935	5600
+ .07	— .07	+24	+2.1	+2.6	+ 7.9	—23		6050	5700
+ .07	— .07	+26	+2.2	+2.6	+ 8.1	—23		6165	5800
+ .08	— .08	+27	+2.2	+2.7	+ 8.3	—24		6275	5900
+ .08	— .08	+28	+2.2	+2.8	+ 8.6	—25		6385	6000
+ .09	— .09	+29	+2.2	+2.9	+ 8.8	—25		6500	6100
+ .09	— .09	+31	+2.2	+3.0	+ 9.1	—26		6615	6200
+ .10	— .10	+32	+2.2	+3.0	+ 9.3	—26		6730	6300
+ .10	— .10	+34	+2.2	+3.1	+ 9.5	—27		6840	6400
+ .11	— .11	+35	+2.2	+3.2	+ 9.8	—27		6955	6500
+ .11	— .11	+36	+2.3	+3.3	+10.0	—28		7070	6600
+ .12	— .12	+37	+2.3	+3.4	+10.3	—28		7185	6700
+ .13	— .12	+39	+2.3	+3.4	+10.5	—29		7300	6800
+ .14	— .13	+40	+2.3	+3.5	+10.8	—29		7415	6900
+ .15	— .14	+41	+2.3	+3.6	+11.1	—30		7530	7000
+ .16	— .14	+42	+2.3	+3.7	+11.4	—30		7640	7100
+ .17	— .15	+44	+2.4	+3.8	+11.7	—31		7760	7200
+ .18	— .16	+45	+2.4	+3.8	+11.9	—31		7870	7300
+ .19	— .17	+47	+2.4	+3.9	+12.2	—32		7985	7400
+ .20	— .18	+48	+2.4	+4.0	+12.5	—33		8095	7500
+ .21	— .19	+49	+2.4	+4.1	+12.8	—33		8205	7600
+ .23	— .20	+51	+2.5	+4.2	+13.1	—34		8315	7700
+ .25	— .21	+52	+2.5	+4.2	+13.4	—34		8425	7800
+ .27	— .23	+54	+2.5	+4.3	+13.7	—35		8535	7900
+ .30	— .25	+55	+2.5	+4.3	+14.0	—36		8640	8000

75-mm. GUN
SHELL, H.E., MK I
FUZE, M46, M47
NORMAL CHARGE
M.V. 1784 f/s

TABLE B

Change in velocity due to change in temperature of powder

Temperature of Powder, F.	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second....	-51	-44	-37	-29	-22	-15	-8	0	+8	+16	+25

TABLE C

Cant of carriage axle, deflection effect in mils due to

Range, yards.	1000	2000	3000	4000	5000	6000	7000
Cant of 10 mils, effect	0.2	0.4	0.8	1.3	1.9	2.6	3.5
Range, yards.	7000	8000	8500	9000	9070	9000	
Cant of 10 mils, effect	3.5	4.8	5.8	7.7	8.9	10.4	

NOTE: Right wheel above left causes left deflection effect.

The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be so cross leveled.

SECTION IV

FIRING TABLES, 105-mm. HOWITZER CHARACTERISTICS

105-mm. Howitzer, M2 and M2A1, Firing Shell HE Mark I 105-mm. Howitzer, M2 and M2A1

Diameter of bore between lands	inches	4.134
Diameter of bore between grooves	inches	4.194
Total length	inches	101.35
Length of rifled portion	inches	77.37
Travel of projectile	inches	81.67
Capacity of powder chamber	cubic inches	153
Number of grooves		36
Character of rifling, uniform twist	1 in 20 calibers	
Maximum pressure for which gun is designed	lb./sq. in.	28,000

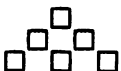
105-mm. Howitzer Carriage, M1, M1A1, and M2

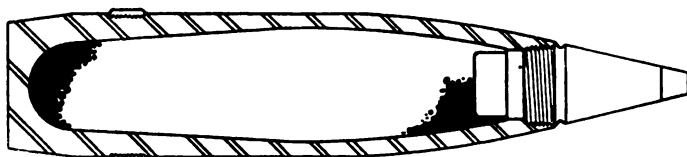
Total traverse (one half on each side)	mils	800
Least possible elevation	mils	-88.9
Greatest possible elevation	mils	+1155.6
Traverse for one turn of traversing handwheel	mils	20
Change in elevation for one turn of elevating handwheel	mils	10

Projectile—mean weight of fuze'd projectile in pounds.

P. D. Fuze M48 and Comb. Time S. Q. Fuze M54.

Variations in weight are indicated by markings stenciled on the projectile as follows:

Marking	Weight
 (Standard)	32.4
.....	33.0
.....	33.6



105-mm. HOWITZER

Characteristics of Charges (HE Shell, Mark I, Fuzes M48 and M54) for which tables are not included herein.

- CHARGE 1** Muzzle Velocity 650 f/s.
Powder Charge Base Section Number 1.
Maximum Range: 3825 yards.
- CHARGE 2** Muzzle Velocity 710 f/s.
Powder Charge: Base Section and Increment Number 2.
Maximum Range: 4475.
- CHARGE 4** Muzzle Velocity 875 f/s.
Powder Charge: Base Section and Increments Number 2, 3, and 4.
Maximum Range: 6430.
- CHARGE 6** Muzzle Velocity 1235 f/s.
Powder Charge: Base Section and Increments Number 2, 3, 4, 5 and 6.
Maximum Range: 10,150.

CHARGE 3

105-mm. HOWITZER

HE Shell, Mark I

Fuze, P. D., M48

Fuze, Comb. Time S. Q., M54

Powder Charge: Base Section and Increments Number 2 and 3.

Maximum Range: 5280.

(Data for ranges 0—5280 included herein.)

NOTE: Columns 9, 21, 22, and 23 are used only when the Shell, M1, is armed with the Combination Fuze, M54, and do not extend beyond the maximum fuze setting, which is 25 seconds.

mm. HOW.
 LL, H. E., MK I
 FUZE, M48, M54
 CHARGE 3
 M. V., 780 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation EI	Fork F	Change in elevation for 100-yd. change in range C	Change in range for 1 mil change in elevation mil	Time of flight Time	Probable Error			Slope of fall Slope	Line number of metro message L	Deflection Effect	
						Range e _{pr}	Deflection e _{pd}	Height of burst			Drift Dft	Lateral wind of MPH (+) W-D
yds.	m	m	m	yds.	sec.	yds.	yds.	m	1/-	No.	m	m
0	- 0.2	1	7.4	14	0.0	4	0	0		1	0	0.0
100	+ 7.2	1	7.6	13	0.4	4	0	0	126	1	0	0.0
200	14.8	1	7.6	13	0.8	5	0	0	62	1	0	0.0
300	22.6	2	7.8	13	1.1	5	0	0	41	1	0	0.0
400	30.4	2	8.0	13	1.5	6	0	0	31	1	0	0.0
500	38.4	2	8.0	13	1.9	6	0	0	24	1	0	0.0
600	46.4	2	8.2	12	2.3	7	0	0	20	1	0	0.0
700	54.6	2	8.2	12	2.7	7	0	0	17.1	1	0	0.0
800	63.0	3	8.4	12	3.1	8	0	1	14.8	1	1	0.0
900	71.4	3	8.6	12	3.5	8	0	1	13.0	1	1	0.0
1000	80.0	3	8.6	12	3.9	9	0	1	11.6	1	1	0.0
1100	88.8	3	8.8	11	4.3	10	0	1	10.5	1	1	0.0
1200	97.6	4	8.8	11	4.8	10	0	1	9.5	1	1	0.0
1300	106.6	4	9.0	11	5.2	11	0	1	8.7	1	1	0.0
1400	115.6	4	9.2	11	5.6	11	0	1	8.0	1	1	0.1
1500	124.6	4	9.2	11	6.0	12	1	1	7.4	1	1	0.1
1600	133.8	5	9.4	11	6.4	13	1	1	6.9	1	1	0.1
1700	143.2	5	9.4	11	6.8	13	1	1	6.5	1	1	0.1
1800	152.8	5	9.6	10	7.3	14	1	1	6.1	1	2	0.1
1900	162.4	6	9.6	10	7.7	14	1	1	5.7	1	2	0.1
2000	172.2	6	9.8	10	8.2	15	1	1	5.4	1	2	0.1
2100	182.0	6	9.8	10	8.6	16	1	2	5.1	1	2	0.1
2200	192.0	7	10.0	10	9.0	16	1	2	4.8	1	2	0.1
2300	202.0	7	10.2	10	9.5	17	1	2	4.5	1	2	0.1
2400	212.2	7	10.2	10	9.9	17	1	2	4.3	1	2	0.1
2500	222.6	8	10.4	10	10.4	18	1	2	4.1	1	3	0.1
2600	233.0	8	10.6	9	10.9	19	1	2	3.9	1	3	0.1
2700	243.6	8	10.6	9	11.3	19	1	2	3.7	1	3	0.1
2800	254.4	9	10.8	9	11.8	20	1	2	3.6	1	3	0.1
2900	265.4	9	11.0	9	12.3	20	1	2	3.4	2	3	0.1
3000	276.4	10	11.2	9	12.8	21	1	2	3.3	2	4	0.1
3100	287.6	10	11.4	9	13.3	22	1	2	3.1	2	4	0.1
3200	299.2	11	11.6	9	13.8	23	1	3	3.0	2	4	0.1
3300	311.0	11	11.8	8	14.3	23	1	3	2.9	2	4	0.1
3400	323.0	12	12.2	8	14.8	24	1	3	2.7	2	5	0.2
3500	335.4	12	12.4	8	15.4	25	1	3	2.6	2	5	0.2
3600	348.2	13	12.8	8	15.9	26	1	3	2.5	2	5	0.2
3700	361.2	14	13.2	8	16.5	26	1	3	2.4	2	5	0.2
3800	374.6	15	13.6	7	17.0	27	1	3	2.3	2	6	0.2
3900	388.4	16	14.2	7	17.6	27	2	3	2.2	2	6	0.2
4000	402.8	16	14.6	7	18.2	28	2	3	2.1	2	6	0.2

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 3
M. V., 780 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	1
Complementary angle of site for each		Range effect of increase of —						Displacement of burst for change of 5 points in fuze setter corrector in		Range R
+ 1 mil of site	— 1 mil of site	One Sq. in wt. of Proj. Two Sq. is standard	One foot per sec. in M.V.	Air temperature 1 degree. Standard is 59° F.	Rear wind of 1 MPH	One percent in air density	Fuze setting for graze burst	Ht.	R	
ϕ	ϕ	Wt yds.	VE yds.	Temp. yds.	W-R yds.	Den. yds.	FS sec.	ϕ	yds.	
.00	.00	0	0.0	0.0	0.0	0	0.6			0
.00	.00	— 1	+ 0.3	0.0	0.0	0	0.9	11.3	139	100
.00	.00	— 3	+ 0.6	0.0	0.0	0	1.3	11.3	138	200
.00	.00	— 4	+ 0.8	0.0	0.0	0	1.6	11.4	137	300
.00	.00	— 6	+ 1.0	0.0	0.0	0	2.0	11.4	137	400
.00	.00	— 7	+ 1.3	0.0	0.0	0	2.3	11.4	136	500
.00	.00	— 8	+ 1.6	0.0	0.0	0	2.7	11.5	136	600
+ .01	— .01	— 9	+ 1.8	0.0	0.0	0	3.1	11.5	135	700
+ .01	— .01	— 10	+ 2.1	0.0	0.0	0	3.5	11.6	134	800
+ .01	— .01	— 12	+ 2.3	— 0.1	0.0	0	3.8	11.6	134	900
+ .01	— .01	— 13	+ 2.6	— 0.1	0.0	0	4.2	11.6	133	1000
+ .01	— .01	— 14	+ 2.8	— 0.1	+ 0.1	— 1	4.6	11.7	132	1100
+ .01	— .01	— 15	+ 3.0	— 0.1	+ 0.1	— 1	5.0	11.7	132	1200
+ .02	— .02	— 16	+ 3.3	— 0.1	+ 0.1	— 1	5.4	11.7	131	1300
+ .02	— .02	— 17	+ 3.5	— 0.1	+ 0.1	— 1	5.8	11.8	130	1400
+ .02	— .02	— 18	+ 3.7	— 0.1	+ 0.1	— 1	6.2	11.8	130	1500
+ .02	— .02	— 19	+ 3.9	— 0.1	+ 0.2	— 1	6.6	11.8	129	1600
+ .02	— .02	— 21	+ 4.2	— 0.1	+ 0.2	— 1	7.0	11.9	128	1700
+ .03	— .03	— 22	+ 4.4	— 0.1	+ 0.2	— 1	7.4	11.9	127	1800
+ .03	— .03	— 23	+ 4.6	— 0.1	+ 0.3	— 1	7.8	11.9	127	1900
+ .03	— .03	— 24	+ 4.9	— 0.1	+ 0.3	— 1	8.2	12.0	126	2000
+ .03	— .03	— 25	+ 5.1	— 0.1	+ 0.3	— 1	8.6	12.0	125	2100
+ .04	— .04	— 26	+ 5.3	— 0.1	+ 0.4	— 2	9.0	12.0	124	2200
+ .04	— .04	— 27	+ 5.5	— 0.1	+ 0.4	— 2	9.5	12.1	123	2300
+ .04	— .04	— 28	+ 5.7	— 0.2	+ 0.5	— 2	9.9	12.1	123	2400
+ .04	— .04	— 29	+ 6.0	— 0.2	+ 0.5	— 2	10.3	12.1	122	2500
+ .05	— .05	— 30	+ 6.2	— 0.2	+ 0.6	— 2	10.7	12.2	121	2600
+ .05	— .05	— 31	+ 6.4	— 0.2	+ 0.7	— 2	11.2	12.2	120	2700
+ .06	— .06	— 31	+ 6.6	— 0.2	+ 0.7	— 2	11.6	12.2	119	2800
+ .06	— .06	— 32	+ 6.8	— 0.2	+ 0.8	— 2	12.1	12.3	119	2900
+ .07	— .06	— 33	+ 7.1	— 0.2	+ 0.8	— 3	12.5	12.3	118	3000
+ .08	— .07	— 34	+ 7.3	— 0.2	+ 0.9	— 3	13.0	12.4	117	3100
+ .09	— .08	— 35	+ 7.5	— 0.2	+ 1.0	— 3	13.5	12.4	116	3200
+ .10	— .09	— 36	+ 7.7	— 0.2	+ 1.0	— 3	13.9	12.4	115	3300
+ .11	— .10	— 37	+ 7.9	— 0.2	+ 1.1	— 3	14.4	12.5	114	3400
+ .13	— .11	— 37	+ 8.1	— 0.2	+ 1.2	— 4	14.9	12.5	113	3500
+ .15	— .12	— 38	+ 8.3	— 0.2	+ 1.2	— 4	15.4	12.6	112	3600
+ .17	— .14	— 39	+ 8.6	— 0.2	+ 1.3	— 4	16.0	12.7	111	3700
+ .19	— .16	— 40	+ 8.8	— 0.2	+ 1.4	— 4	16.5	12.8	110	3800
+ .21	— .18	— 40	+ 9.0	— 0.2	+ 1.4	— 4	17.0	12.9	109	3900
+ .23	— .21	— 41	+ 9.2	— 0.3	+ 1.5	— 5	17.6	12.9	108	4000

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 3
M. V., 780 f/s

TABLE A[illegible]

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 3
M. V., 780 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	1
Complementary angle of site for each		Range effect of increase of —						Displacement of burst for change of 5 points in fuze setter corrector in		Range R
+ 1 mil of site	— 1 mil of site	One Sq. in wt. of proj. Two Sq. is standard	One foot per sec. in M.V.	Air temperature 1 degree Standard is 59° F.	Rear wind of 1 MPH	One percent in air density	Fuze setting for graze burst	Ht.	R	
<i>m</i>	<i>m</i>	yds.	yds.	yds.	yds.	yds.	sec.	<i>m</i>	yds.	yds.
+ .23	— .21	—41	+ 9.2	—0.3	+1.5	— 5	17.6	12.9	108	4000
+ .26	— .24	—42	+ 9.4	—0.3	+1.6	— 5	18.2	13.0	106	4100
+ .29	— .27	—42	+ 9.6	—0.3	+1.7	— 5	18.8	13.2	105	4200
+ .33	— .31	—43	+ 9.8	—0.3	+1.7	— 5	19.4	13.3	104	4300
+ .37	— .35	—44	+10.0	—0.3	+1.8	— 6	20.1	13.4	102	4400
+ .42	— .39	—44	+10.2	—0.3	+1.9	— 6	20.7	13.6	101	4500
+ .48	— .44	—45	+10.4	—0.3	+2.0	— 6	21.4	13.7	99	4600
+ .55	— .49	—45	+10.6	—0.3	+2.1	— 7	22.1	13.9	97	4700
+ .63	— .55	—46	+10.9	—0.3	+2.2	— 7	22.9	14.1	95	4800
+ .78	— .64	—47	+11.1	—0.3	+2.3	— 7	23.8	14.3	94	4900
+1.13	— .79	—47	+11.3	—0.3	+2.4	— 7	24.8	14.6	92	5000
+1.88	—1.08	—48	+11.5	—0.3	+2.6	— 8	26.0	14.9	89	5100
	—1.69	—48	+11.7	—0.3	+2.7	— 8				5200
	—4.40*	—48	+11.8	—0.3	+3.0	— 9				5280
	+2.64	—47	+11.7	—0.3	+3.3	— 9				5200
—2.93	+2.06	—45	+11.4	—0.3	+3.4	— 9				5100
—2.19	+1.80	—44	+11.1	—0.3	+3.4	— 9				5000
—1.90	+1.66	—43	+10.8	—0.3	+3.5	— 8				4900
—1.74	+1.56	—42	+10.6	—0.3	+3.5	— 8				4800
—1.62	+1.48	—41	+10.4	—0.3	+3.5	— 8				4700
—1.53	+1.41	—40	+10.2	—0.3	+3.6	— 8				4600
—1.46	+1.36	—39	+ 9.9	—0.3	+3.6	— 8				4500
—1.40	+1.32	—38	+ 9.7	—0.3	+3.6	— 8				4400
—1.35	+1.29	—38	+ 9.5	—0.3	+3.6	— 7				4300
—1.31	+1.27	—37	+ 9.3	—0.3	+3.6	— 7				4200
—1.28	+1.25	—36	+ 9.0	—0.3	+3.6	— 7				4100
—1.25	+1.24	—35	+ 8.8	—0.3	+3.6	— 7				4000
—1.23	+1.23	—34	+ 8.6	—0.2	+3.6	— 7				3900

* Positive value used when elevation is greater than 771.6 *m*.

105-mm. HOW.
SHELL, H. E. MK I
FUZE M48, M54
CHARGE 3
M. V. 780 f/s

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, F-----	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second-----	-22	-19	-16	-13	-9	-6	-3	0	+3	+6	+9

TABLE C

Cant of carriage axle, deflection effects in mils due to.

Range, yards-----	1000	2000	3000	4000	4500	5000	5100	5200
Cant of 10 mils, effect-----	0.8	1.7	2.8	4.2	5.2	6.7	7.2	7.9

Range, yards-----	5200	5280	5200	5100	5000	4500	4000	3900
Cant of 10 mils, effect-----	7.9	9.5	11.4	12.5	13.4	17.2	21.1	22.0

Note: Right wheel above left causes left deflection effect.

The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be used.

CHARGE 5

105-mm. HOWITZER

HE Shell, Mark I

Fuze, P. D., M48

Fuze, Comb. Time S. Q., M54

Powder Charge: Base Section and Increments Number 2, 3, 4, and 5.

Maximum Range: 8295.

(Data for ranges 0—8000 included herein.)

NOTE: Columns 9, 21, 22, and 23 are used only when the Shell, M1, is armed with the Combination Fuze, M54, and do not extend beyond the maximum fuze setting, which is 25 seconds.

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 5
M. V., 1020 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range C	Change in range for 1 mil change in elevation mil	Time of flight Time	Range e _{pr}	Deflection e _{pd}	Height of burst	Slope of fall Slope	Line number of metro message L	Drift Dft	Deflection Error Lateral wind of 1 MPH (+) W-D
yds.	m	m	m	yds.	sec.	yds.	yds.	m	1/-	No.	m	m
0	1.0	2	4.4	23	0.0	14	0	0		1	0	0.0
100	5.4	2	4.4	22	0.3	14	0	0	212	1	0	0.0
200	9.8	2	4.6	22	0.6	14	0	0	106	1	0	0.0
300	14.4	2	4.6	22	0.9	14	0	0	70	1	0	0.0
400	19.2	2	4.6	21	1.2	15	0	0	52	1	0	0.0
500	23.8	3	4.8	21	1.5	15	0	0	41	1	0	0.0
600	28.6	3	4.8	21	1.8	15	0	0	34	1	0	0.0
700	33.4	3	4.8	21	2.1	15	0	0	29	1	0	0.0
800	38.4	3	5.0	20	2.4	15	0	0	25	1	0	0.0
900	43.2	3	5.0	20	2.7	16	0	0	22	1	0	0.0
1000	48.2	3	5.0	20	3.0	16	0	0	20.0	1	0	0.0
1100	53.2	3	5.0	20	3.3	16	0	0	18.0	1	0	0.0
1200	58.4	3	5.2	20	3.6	16	0	0	16.4	1	1	0.0
1300	63.6	3	5.2	19	3.9	17	0	0	15.1	1	1	0.0
1400	68.8	3	5.2	19	4.2	17	0	0	13.9	1	1	0.0
1500	74.0	4	5.2	19	4.6	17	1	0	12.9	1	1	0.1
1600	79.2	4	5.4	19	4.9	17	1	1	12.0	1	1	0.1
1700	84.6	4	5.4	19	5.2	18	1	1	11.2	1	1	0.1
1800	90.0	4	5.4	19	5.5	18	1	1	10.5	1	1	0.1
1900	95.4	4	5.4	18	5.9	18	1	1	9.9	1	1	0.1
2000	100.8	4	5.4	18	6.2	19	1	1	9.3	1	1	0.1
2100	106.2	4	5.6	18	6.5	19	1	1	8.8	1	1	0.1
2200	111.8	4	5.6	18	6.9	19	1	1	8.3	1	1	0.1
2300	117.4	5	5.6	18	7.2	20	1	1	7.9	1	1	0.1
2400	123.0	5	5.6	18	7.5	20	1	1	7.6	1	1	0.1
2500	128.8	5	5.8	17	7.9	21	1	1	7.2	1	1	0.1
2600	134.4	5	5.8	17	8.2	21	1	1	6.9	1	1	0.1
2700	140.2	5	5.8	17	8.5	22	1	1	6.6	1	1	0.1
2800	146.0	5	5.8	17	8.9	22	1	1	6.3	1	1	0.1
2900	151.8	5	5.8	17	9.2	23	2	1	6.0	1	1	0.1
3000	157.8	6	6.0	17	9.6	23	2	1	5.8	1	2	0.1
3100	163.8	6	6.0	17	9.9	23	2	1	5.6	1	2	0.1
3200	169.8	6	6.0	17	10.3	24	2	1	5.4	1	2	0.1
3300	175.8	6	6.0	16	10.6	24	2	1	5.2	1	2	0.1
3400	182.0	6	6.2	16	11.0	24	2	1	5.0	1	2	0.1
3500	188.2	6	6.2	16	11.4	25	2	1	4.8	1	2	0.1
3600	194.4	7	6.2	16	11.7	25	2	2	4.6	1	2	0.1
3700	200.6	7	6.4	16	12.1	26	2	2	4.5	1	2	0.1
3800	207.0	7	6.4	16	12.4	26	2	2	4.3	2	2	0.1
3900	213.4	7	6.4	16	12.8	27	2	2	4.2	2	2	0.1
4000	220.0	7	6.6	15	13.2	27	2	2	4.0	2	2	0.1

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 5
M. V., 1020 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	1
Complementary angle of site for each		Range effect of increase of —					Fuze setting for graze burst FS	Displacement of burst for change of 5 points in fuze setter correction in —		Range R
+ 1 mil of site	— 1 mil of site	One Sq. in wt. of proj. Two Sq. is standard Wt.	One foot per sec. in M.V. VE	Air temperature 1 degree, Standard is 59° F. Temp.	Rear wind of 1 MPH W-R	One percent in air density Den.				
η	η	yds.	yds.	yds.	yds.	yds.	sec.	η	yds.	yds.
.00	.00	0	0.0	0.0	0.0	0	0.6			0
.00	.00	— 2	+ 0.2	0.0	0.0	0	0.9	8.6	179	100
.00	.00	— 3	+ 0.4	0.0	0.0	0	1.2	8.6	178	200
.00	.00	— 5	+ 0.6	0.0	0.0	0	1.5	8.6	178	300
.00	.00	— 6	+ 0.8	+0.1	0.0	0	1.8	8.6	177	400
.00	.00	— 7	+ 1.0	+0.1	0.0	0	2.0	8.7	176	500
.00	.00	— 8	+ 1.1	+0.1	0.0	0	2.3	8.7	175	600
.00	.00	—10	+ 1.3	+0.1	+0.1	0	2.6	8.7	174	700
.00	.00	—11	+ 1.5	+0.2	+0.1	0	2.9	8.7	174	800
.00	.00	—12	+ 1.6	+0.2	+0.2	0	3.2	8.7	173	900
.00	.00	—13	+ 1.8	+0.2	+0.2	0	3.5	8.7	172	1000
.00	.00	—14	+ 1.9	+0.2	+0.3	0	3.8	8.8	171	1100
.00	.00	—15	+ 2.1	+0.2	+0.4	0	4.1	8.8	170	1200
.00	.00	—16	+ 2.2	+0.3	+0.4	0	4.4	8.8	170	1300
.00	.00	—17	+ 2.4	+0.3	+0.5	0	4.7	8.8	169	1400
.00	.00	—18	+ 2.5	+0.3	+0.6	0	5.0	8.8	168	1500
.00	.00	—19	+ 2.6	+0.3	+0.6	0	5.3	8.9	167	1600
.00	.00	—20	+ 2.8	+0.3	+0.7	— 1	5.6	8.9	166	1700
+ .01	— .01	—21	+ 2.9	+0.4	+0.8	— 1	5.9	8.9	166	1800
+ .01	— .01	—21	+ 3.1	+0.4	+0.9	— 1	6.2	8.9	165	1900
+ .01	— .01	—22	+ 3.2	+0.4	+1.0	— 1	6.5	8.9	164	2000
+ .01	— .01	—23	+ 3.3	+0.4	+1.1	— 1	6.8	8.9	163	2100
+ .01	— .01	—24	+ 3.5	+0.4	+1.2	— 1	7.1	9.0	162	2200
+ .01	— .01	—25	+ 3.6	+0.5	+1.3	— 1	7.4	9.0	162	2300
+ .01	— .01	—25	+ 3.7	+0.5	+1.4	— 1	7.7	9.0	161	2400
+ .01	— .01	—26	+ 3.9	+0.5	+1.4	— 1	8.1	9.0	160	2500
+ .01	— .01	—27	+ 4.0	+0.5	+1.5	— 2	8.4	9.0	159	2600
+ .02	— .02	—28	+ 4.1	+0.5	+1.6	— 2	8.7	9.1	158	2700
+ .02	— .02	—28	+ 4.3	+0.5	+1.7	— 2	9.0	9.1	158	2800
+ .02	— .02	—29	+ 4.4	+0.6	+1.8	— 2	9.4	9.1	157	2900
+ .02	— .02	—30	+ 4.5	+0.6	+1.9	— 2	9.7	9.1	156	3000
+ .02	— .02	—30	+ 4.7	+0.6	+2.0	— 2	10.0	9.1	155	3100
+ .02	— .02	—31	+ 4.8	+0.6	+2.1	— 3	10.3	9.1	154	3200
+ .03	— .03	—31	+ 4.9	+0.6	+2.2	— 3	10.7	9.2	154	3300
+ .03	— .03	—32	+ 5.0	+0.6	+2.3	— 3	11.0	9.2	153	3400
+ .03	— .03	—33	+ 5.2	+0.7	+2.4	— 3	11.4	9.2	152	3500
+ .03	— .03	—33	+ 5.3	+0.7	+2.5	— 3	11.7	9.2	151	3600
+ .04	— .04	—34	+ 5.4	+0.7	+2.6	— 4	12.0	9.2	150	3700
+ .04	— .04	—34	+ 5.6	+0.7	+2.7	— 4	12.4	9.3	150	3800
+ .04	— .04	—35	+ 5.7	+0.7	+2.8	— 4	12.7	9.3	149	3900
+ .05	— .04	—36	+ 5.8	+0.7	+2.9	— 4	13.1	9.3	148	4000

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 5
M. V., 1020 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range C	Change in range for 1 mil change in elevation 1mil	Time of flight Time	Probable error			Slope of fall Slope	Line number of metro message L	Deflection error	
						Range e _{pr}	Deflection e _{pd}	Height of burst			Drift Dft	Lateral wind of 1 MPH (+) W-D
Yds.	m	m	m	yds.	sec.	yds.	yds.	m	1/-	No.	m	m
4000	220.0	7	6.6	15	13.2	27	2	2	4.0	2	2	0.1
4100	226.4	8	6.6	15	13.5	27	2	2	3.9	2	3	0.1
4200	233.0	8	6.6	15	13.9	28	2	2	3.8	2	3	0.1
4300	239.8	8	6.8	15	14.3	28	3	2	3.7	2	3	0.2
4400	246.6	8	6.8	15	14.7	29	3	2	3.6	2	3	0.2
4500	253.6	8	7.0	14	15.1	29	3	2	3.4	2	3	0.2
4600	260.6	9	7.0	14	15.5	30	3	2	3.3	2	3	0.2
4700	267.6	9	7.2	14	15.9	30	3	2	3.2	2	3	0.2
4800	274.8	9	7.2	14	16.3	30	3	2	3.1	2	3	0.2
4900	282.0	9	7.4	14	16.7	31	3	2	3.0	2	4	0.2
5000	289.4	9	7.4	13	17.1	31	3	2	3.0	2	4	0.2
5100	296.8	10	7.6	13	17.5	32	3	2	2.9	2	4	0.2
5200	304.4	10	7.6	13	17.9	32	3	3	2.8	2	4	0.2
5300	312.0	10	7.8	13	18.3	32	3	3	2.7	2	4	0.2
5400	319.8	10	7.8	13	18.8	33	3	3	2.6	2	4	0.2
5500	327.8	10	8.0	12	19.2	33	3	3	2.6	2	5	0.2
5600	335.8	11	8.2	12	19.6	34	4	3	2.5	3	5	0.2
5700	344.0	11	8.2	12	20.1	34	4	3	2.4	3	5	0.2
5800	352.4	11	8.4	12	20.5	34	4	3	2.4	3	5	0.2
5900	361.0	12	8.6	12	21.0	35	4	3	2.3	3	5	0.2
6000	369.6	12	8.8	11	21.4	35	4	3	2.3	3	6	0.2
6100	378.4	13	9.0	11	21.9	36	4	3	2.2	3	6	0.2
6200	387.4	13	9.2	11	22.4	36	4	3	2.1	3	6	0.2
6300	396.8	14	9.4	11	22.9	37	4	3	2.0	3	6	0.2
6400	406.4	14	9.6	10	23.4	37	4	3	1.99	3	7	0.2
6500	416.2	15	10.0	10	23.9	38	4	3	1.93	3	7	0.2
6600	426.2	15	10.2	10	24.4	38	4	4	1.87	3	7	0.2
6700	436.6	16	10.6	10	25.0	39	5	4	1.82	3	7	0.2
6800	447.4	17	10.8	9	25.5	39	5	4	1.76	3	8	0.3
6900	458.4	18	11.2	9	26.1	40	5	4	1.71	3	8	0.3
7000	469.8	19	11.8	9	26.7	40	5		1.66	3	8	0.3
7100	481.8	20	12.2	8	27.3	40	5		1.61	3	9	0.3
7200	494.2	21	12.6	8	27.9	41	5		1.56	4	9	0.3
7300	507.2	22	13.2	8	28.5	41	5		1.51	4	10	0.3
7400	520.8	23	14.0	7	29.2	41	5		1.46	4	10	0.3
7500	535.2	25	14.8	7	29.9	42	5		1.41	4	10	0.3
7600	550.6	27	15.8	6	30.7	42	6		1.36	4	11	0.3
7700	567.2	29	17.2	6	31.5	42	6		1.30	4	11	0.3
7800	585.0	32	18.8	5	32.3	43	6		1.25	4	12	0.3
7900	604.8	36	20.8	5	33.2	43	6		1.20	4	13	0.3
8000	627.4	41	23.8	4	34.2	43	6		1.14	5	14	0.3

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 5
M. V., 1020 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	1
Complementary angle of site for each		Range effect of increase of —					Fuze setting for graze burst	Displacement of burst for change of 5 points in fuze setter corrector in		Range R
+ 1 mil of site	— 1 mil of site	One Sq. in wt. of proj. Two Sq. is standard	One foot per sec. in M.V.	Air temperature 1 degree. Standard is 59° F.	Rear wind of 1 Mi/H	One percent in air density	FS	Ht	R	
η	η	yds.	yds.	yds.	yds.	yds.	sec.	η	yds.	yds.
+ .05	— .04	—36	+ 5.8	+0.7	+2.9	— 4	13.1	9.3	148	4000
+ .05	— .04	—36	+ 5.9	+0.7	+3.1	— 4	13.4	9.3	147	4100
+ .05	— .05	—36	+ 6.1	+0.7	+3.2	— 5	13.8	9.4	146	4200
+ .06	— .05	—37	+ 6.2	+0.8	+3.3	— 5	14.1	9.4	146	4300
+ .06	— .05	—37	+ 6.3	+0.8	+3.4	— 5	14.5	9.4	145	4400
+ .06	— .05	—38	+ 6.4	+0.8	+3.5	— 5	14.9	9.4	144	4500
+ .07	— .06	—38	+ 6.6	+0.8	+3.6	— 6	15.2	9.5	143	4600
+ .07	— .06	—39	+ 6.7	+0.8	+3.7	— 6	15.6	9.5	142	4700
+ .07	— .06	—39	+ 6.8	+0.8	+3.8	— 6	16.0	9.5	142	4800
+ .08	— .07	—40	+ 6.9	+0.8	+3.9	— 7	16.4	9.6	141	4900
+ .08	— .07	—40	+ 7.1	+0.8	+4.0	— 7	16.8	9.6	140	5000
+ .09	— .07	—41	+ 7.2	+0.9	+4.1	— 7	17.1	9.6	139	5100
+ .09	— .08	—41	+ 7.3	+0.9	+4.2	— 7	17.5	9.7	138	5200
+ .10	— .08	—41	+ 7.4	+0.9	+4.3	— 8	17.9	9.7	137	5300
+ .11	— .09	—42	+ 7.6	+0.9	+4.4	— 8	18.3	9.7	136	5400
+ .12	— .10	—42	+ 7.7	+0.9	+4.5	— 8	18.8	9.8	135	5500
+ .13	— .11	—42	+ 7.8	+0.9	+4.6	— 9	19.2	9.8	134	5600
+ .14	— .12	—43	+ 7.9	+0.9	+4.7	— 9	19.6	9.9	133	5700
+ .15	— .13	—43	+ 8.1	+0.9	+4.8	— 9	20.0	9.9	133	5800
+ .16	— .14	—44	+ 8.2	+0.9	+4.9	—10	20.4	9.9	132	5900
+ .18	— .15	—44	+ 8.3	+0.9	+5.1	—10	20.9	10.0	131	6000
+ .19	— .16	—44	+ 8.4	+0.9	+5.2	—10	21.3	10.0	130	6100
+ .20	— .17	—45	+ 8.6	+0.9	+5.3	—11	21.8	10.1	129	6200
+ .22	— .18	—45	+ 8.7	+0.9	+5.4	—11	22.3	10.1	128	6300
+ .24	— .20	—45	+ 8.8	+0.9	+5.5	—11	22.7	10.2	127	6400
+ .26	— .22	—46	+ 8.9	+0.9	+5.6	—11	23.2	10.2	126	6500
+ .28	— .24	—46	+ 9.1	+0.9	+5.7	—12	23.7	10.3	125	6600
+ .30	— .26	—46	+ 9.2	+0.9	+5.8	—12	24.2	10.4	124	6700
+ .32	— .28	—47	+ 9.3	+0.9	+5.9	—12	24.8	10.4	122	6800
+ .34	— .30	—47	+ 9.4	+0.9	+6.0	—13	25.3	10.5	121	6900
+ .37	— .32	—47	+ 9.6	+0.9	+6.2	—13				7000
+ .40	— .34	—48	+ 9.7	+0.9	+6.3	—13				7100
+ .44	— .36	—48	+ 9.8	+0.9	+6.4	—14				7200
+ .48	— .39	—48	+ 9.9	+0.9	+6.5	—14				7300
+ .53	— .43	—48	+10.0	+0.9	+6.6	—14				7400
+ .60	— .49	—49	+10.2	+0.9	+6.7	—15				7500
+ .68	— .56	—49	+10.3	+0.9	+6.8	—15				7600
+ .78	— .64	—49	+10.4	+0.9	+7.0	—15				7700
+ .91	— .73	—50	+10.5	+0.9	+7.1	—16				7800
+1.10	— .84	—50	+10.7	+0.9	+7.2	—16				7900
+1.49	— .98	—50	+10.8	+0.8	+7.3	—16				8000

105-mm. HOW.
SHELL, H. E. MK I
FUZE M48, M54
CHARGE 5
M. V. 1020 f/s

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, F-----	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second-----	-22	-19	-16	-13	-9	-6	-3	0	+ 3	+ 6	+ 9

TABLE C

Cant of carriage axle, deflection effects in mils due to.

Range, yards..	1000	2000	3000	4000	5000	6000	7000	7500	8000	8100
Cant of 10 mils, effect-----	0.5	1.0	1.6	2.2	2.9	3.8	5.0	5.8	7.1	7.5

Range, yards..	8100	8200	8295	8200	8100	8000	7500	7000	6500	6100
Cant of 10 mils, effect-----	7.5	8.0	9.4	11.0	11.8	12.4	15.0	17.4	19.8	21.8

Note: Right wheel above left causes left deflection effect.

The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be used.

CHARGE 7

105-mm. HOWITZER

HE Shell, Mark I

Fuze, P. D., M48

Fuze, Comb. Time S. Q., M54

**Powder Charge: Base Section and Increments Number 2, 3, 4, 5,
6 and 7.**

Maximum Range: 12,205.

(Data for ranges 0—12,000 included herein.)

NOTE: Columns 9, 21, 22, and 23 are used only when the Shell, M1, is armed with the Combination Fuze, M54, and do not extend beyond the maximum fuze setting, which is 25 seconds.

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 7
M. V., 1550 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range C	Change in range for 1 mil change in elevation l mil	Time of flight Time	Probable error			Slope of fall Slope	Line number of metro message L	Deflection Dft	Deflection effect Lateral wind of 1 MPH (+) W - D
yds.	m	m	m	yds.	sec.	Range e pr	Deflection e pd	Height of burst m	1/—	No.	m	m
0	1.6	1	1.8	56	0.0	8	0	0		1	0	0.0
100	3.4	1	1.8	55	0.2	8	0	0	485	1	0	0.0
200	5.4	1	2.0	53	0.4	8	0	0	243	1	0	0.0
300	7.2	1	2.0	52	0.6	8	0	0	159	1	0	0.0
400	9.2	1	2.0	50	0.8	8	0	0	116	1	0	0.0
500	11.2	1	2.0	49	1.0	8	0	0	91	1	0	0.0
600	13.4	1	2.2	48	1.2	8	0	0	74	1	0	0.0
700	15.6	1	2.2	47	1.4	9	0	0	62	1	0	0.0
800	17.8	1	2.2	45	1.6	9	0	0	53	1	0	0.0
900	20.0	1	2.2	44	1.8	9	0	0	46	1	0	0.0
1000	22.2	1	2.4	43	2.0	9	0	0	41	1	0	0.0
1100	24.6	1	2.4	42	2.2	9	0	0	36	1	0	0.0
1200	27.0	1	2.4	41	2.5	10	0	0	32	1	0	0.1
1300	29.6	1	2.6	40	2.7	10	0	0	29	1	0	0.1
1400	32.0	1	2.6	39	2.9	10	0	0	27	1	0	0.1
1500	34.6	1	2.6	38	3.1	10	0	0	25	1	0	0.1
1600	37.4	1	2.6	37	3.4	10	0	0	23	1	0	0.1
1700	40.0	1	2.8	37	3.6	11	0	0	21	1	0	0.1
1800	42.8	1	2.8	36	3.9	11	0	0	19.5	1	0	0.1
1900	45.6	1	2.8	36	4.1	11	0	0	18.1	1	0	0.1
2000	48.4	1	2.8	35	4.4	11	1	0	16.9	1	0	0.1
2100	51.4	1	3.0	34	4.6	11	1	0	15.9	1	0	0.1
2200	54.4	1	3.0	34	4.9	12	1	0	14.9	1	0	0.1
2300	57.4	1	3.0	33	5.1	12	1	0	14.0	1	0	0.1
2400	60.4	1	3.0	33	5.4	12	1	0	13.2	1	0	0.2
2500	63.4	2	3.2	32	5.7	12	1	0	12.5	1	0	0.2
2600	66.6	2	3.2	31	5.9	12	1	0	11.9	1	1	0.2
2700	69.8	2	3.2	31	6.2	13	1	0	11.3	1	1	0.2
2800	73.2	2	3.2	30	6.5	13	1	0	10.7	1	1	0.2
2900	76.4	2	3.4	30	6.7	13	1	0	10.2	1	1	0.2
3000	79.8	2	3.4	29	7.0	13	1	0	9.7	1	1	0.2
3100	83.2	2	3.4	29	7.3	13	1	0	9.3	1	1	0.2
3200	86.6	2	3.4	29	7.6	14	1	0	8.9	1	1	0.2
3300	90.2	2	3.6	28	7.9	14	1	0	8.5	1	1	0.2
3400	93.6	2	3.6	28	8.1	14	1	0	8.2	1	1	0.2
3500	97.2	2	3.6	27	8.4	14	1	0	7.9	1	1	0.2
3600	101.0	2	3.6	27	8.7	14	1	0	7.6	1	1	0.2
3700	104.6	2	3.8	27	9.0	15	1	0	7.3	1	1	0.2
3800	108.4	2	3.8	26	9.3	15	1	0	7.0	1	1	0.2
3900	112.2	2	3.8	26	9.6	15	1	0	6.7	1	1	0.3
4000	116.0	2	3.8	26	9.9	15	1	0	6.5	1	1	0.3

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 7
M. V., 1550 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	1
Complementary angle of site for each		Range effect of increase of —					Fuze setting for graze burst	Displacement of burst for change of 5 points in fuze setter corrector		Range R
+ 1 mil of site	— 1 mil of site	One Sq. in wt. of proj. Two Sq. is standard Wt	One foot per sec. in M.V. VE	Air temperature 1 degree. Standard is 59° F. Temp.	Rear wind of 1 MPH W-R	One percent in air density Den.				
<i>m</i>	<i>m</i>	yds.	yds.	yds.	yds.	yds.	sec.	<i>m</i>	yds.	yds.
.00	.00	0	0.0	0.0	0.0	0	0.8			0
.00	.00	— 1	+0.1	0.0	0.0	0	0.9	5.4	269	100
.00	.00	— 2	+0.2	—0.1	0.0	0	1.0	5.4	265	200
.00	.00	— 3	+0.3	—0.1	0.0	0	1.3	5.5	262	300
.00	.00	— 4	+0.4	—0.1	0.0	— 1	1.5	5.6	258	400
.00	.00	— 5	+0.6	—0.1	0.0	— 1	1.7	5.7	255	500
.00	.00	— 6	+0.7	—0.2	0.0	— 1	1.9	5.8	252	600
.00	.00	— 7	+0.8	—0.2	0.0	— 1	2.1	5.8	249	700
.00	.00	— 8	+0.9	—0.2	+ 0.1	— 1	2.3	5.9	246	800
.00	.00	— 9	+1.0	—0.2	+ 0.1	— 1	2.5	5.9	243	900
.00	.00	—10	+1.2	—0.2	+ 0.1	— 2	2.7	6.0	240	1000
.00	.00	—10	+1.3	—0.2	+ 0.1	— 2	2.9	6.0	237	1100
.00	.00	—11	+1.4	—0.3	+ 0.1	— 2	3.1	6.1	234	1200
.00	.00	—12	+1.5	—0.3	+ 0.2	— 2	3.3	6.1	232	1300
.00	.00	—13	+1.6	—0.3	+ 0.2	— 2	3.5	6.2	229	1400
.00	.00	—14	+1.7	—0.3	+ 0.2	— 3	3.8	6.2	226	1500
.00	.00	—14	+1.8	—0.3	+ 0.2	— 3	4.0	6.2	223	1600
.00	.00	—15	+1.9	—0.3	+ 0.3	— 3	4.2	6.3	220	1700
.00	.00	—16	+2.0	—0.3	+ 0.3	— 3	4.5	6.3	218	1800
.00	.00	—16	+2.1	—0.3	+ 0.4	— 3	4.7	6.3	215	1900
.00	.00	—17	+2.2	—0.3	+ 0.4	— 4	4.9	6.4	212	2000
.00	.00	—18	+2.2	—0.3	+ 0.5	— 4	5.2	6.4	210	2100
.00	.00	—18	+2.3	—0.3	+ 0.5	— 4	5.4	6.4	207	2200
.00	.00	—19	+2.4	—0.3	+ 0.6	— 4	5.7	6.4	205	2300
.00	.00	—19	+2.5	—0.3	+ 0.7	— 5	5.9	6.5	202	2400
.00	.00	—19	+2.6	—0.3	+ 0.7	— 5	6.1	6.5	200	2500
.00	.00	—20	+2.7	—0.3	+ 0.8	— 5	6.4	6.5	198	2600
.00	.00	—20	+2.8	—0.3	+ 0.9	— 5	6.7	6.5	196	2700
.00	.00	—21	+2.9	—0.3	+ 1.0	— 6	6.9	6.6	194	2800
.00	.00	—21	+2.9	—0.3	+ 1.1	— 6	7.2	6.6	192	2900
.00	.00	—21	+3.0	—0.2	+ 1.1	— 7	7.4	6.6	190	3000
.00	.00	—22	+3.1	—0.2	+ 1.2	— 7	7.7	6.6	189	3100
.00	.00	—22	+3.2	—0.2	+ 1.3	— 7	8.0	6.7	187	3200
.00	.00	—22	+3.2	—0.2	+ 1.4	— 7	8.3	6.7	186	3300
.00	.00	—22	+3.3	—0.1	+ 1.5	— 8	8.5	6.7	184	3400
.00	.00	—22	+3.3	—0.1	+ 1.6	— 8	8.8	6.8	183	3500
.00	.00	—22	+3.4	—0.1	+ 1.8	— 8	9.1	6.8	182	3600
.00	.00	—22	+3.5	0.0	+ 1.9	— 9	9.4	6.8	181	3700
.00	.00	—22	+3.5	0.0	+ 2.0	— 9	9.6	6.9	179	3800
.00	.00	—22	+3.6	0.0	+ 2.2	— 9	9.9	6.9	178	3900
.00	.00	—22	+3.6	+0.1	+ 2.3	—10	10.2	6.9	177	4000

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 7
M. V., 1550 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range C	Change in range for 1 mil change in elevation 1mil	Time of flight Time	Probable Error			Slope of fall Slope	Line number of metro message L	Deflection effect	
						Range e _{pr}	Deflection e _{pd}	Height of burst			Drift Dft	Lateral wind of 1 MPH (+) W-D
yds.	m	m	m	yds.	sec.	yds.	yds.	m	1/-	No.	m	m
4000	116.0	2	3.8	26	9.9	15	1	0	6.5	1	1	0.3
4100	120.0	2	4.0	25	10.2	16	1	1	6.3	1	1	0.3
4200	123.8	2	4.0	25	10.5	16	1	1	6.1	1	1	0.3
4300	127.8	2	4.0	25	10.8	16	1	1	5.9	1	1	0.3
4400	131.8	3	4.0	25	11.1	16	1	1	5.7	1	1	0.3
4500	136.0	3	4.2	24	11.4	17	1	1	5.5	1	1	0.3
4600	140.0	3	4.2	24	11.7	17	1	1	5.3	1	1	0.3
4700	144.2	3	4.2	24	12.0	17	1	1	5.1	1	1	0.3
4800	148.4	3	4.2	24	12.3	17	2	1	5.0	2	1	0.3
4900	152.8	3	4.4	23	12.6	17	2	1	4.8	2	2	0.3
5000	157.0	3	4.4	23	13.0	18	2	1	4.7	2	2	0.3
5100	161.4	3	4.4	23	13.3	18	2	1	4.6	2	2	0.3
5200	165.8	3	4.4	23	13.6	18	2	1	4.5	2	2	0.3
5300	170.2	3	4.4	22	13.9	18	2	1	4.4	2	2	0.3
5400	174.6	3	4.6	22	14.2	19	2	1	4.3	2	2	0.3
5500	179.2	3	4.6	22	14.6	19	2	1	4.2	2	2	0.3
5600	184.0	3	4.6	22	14.9	19	2	1	4.0	2	2	0.3
5700	188.6	4	4.6	21	15.2	20	2	1	3.9	2	2	0.3
5800	193.2	4	4.8	21	15.6	20	2	1	3.8	2	2	0.4
5900	197.8	4	4.8	21	15.9	20	2	1	3.7	2	2	0.4
6000	202.6	4	4.8	21	16.2	21	2	1	3.6	2	2	0.4
6100	207.6	4	4.8	21	16.6	21	2	1	3.5	2	2	0.4
6200	212.4	4	4.8	20	16.9	21	2	1	3.4	2	2	0.4
6300	217.4	4	5.0	20	17.2	22	2	1	3.3	2	2	0.4
6400	222.2	4	5.0	20	17.6	22	2	1	3.2	2	3	0.4
6500	227.2	4	5.0	20	17.9	22	2	1	3.2	2	3	0.4
6600	232.4	4	5.0	20	18.3	23	2	2	3.1	2	3	0.4
6700	237.4	5	5.2	19	18.6	23	2	2	3.0	2	3	0.4
6800	242.6	5	5.2	19	19.0	23	2	2	2.9	3	3	0.4
6900	247.8	5	5.2	19	19.3	24	3	2	2.9	3	3	0.4
7000	253.0	5	5.2	19	19.7	24	3	2	2.8	3	3	0.4
7100	258.4	5	5.4	19	20.0	24	3	2	2.8	3	3	0.4
7200	263.6	5	5.4	19	20.4	24	3	2	2.7	3	3	0.4
7300	269.0	5	5.4	18	20.8	25	3	2	2.6	3	3	0.4
7400	274.6	5	5.4	18	21.1	25	3	2	2.6	3	3	0.4
7500	280.0	6	5.6	18	21.5	25	3	2	2.5	3	4	0.4
7600	285.6	6	5.6	18	21.9	26	3	2	2.5	3	4	0.4
7700	291.2	6	5.6	18	22.3	26	3	2	2.4	3	4	0.4
7800	297.0	6	5.8	17	22.6	26	3	2	2.4	3	4	0.4
7900	302.6	6	5.8	17	23.0	27	3	2	2.3	3	4	0.4
8000	308.4	6	5.8	17	23.4	27	3	2	2.3	3	4	0.4

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 7
M. V., 1550 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	1
Complementary angle of site ^c for each		Range effect of increase of —					Fuze setting for graze burst FS	Displacement of burst for change of 5 points in fuze setter corrector in — Ht. R		Range R
+ 1 mil of site	— 1 mil of site	One Sq. in wt. of proj. Two Sq. is standard Wt	One foot per sec. in M.V. VE	Air temperature 1 degree Standard is 59 F. Temp.	Rear wind of 1 MPH W-R	One percent in air density Den.				
‰	‰	yds.	yds.	yds.	yds.	yds.	sec.	‰	yds.	yds.
.00	.00	-22	+3.6	+0.1	+ 2.3	-10	10.2	6.9	177	4000
.00	.00	-22	+3.7	+0.1	+ 2.5	-10	10.5	7.0	176	4100
.00	.00	-22	+3.7	+0.2	+ 2.6	-10	10.8	7.0	175	4200
.00	.00	-22	+3.8	+0.2	+ 2.7	-11	11.1	7.0	174	4300
.00	.00	-22	+3.8	+0.2	+ 2.9	-11	11.4	7.1	173	4400
.00	.00	-21	+3.8	+0.3	+ 3.1	-11	11.6	7.1	172	4500
.00	.00	-21	+3.9	+0.3	+ 3.2	-12	11.9	7.1	172	4600
.00	.00	-21	+3.9	+0.4	+ 3.4	-12	12.2	7.2	171	4700
.00	.00	-21	+4.0	+0.4	+ 3.5	-12	12.5	7.2	170	4800
.00	.00	-21	+4.0	+0.5	+ 3.7	-13	12.8	7.2	169	4900
.00	.00	-20	+4.0	+0.6	+ 3.9	-13	13.1	7.3	168	5000
.00	.00	-20	+4.1	+0.6	+ 4.0	-13	13.4	7.3	168	5100
+	.01	.00	-20	+4.1	+0.7	+ 4.2	13.7	7.3	167	5200
+	.01	.00	-19	+4.1	+0.7	+ 4.4	14.0	7.3	166	5300
+	.01	.00	-19	+4.1	+0.8	+ 4.6	14.3	7.4	165	5400
+	.01	.00	-18	+4.2	+0.9	+ 4.8	14.6	7.4	164	5500
+	.01	.00	-18	+4.2	+0.9	+ 5.0	15.0	7.4	164	5600
+	.01	.00	-17	+4.2	+1.0	+ 5.2	15.3	7.5	163	5700
+	.01	.00	-17	+4.2	+1.1	+ 5.4	15.6	7.5	162	5800
+	.01	.00	-17	+4.2	+1.1	+ 5.6	15.9	7.5	161	5900
+	.01	.00	-16	+4.3	+1.2	+ 5.8	16.2	7.6	160	6000
+	.01	— .01	-16	+4.3	+1.3	+ 6.0	16.6	7.6	160	6100
+	.01	— .01	-15	+4.3	+1.4	+ 6.2	16.9	7.6	159	6200
+	.01	— .01	-15	+4.3	+1.4	+ 6.4	17.2	7.6	158	6300
+	.02	— .01	-14	+4.3	+1.5	+ 6.6	17.5	7.7	157	6400
+	.02	— .01	-13	+4.3	+1.6	+ 6.8	17.8	7.7	156	6500
+	.02	— .01	-13	+4.3	+1.6	+ 7.0	18.2	7.7	156	6600
+	.02	— .01	-12	+4.3	+1.7	+ 7.2	18.5	7.8	155	6700
+	.02	— .02	-12	+4.4	+1.8	+ 7.4	18.9	7.8	154	6800
+	.02	— .02	-11	+4.4	+1.9	+ 7.6	19.2	7.8	153	6900
+	.03	— .02	-10	+4.4	+1.9	+ 7.9	19.5	7.8	152	7000
+	.03	— .02	-10	+4.4	+2.0	+ 8.1	19.9	7.9	152	7100
+	.03	— .02	- 9	+4.4	+2.1	+ 8.3	20.2	7.9	151	7200
+	.03	— .03	- 8	+4.4	+2.1	+ 8.5	20.6	7.9	150	7300
+	.03	— .03	- 8	+4.4	+2.2	+ 8.7	20.9	7.9	149	7400
+	.04	— .03	- 7	+4.4	+2.3	+ 8.9	21.3	8.0	148	7500
+	.04	— .04	- 6	+4.4	+2.3	+ 9.2	21.6	8.0	148	7600
+	.04	— .04	- 6	+4.4	+2.4	+ 9.4	22.0	8.0	147	7700
+	.05	— .04	- 5	+4.5	+2.5	+ 9.6	22.4	8.1	146	7800
+	.05	— .05	- 4	+4.5	+2.5	+ 9.8	22.7	8.1	145	7900
+	.06	— .05	- 4	+4.5	+2.6	+10.0	23.1	8.1	144	8000

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 7
M. V., 1550 f/s

TABLE A

1	2	3	4	5	6	7	8	9	10	11	12	13
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range C	Change in range for 1 mil change in elevation 1 mil	Time of flight Time	Probable Error			Slope of fall Slope	Line number of metro message L	Deflection Error	
yds.	ft	ft	ft	yds.	sec.	Range e _{pr}	Deflection e _{pd}	Height of burst ft	1/-	No.	Drift ft	Lateral wind of 1 MPH (+) W-D
8000	308.4	6	5.8	17	23.4	27	3	2	2.3	3	4	0.4
8100	314.4	7	6.0	17	23.8	27	3	2	2.2	3	4	0.5
8200	320.4	7	6.0	17	24.2	28	3	2	2.2	3	4	0.5
8300	326.4	7	6.0	17	24.6	28	3	2	2.1	3	5	0.5
8400	332.4	7	6.2	16	25.0	28	3	2	2.1	3	5	0.5
8500	338.6	7	6.2	16	25.4	29	3	2	2.0	3	5	0.5
8600	344.8	7	6.4	16	25.8	29	3	3	2.00	3	5	0.5
8700	351.2	8	6.4	16	26.2	29	4		1.96	3	5	0.5
8800	357.6	8	6.4	15	26.6	29	4		1.92	3	5	0.5
8900	364.0	8	6.6	15	27.0	30	4		1.88	4	5	0.5
9000	370.6	8	6.6	15	27.5	30	4		1.84	4	6	0.5
9100	377.4	8	6.8	15	27.9	30	4		1.81	4	6	0.5
9200	384.2	9	6.8	15	28.3	31	4		1.77	4	6	0.5
9300	391.0	9	7.0	14	28.8	31	4		1.73	4	6	0.5
9400	398.0	9	7.2	14	29.2	31	4		1.70	4	6	0.5
9500	405.2	9	7.2	14	29.7	32	4		1.67	4	6	0.5
9600	412.6	10	7.4	14	30.1	32	4		1.63	4	7	0.5
9700	420.0	10	7.6	13	30.6	32	4		1.60	4	7	0.5
9800	427.6	10	7.6	13	31.1	33	4		1.57	4	7	0.5
9900	435.2	11	7.8	13	31.6	33	4		1.53	4	7	0.5
10000	443.0	11	8.0	13	32.0	33	4		1.50	4	7	0.5
10100	451.0	11	8.2	12	32.5	34	4		1.47	4	8	0.5
10200	459.2	12	8.4	12	33.0	34	4		1.44	5	8	0.5
10300	467.6	12	8.6	12	33.5	34	5		1.41	5	8	0.5
10400	476.2	12	8.8	12	34.0	35	5		1.38	5	9	0.5
10500	485.0	13	9.0	11	34.6	35	5		1.35	5	9	0.5
10600	494.0	13	9.2	11	35.1	35	5		1.32	5	9	0.6
10700	503.2	13	9.4	11	35.7	35	5		1.29	5	10	0.6
10800	512.6	14	9.8	10	36.3	36	5		1.26	5	10	0.6
10900	522.4	14	10.0	10	36.9	36	5		1.23	5	10	0.6
11000	532.6	15	10.4	10	37.5	36	5		1.20	5	11	0.6
11100	543.0	16	10.8	9	38.1	36	5		1.18	6	11	0.6
11200	554.0	17	11.2	9	38.7	37	5		1.15	6	11	0.6
11300	565.6	18	11.8	8	39.4	37	5		1.12	6	12	0.6
11400	577.8	19	12.6	8	40.1	37	5		1.09	6	12	0.6
11500	590.8	20	13.4	8	40.9	38	6		1.06	6	13	0.6
11600	604.6	22	14.4	7	41.7	38	6		1.03	6	13	0.6
11700	619.4	24	15.6	6	42.5	38	6		0.99	6	14	0.6
11800	635.6	26	17.2	6	43.4	38	6		0.96	6	15	0.7
11900	654.0	30	19.6	5	44.4	38	6		0.92	6	15	0.7
12000	676.0	37	23.8	4	45.6	39	6		0.88	6	16	0.7

105-mm. HOW.
SHELL, H. E., MK I
FUZE, M48, M54
CHARGE 7
M. V., 1550 f/s

TABLE A

14	15	16	17	18	19	20	21	22	23	1
Complementary angle of site for each		Range effect of increase of —					Fuze setting for graze burst	Displacement of burst for change of 5 points in fuze setter corrector in		Range R
+ 1 mil of site	— 1 mil of site	One Sq. in wt. of proj. Two Sq. is standard Wt.	One foot per sec. in M.V. VE	Air temperature 1 degree. Standard is 59° F. Temp.	Rear wind of 1 MPH W-R	One percent in air density Den.				
<i>m</i>	<i>m</i>	yds.	yds.	yds.	yds.	yds.	sec.	<i>m</i>	R yds.	yds.
+ .06	— .05	— 4	+4.5	+2.6	+10.0	—25	23.1	8.1	144	8000
+ .06	— .05	— 3	+4.5	+2.7	+10.3	—26	23.5	8.1	144	8100
+ .07	— .06	— 2	+4.5	+2.7	+10.5	—26	23.8	8.2	143	8200
+ .07	— .06	— 2	+4.5	+2.8	+10.7	—26	24.2	8.2	142	8300
+ .08	— .06	— 1	+4.5	+2.9	+10.9	—27	24.6	8.2	141	8400
+ .08	— .07	0	+4.5	+2.9	+11.1	—27	25.0	8.3	140	8500
+ .09	— .07	+ 1	+4.6	+3.0	+11.4	—28	25.4	8.3	140	8600
+ .09	— .08	+ 1	+4.6	+3.0	+11.6	—28				8700
+ .09	— .08	+ 2	+4.6	+3.1	+11.8	—29				8800
+ .10	— .09	+ 3	+4.6	+3.2	+12.0	—29				8900
+ .10	— .09	+ 3	+4.6	+3.2	+12.2	—30				9000
+ .11	— .10	+ 4	+4.6	+3.3	+12.5	—30				9100
+ .11	— .10	+ 5	+4.6	+3.3	+12.7	—31				9200
+ .12	— .11	+ 6	+4.6	+3.4	+12.9	—31				9300
+ .12	— .11	+ 6	+4.7	+3.5	+13.1	—32				9400
+ .13	— .12	+ 7	+4.7	+3.5	+13.3	—32				9500
+ .13	— .12	+ 8	+4.7	+3.6	+13.6	—33				9600
+ .14	— .13	+ 9	+4.7	+3.6	+13.8	—33				9700
+ .15	— .14	+ 9	+4.7	+3.7	+14.0	—34				9800
+ .16	— .15	+10	+4.7	+3.7	+14.2	—34				9900
+ .17	— .16	+11	+4.7	+3.8	+14.5	—35				10000
+ .18	— .17	+12	+4.8	+3.9	+14.7	—35				10100
+ .19	— .18	+13	+4.8	+3.9	+14.9	—36				10200
+ .20	— .19	+14	+4.8	+4.0	+15.1	—36				10300
+ .21	— .20	+14	+4.8	+4.0	+15.3	—37				10400
+ .23	— .21	+15	+4.8	+4.1	+15.6	—38				10500
+ .25	— .22	+16	+4.8	+4.1	+15.8	—38				10600
+ .28	— .24	+17	+4.9	+4.2	+16.0	—39				10700
+ .31	— .26	+18	+4.9	+4.2	+16.2	—39				10800
+ .35	— .28	+19	+4.9	+4.2	+16.5	—40				10900
+ .39	— .30	+20	+4.9	+4.3	+16.7	—41				11000
+ .44	— .33	+21	+4.9	+4.3	+16.9	—41				11100
+ .50	— .37	+22	+5.0	+4.4	+17.1	—42				11200
+ .56	— .42	+23	+5.0	+4.4	+17.3	—43				11300
+ .63	— .49	+24	+5.0	+4.4	+17.6	—43				11400
+ .70	— .56	+25	+5.0	+4.4	+17.8	—44				11500
+ .79	— .63	+26	+5.1	+4.4	+18.0	—45				11600
+ .91	— .71	+28	+5.1	+4.5	+18.2	—46				11700
+1.09	— .80	+29	+5.1	+4.5	+18.4	—47				11800
+1.52	— .91	+30	+5.1	+4.5	+18.7	—48				11900
	—1.07	+32	+5.1	+4.5	+18.9	—49				12000

105-mm. HOW.
SHELL, H. E. MK I
FUZE M48, M54
CHARGE 7
M. V. 1550

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, F.....	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second.....	-22	-19	-16	-13	-9	-6	-3	0	+3	+6	+9

TABLE C

Range, yards.....	1000	2000	3000	4000	5000	6000	7000	8000	9000
Cant of 10 mils, effect	0.2	0.5	0.8	1.1	1.5	2.0	2.5	3.1	3.8

Range, yards.....	9000	10000	11000	11500	12000	12100	12200	12205	12200
Cant of 10 mils, effect	3.8	4.6	5.7	6.5	7.8	8.2	9.2	9.6	9.8

Range, yards.....	12200	12100	12000	11500	11000	10000	9000		
Cant of 10 mils, effect	9.8	10.9	11.5	13.6	15.3	18.5	21.9		

Note: Right wheel above left causes left deflection effect.

The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be used.

SECTION V

FIRING TABLES, 155-mm. HOWITZER CHARACTERISTICS

**155-mm. (Schneider) Howitzer, M1917, M1917A1, M1918, and
M1918A1, Firing HE Shell Mark I—Narrow Band**

155-mm. howitzer

Diameter of the bore between lands.....	inches	6.102
Diameter of the bore between grooves.....	inches	6.178
Total length of howitzer.....	inches	91.81
Length of rifled portion.....	inches	68.39
Travel of projectile.....	inches	69.88
Capacity of powder chamber	cubic inches	425
Number of grooves		48
Character of rifling	{ uniform twist } 1 in 25.5 calibers	
Maximum pressure for which howitzer is designed.....	lb./sq. in.	30,000

155-mm. howitzer carriage

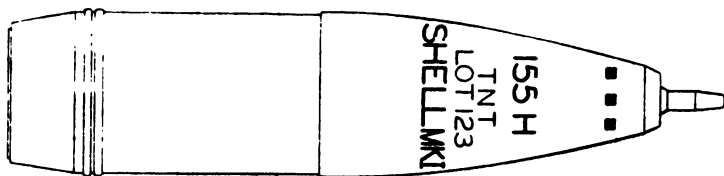
Total traverse (one half on each side)	mils	105
Least possible elevation.....	mils	0
Greatest possible elevation.....	mils	752.6
Traverse for one turn of traversing handwheel	mils	0.41
Change in elevation for one turn of elevating handwheel.....	mils	4.0
Horizontal site setting.....	mils	300

Projectile—mean weight of fuzed projectile in pounds

P. D. Fuzes, M46 and M47

Variations in weight are indicated by markings stenciled on the projectile as follows:

Marking	Weight
□ □	92.5
□ □ □	93.6
□ □ □ □ (Standard)	94.7
□ □ □ □ □	95.8
□ □ □ □ □ □	96.9



155-mm. HOWITZER

Characteristics of Charges (HE Shell, Mark I, Narrow Band; Fuzes, M46 and M47¹) for which tables are not included herein.

CHARGE 1 Muzzle Velocity 680 f/s.

Powder Charge: Charge consists of base section Number 1. The cartridge cloth is dyed green. The igniter pad is on the rear end of the base section.

Maximum Range 4255 yards.

CHARGE 2 Muzzle Velocity 742 f/s.

Powder Charge: Charge consists of base section and one increment, numbered respectively 1 and 2. The cartridge cloth is dyed green. The igniter pad is on the rear end of the base section.

Maximum Range 4910 yards.

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¹Fuzes Mark III and Mark IV, and shrapnel and chemical shell may also be used with this weapon. Tables and characteristics for these are not included herein.

CHARGE 3

155-mm. HOWITZER

HE Shell, Mark I, Narrow Band;

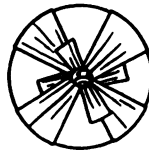
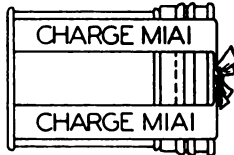
Fuze M46 and M47

Powder Charge: Base Section and Increments Number 2 and 3.

Charge consists of base section and two increments, numbered respectively 1, 2, and 3. The cartridge cloth is dyed green. The igniter pad is on the rear end of the base section.

Maximum Range: 5990 yards.

(Data for ranges 0—5990 included herein.)



155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 3
M.V. 832 f/s

TABLE A

1	2	3	4	5	6	7 8		9	10
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range c	Change in range for 1-mil change in elevation 1 mil	Time of flight Time	Probable Error		Slope of fall Slope	Line number of metro message Line
						Range e _{pr}	Deflection e _{pd}		
yds.	m	m	m	yds.	sec.	yds.	yds.	1/-	No.
0	-5.8	2	7.0	14	0.0	8	0		1
100	+1.2	2	7.2	14	0.4	8	0	140	1
200	8.4	2	7.2	14	0.7	8	0	70	1
300	15.6	2	7.2	14	1.1	8	0	46	1
400	23.0	2	7.2	14	1.4	8	0	35	1
500	30.2	2	7.2	14	1.8	8	0	28	1
600	37.6	2	7.4	14	2.2	9	0	23	1
700	44.8	2	7.4	13	2.6	9	0	19.7	1
800	52.2	3	7.4	13	2.9	9	0	17.2	1
900	59.8	3	7.4	13	3.3	9	0	15.2	1
1000	67.2	3	7.4	13	3.7	9	0	13.6	1
1100	74.8	3	7.4	13	4.1	10	0	12.3	1
1200	82.2	3	7.6	13	4.5	10	0	11.2	1
1300	89.8	3	7.6	13	4.8	10	0	10.3	1
1400	97.6	3	7.6	13	5.2	10	1	9.5	1
1500	105.4	3	7.8	13	5.6	10	1	8.9	1
1600	113.2	4	7.8	13	6.0	11	1	8.3	1
1700	121.0	4	7.8	13	6.4	11	1	7.7	1
1800	128.8	4	8.0	12	6.7	11	1	7.2	1
1900	136.8	4	8.0	12	7.1	12	1	6.8	1
2000	145.0	4	8.2	12	7.5	12	1	6.4	1
2100	153.0	5	8.2	12	7.9	12	1	6.1	1
2200	161.2	5	8.4	12	8.3	13	1	5.8	1
2300	169.6	5	8.4	12	8.7	13	1	5.5	1
2400	178.0	5	8.6	12	9.1	13	1	5.2	1
2500	186.6	5	8.6	12	9.5	14	1	5.0	1
2600	195.2	6	8.8	11	9.9	14	1	4.7	1
2700	204.0	6	8.8	11	10.3	15	1	4.5	1
2800	212.8	6	9.0	11	10.8	15	1	4.3	1
2900	221.8	6	9.0	11	11.2	16	1	4.1	1
3000	230.8	6	9.2	11	11.6	16	1	4.0	1
3100	240.0	7	9.2	11	12.1	17	1	3.8	1
3200	249.4	7	9.4	10	12.5	17	1	3.7	2
3300	259.0	7	9.6	10	13.0	17	1	3.5	2
3400	268.8	7	9.8	10	13.4	18	1	3.4	2
3500	278.6	7	10.0	10	13.9	18	1	3.2	2
3600	288.6	8	10.2	10	14.4	19	2	3.1	2
3700	299.0	8	10.4	10	14.9	19	2	2.9	2
3800	309.6	8	10.6	9	15.4	19	2	2.8	2
3900	320.4	9	10.8	9	15.9	20	2	2.7	2
4000	331.4	9	11.2	9	16.4	20	2	2.6	2

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 8
M.V. 832 f/s

TABLE A

11	12	13	14	15	16	17	18	19	1
Deflection effect		Complementary angle of site for each		Range Effect of Increase of—					Range R
Drift Dft.*	Lateral wind of 1 mile per hour(+)	+ 1 mil of site	- 1 mil of site	One Sq. in wt. of proj. Four Sq. is standard	One foot per sec. in M.V.	Air temperature 1 degree. Standard is 59° F.	Rear wind 1 mile per hour	One percent in air density	
W-D	W-D	W-D	W-D	Wt. yds.	VE yds.	Temp. yds.	W-R yds.	Den. yds.	
L 4	0.0	.00	.00	0	0.0	0.0	0.0	0	0
L 4	0.0	.00	.00	- 1	+0.2	0.0	0.0	0	100
L 4	0.0	.00	.00	- 2	+0.4	0.0	0.0	0	200
L 4	0.0	.00	.00	- 3	+0.7	0.0	0.0	0	300
L 4	0.0	.00	.00	- 4	+0.9	0.0	0.0	0	400
L 3	0.0	.00	.00	- 5	+1.1	0.0	+0.1	0	500
L 3	0.0	.00	.00	- 6	+1.3	0.0	+0.1	0	600
L 3	0.0	.00	.00	- 7	+1.6	0.0	+0.1	0	700
L 3	0.0	.00	.00	- 8	+1.8	0.0	+0.1	0	800
L 3	0.0	.00	.00	- 9	+2.1	0.0	+0.2	0	900
L 2	0.0	+0.1	-.01	-10	+2.3	0.0	+0.2	0	1000
L 2	0.0	+0.1	-.01	-11	+2.5	0.0	+0.2	0	1100
L 2	0.0	+0.1	-.01	-12	+2.7	0.0	+0.2	0	1200
L 2	0.0	+0.2	-.01	-12	+3.0	0.0	+0.3	0	1300
L 2	0.0	+0.2	-.01	-13	+3.2	0.0	+0.3	0	1400
L 1	0.1	+0.2	-.01	-14	+3.4	0.0	+0.3	0	1500
L 1	0.1	+0.2	-.01	-15	+3.6	0.0	+0.3	-1	1600
L 1	0.1	+0.2	-.01	-16	+3.8	0.0	+0.4	-1	1700
L 1	0.1	+0.3	-.02	-17	+4.1	0.0	+0.4	-1	1800
0	0.1	+0.3	-.02	-18	+4.3	0.0	+0.4	-1	1900
0	0.1	+0.3	-.02	-19	+4.5	0.0	+0.4	-1	2000
0	0.1	+0.4	-.02	-20	+4.7	0.0	+0.5	-1	2100
0	0.1	+0.4	-.03	-21	+4.9	0.0	+0.5	-1	2200
R 1	0.1	+0.4	-.03	-21	+5.2	0.0	+0.5	-1	2300
R 1	0.1	+0.5	-.03	-22	+5.4	0.0	+0.6	-1	2400
R 1	0.1	+0.5	-.04	-23	+5.6	0.0	+0.6	-1	2500
R 1	0.1	+0.6	-.04	-24	+5.8	0.0	+0.6	-2	2600
R 2	0.1	+0.6	-.05	-25	+6.0	0.0	+0.7	-2	2700
R 2	0.1	+0.7	-.06	-25	+6.2	0.0	+0.7	-2	2800
R 2	0.1	+0.7	-.06	-26	+6.4	0.0	+0.8	-2	2900
R 2	0.1	+0.8	-.07	-27	+6.6	0.0	+0.8	-2	3000
R 2	0.1	+0.8	-.08	-28	+6.8	0.0	+0.8	-2	3100
R 3	0.1	+0.9	-.08	-28	+7.0	0.0	+0.9	-2	3200
R 3	0.1	+0.9	-.09	-29	+7.2	0.0	+0.9	-2	3300
R 3	0.1	+1.0	-.10	-29	+7.4	0.0	+1.0	-3	3400
R 3	0.1	+1.1	-.11	-30	+7.6	0.0	+1.0	-3	3500
R 4	0.1	+1.2	-.12	-31	+7.8	0.0	+1.1	-3	3600
R 4	0.1	+1.3	-.13	-31	+8.0	0.0	+1.1	-3	3700
R 4	0.1	+1.4	-.14	-32	+8.2	0.0	+1.2	-3	3800
R 5	0.1	+1.5	-.15	-32	+8.4	0.0	+1.2	-4	3900
R 5	0.1	+1.6	-.16	-33	+8.6	-0.1	+1.3	-4	4000

*Drift includes side jump.

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 3
M.V. 832 f/s

TABLE A[illegible]

TABLE A

*Drift includes side jump.

155-mm. HOW.
 SHELL, H.E., MK I
 FUZE M46 AND M47
 CHARGE 3
 M.V. 832 f/s

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, F.-----	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second--	-19	-17	-14	-12	-9	-6	-3	0	+3	+6	+9

TABLE C

Cant of carriage axle, deflection effect in mils due to.

Range, yards	1000	2000	3000	4000	4500	5000
Cant of 10 mils, effect--	0.7	1.5	2.4	3.4	4.1	4.9

Range, yards	5000	5500	5600	5700	5800	5900	5990
Cant of 10 mils, effect--	4.9	6.1	6.5	6.9	7.3	8.0	9.7

Note: Right wheel above left causes left deflection effect.
 The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be so cross leveled.

CHARGE 4

155-mm. HOWITZER

HE Shell, Mark I, Narrow Band;

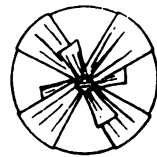
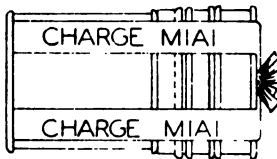
Fuze M46 and M47

Powder Charge: Base Section and Increments Number 2, 3, and 4.

Charge consists of base section and three increments, numbered respectively 1, 2, 3, and 4. The cartridge cloth is dyed green. The igniter pad is on the rear end of the base section.

Maximum Range: 7360 yards.

(Data for ranges 3000—7000 included herein.)



155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 4
M.V. 939 f/s

TABLE A

1	2	3	4	5	6	Probable Error		9	10
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range c	Change in range for 1-mil change in elevation 1 mil	Time of flight Time	Range e _{pr}	Deflection e _{pd}	Slope of fall Slope	Line number of metro message Line
yds.	m	m	m	yds.	sec.	yds.	yds.	1/-	No.
3000	178.2	4	6.8	15	10.3	16	2	5.1	1
3100	185.0	5	7.0	14	10.7	17	2	4.9	1
3200	192.0	5	7.0	14	11.1	17	2	4.7	1
3300	199.0	5	7.0	14	11.4	17	2	4.6	1
3400	206.2	5	7.2	14	11.8	18	2	4.4	1
3500	213.4	5	7.2	14	12.2	18	2	4.2	1
3600	220.6	6	7.4	14	12.6	19	2	4.1	2
3700	228.0	6	7.4	13	13.0	19	2	3.9	2
3800	235.4	6	7.6	13	13.4	19	2	3.8	2
3900	243.0	6	7.6	13	13.8	20	2	3.7	2
4000	250.8	6	7.8	13	14.2	20	2	3.5	2
4100	258.6	7	7.8	13	14.6	21	2	3.4	2
4200	266.4	7	8.0	13	15.0	21	2	3.3	2
4300	274.4	7	8.0	12	15.4	21	2	3.2	2
4400	282.6	7	8.2	12	15.8	22	2	3.1	2
4500	291.0	7	8.4	12	16.3	22	2	3.0	2
4600	299.4	8	8.4	12	16.7	23	2	2.9	2
4700	308.0	8	8.6	12	17.2	23	2	2.8	2
4800	316.6	8	8.8	11	17.6	23	2	2.7	2
4900	325.4	9	9.0	11	18.1	24	2	2.7	2
5000	334.4	9	9.2	11	18.5	24	2	2.6	2
5100	343.6	10	9.2	11	19.0	25	3	2.5	2
5200	352.8	10	9.4	11	19.5	25	3	2.4	3
5300	362.4	10	9.6	10	19.9	25	3	2.3	3
5400	372.2	11	10.0	10	20.4	26	3	2.3	3
5500	382.2	11	10.2	10	20.9	26	3	2.2	3
5600	392.4	12	10.4	10	21.4	27	3	2.1	3
5700	403.0	12	10.6	9	21.9	27	3	2.1	3
5800	413.8	12	11.0	9	22.4	27	3	2.00	3
5900	425.0	13	11.4	9	22.9	28	3	1.94	3
6000	436.6	13	11.8	9	23.5	28	3	1.88	3
6100	448.6	14	12.2	8	24.1	29	3	1.82	3
6200	461.0	14	12.6	8	24.7	29	3	1.76	3
6300	473.8	15	13.2	8	25.3	29	3	1.70	3
6400	487.2	16	13.8	7	25.9	30	3	1.64	3
6500	501.4	17	14.6	7	26.6	30	3	1.58	3
6600	516.4	18	15.6	6	27.3	31	3	1.52	4
6700	532.4	20	16.8	6	28.1	31	3	1.46	4
6800	549.6	22	18.2	6	28.9	31	4	1.40	4
6900	568.6	25	20.0	5	29.7	32	4	1.34	4
7000	590.0	29	22.4	5	30.6	32	4	1.28	4

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 4
M.V. 939 f/s

TABLE A

11	12	13	14	15	16	17	18	19	1
Deflection effect		Complementary angle of site for each		Range Effect of Increase of—					
Drift Dft.*	Lateral wind of 1 mile per hour (+) W-D	+ 1 mil of site	- 1 mil of site	One Sq. in wt. of proj. Four Sq. is standard Wt.	One foot per sec. in M.V. VE	Air temperature 1 degree. Standard is 59° F. Temp.	Rear wind 1 mile per hour W-R	One percent in air density Den.	Range R
m	m	m	m	yds.	yds.	yds.	yds.	yds.	yds.
R 1	0.1	+ .04	- .04	-26	+ 5.6	+0.2	+1.2	- 2	3000
R 1	0.1	+ .04	- .04	-27	+ 5.8	+0.2	+1.3	- 2	3100
R 1	0.1	+ .05	- .05	-27	+ 5.9	+0.2	+1.3	- 2	3200
R 2	0.1	+ .05	- .05	-28	+ 6.1	+0.2	+1.4	- 2	3300
R 2	0.1	+ .06	- .06	-28	+ 6.2	+0.2	+1.4	- 2	3400
R 2	0.1	+ .06	- .06	-29	+ 6.4	+0.2	+1.5	- 3	3500
R 2	0.1	+ .06	- .06	-30	+ 6.6	+0.2	+1.6	- 3	3600
R 3	0.1	+ .07	- .07	-30	+ 6.7	+0.2	+1.6	- 3	3700
R 3	0.1	+ .07	- .07	-31	+ 6.9	+0.2	+1.7	- 3	3800
R 3	0.1	+ .08	- .08	-31	+ 7.0	+0.2	+1.7	- 3	3900
R 3	0.1	+ .08	- .08	-32	+ 7.2	+0.3	+1.8	- 3	4000
R 3	0.1	+ .09	- .09	-33	+ 7.4	+0.3	+1.9	- 4	4100
R 4	0.1	+ .09	- .09	-33	+ 7.5	+0.3	+2.0	- 4	4200
R 4	0.1	+ .10	- .10	-34	+ 7.7	+0.3	+2.0	- 4	4300
R 4	0.1	+ .10	- .10	-34	+ 7.8	+0.3	+2.1	- 4	4400
R 4	0.1	+ .11	- .11	-35	+ 8.0	+0.3	+2.2	- 4	4500
R 5	0.1	+ .12	- .12	-35	+ 8.2	+0.3	+2.3	- 5	4600
R 5	0.1	+ .13	- .13	-36	+ 8.3	+0.3	+2.3	- 5	4700
R 5	0.1	+ .14	- .13	-36	+ 8.5	+0.3	+2.4	- 5	4800
R 6	0.1	+ .15	- .14	-37	+ 8.6	+0.3	+2.4	- 5	4900
R 6	0.2	+ .16	- .15	-37	+ 8.8	+0.3	+2.5	- 5	5000
R 6	0.2	+ .17	- .16	-38	+ 9.0	+0.3	+2.6	- 6	5100
R 7	0.2	+ .18	- .17	-38	+ 9.2	+0.4	+2.6	- 6	5200
R 7	0.2	+ .19	- .18	-39	+ 9.3	+0.4	+2.7	- 6	5300
R 7	0.2	+ .20	- .19	-39	+ 9.5	+0.4	+2.7	- 6	5400
R 7	0.2	+ .22	- .20	-40	+ 9.7	+0.4	+2.8	- 7	5500
R 7	0.2	+ .24	- .21	-41	+ 9.9	+0.4	+2.9	- 7	5600
R 8	0.2	+ .26	- .22	-41	+10.1	+0.4	+2.9	- 7	5700
R 8	0.2	+ .28	- .24	-42	+10.2	+0.4	+3.0	- 7	5800
R 9	0.2	+ .30	- .26	-42	+10.4	+0.4	+3.0	- 8	5900
R 9	0.2	+ .32	- .28	-43	+10.6	+0.4	+3.1	- 8	6000
R 9	0.2	+ .35	- .30	-44	+10.8	+0.4	+3.2	- 8	6100
R 10	0.2	+ .38	- .33	-44	+11.0	+0.4	+3.2	- 9	6200
R 10	0.2	+ .42	- .36	-45	+11.1	+0.4	+3.3	- 9	6300
R 11	0.2	+ .46	- .40	-45	+11.3	+0.4	+3.3	- 9	6400
R 11	0.2	+ .51	- .44	-46	+11.5	+0.4	+3.4	-10	6500
R 12	0.2	+ .58	- .49	-46	+11.7	+0.4	+3.5	-10	6600
R 13	0.2	+ .66	- .56	-47	+11.9	+0.4	+3.5	-10	6700
R 13	0.3	+ .77	- .65	-47	+12.1	+0.3	+3.6	-11	6800
R 14	0.3	+ .92	- .76	-48	+12.3	+0.3	+3.6	-11	6900
R 15	0.3	+1.19	- .87	-48	+12.5	+0.3	+3.7	-12	7000

*Drift includes side jump.

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 4
M.V. 939 f/s

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, F.-----	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second..	-18	-16	-14	-12	-9	-6	-3	0	+3	+6	+9

TABLE C

Cant of carriage axle, deflection effect in mils due to.

Range, yards	1000	2000	3000	4000	5000	6000	6500
Cant of 10 mils, effect..	0.6	1.2	1.8	2.6	3.5	4.6	5.4

Range, yards	6500	7000	7100	7200	7300	7360
Cant of 10 mils, effect..	5.4	6.6	7.0	7.4	8.1	9.3

Note: Right wheel above left causes left deflection effect.
The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be so cross leveled.

CHARGE 5
155-mm. HOWITZER
HE Shell, Mark I, Narrow Band;
Fuzeze M46 and M47

Powder Charge: M1A1, Base Section and Increments Number 2, 3, 4, and 5.

Powder Charge: M2, Base Section Number 5.

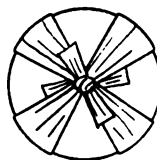
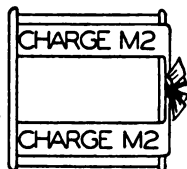
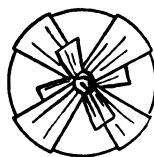
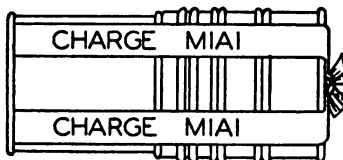
Charge M1A1 consists of base section and four increments, numbered respectively 1, 2, 3, 4, and 5. The cartridge cloth is dyed green. The igniter pad is on the rear end of the base section.

Charge M2 consists of base section Number 5. (Some M2 charges in storage are in 7 sections.) The cartridge cloth is white. The igniter pad is on the rear end of the base section.

For firing Charge 5, either Charge M1A1 or Charge M2, prepared as shown in the figure, may be used; but preferably the former.

Maximum Range: 9295 yards.

(Data for ranges 5000—9000 included herein.)



155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 5
M.V. 1083 f/s

TABLE A

1	2	3	4	5	6	7 8		9	10
Range R	Elevation El	Fork F	Change in elevation for 100-yd. change in range c	Change in range for 1-mil change in elevation 1 mil	Time of flight Time	Probable Error		Slope of fall Slope	Line number of metro message Line
						Range e _{pr}	Deflection e _{pd}		
yds.	m	m	m	yds.	sec.	yds.	yds.	1/-	No.
5000	250.2	8	6.4	16	16.0	30	3	3.4	2
5100	256.6	8	6.4	16	16.4	31	3	3.3	2
5200	263.0	8	6.6	15	16.7	31	3	3.3	2
5300	269.6	8	6.6	15	17.1	31	3	3.2	2
5400	276.2	8	6.8	15	17.5	32	3	3.1	2
5500	282.8	9	6.8	15	17.9	32	3	3.0	2
5600	289.6	9	6.8	15	18.3	33	3	2.9	2
5700	296.4	9	7.0	14	18.7	33	3	2.8	2
5800	303.4	10	7.0	14	19.1	34	3	2.8	2
5900	310.4	10	7.2	14	19.5	34	3	2.7	3
6000	317.6	10	7.2	14	19.9	34	3	2.6	3
6100	324.8	10	7.4	14	20.3	35	3	2.6	3
6200	332.2	10	7.4	14	20.7	35	3	2.5	3
6300	339.8	11	7.6	13	21.2	36	3	2.4	3
6400	347.4	11	7.6	13	21.6	36	3	2.4	3
6500	355.0	11	7.8	13	22.0	37	3	2.3	3
6600	362.8	12	7.8	13	22.5	37	3	2.3	3
6700	370.8	12	8.0	12	22.9	37	3	2.2	3
6800	379.0	12	8.2	12	23.4	38	4	2.1	3
6900	387.2	13	8.2	12	23.8	38	4	2.1	3
7000	395.6	13	8.4	12	24.3	39	4	2.0	3
7100	404.2	14	8.6	12	24.7	39	4	1.99	3
7200	413.0	14	8.8	11	25.2	40	4	1.94	3
7300	421.8	15	9.0	11	25.7	40	4	1.89	3
7400	431.0	15	9.2	11	26.2	41	4	1.85	3
7500	440.4	16	9.4	11	26.7	41	4	1.80	3
7600	450.0	16	9.6	10	27.2	42	4	1.75	3
7700	459.8	17	9.8	10	27.8	42	4	1.71	4
7800	470.0	17	10.2	10	28.3	42	4	1.66	4
7900	480.4	18	10.6	9	28.9	43	4	1.62	4
8000	491.2	19	11.0	9	29.4	43	4	1.57	4
8100	502.4	20	11.4	9	30.0	44	4	1.53	4
8200	514.0	21	11.8	8	30.6	44	4	1.49	4
8300	526.0	22	12.4	8	31.3	45	4	1.44	4
8400	538.6	23	13.0	8	31.9	45	5	1.40	4
8500	552.0	25	13.8	7	32.6	46	5	1.36	4
8600	566.2	27	14.6	7	33.3	46	5	1.32	4
8700	581.2	29	15.6	6	34.1	46	5	1.27	5
8800	597.6	32	17.0	6	34.9	47	5	1.23	5
8900	615.4	36	18.8	5	35.7	47	5	1.18	5
9000	635.6	41	21.6	5	36.7	48	5	1.13	5

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 5
M.V. 1083 f/s

TABLE A

11	12	13	14	15	16	17	18	19	20	1
Deflection effect		Complementary angle of site for each		Range Effect of Increase of—						
Drift	Lateral wind of 1 mile per hour (+)	+ 1 mil o site	- 1 mil of site	One Sq. in wt. of proj. Four Sq. is standard	One Sq. in wt. of proj. Four Sq. is standard	One foot per sec. in M.V.	Air temperature 1 degree. Standard is 59° F.	Rear wind 1 mile per hour	One percent in air density	Range
Dft.*	W-D			**Wt.	***Wt.	VE	Temp.	W-R	Den.	R
m	m	m	m	yds.	yds.	yds.	yds.	yds.	yds.	yds.
R 5	0.2	+ .08	-.08	-28	-12	+ 6.2	+3.0	+4.4	- 6	5000
R 5	0.2	+ .08	-.08	-29	-12	+ 6.3	+3.1	+4.5	- 6	5100
R 6	0.2	+ .09	-.09	-29	-12	+ 6.4	+3.1	+4.6	- 7	5200
R 6	0.2	+ .09	-.09	-29	-12	+ 6.5	+3.2	+4.8	- 7	5300
R 6	0.2	+ .10	-.10	-30	-12	+ 6.6	+3.2	+4.9	- 7	5400
R 6	0.2	+ .10	-.10	-30	-12	+ 6.7	+3.3	+5.0	- 7	5500
R 6	0.2	+ .10	-.10	-30	-12	+ 6.8	+3.4	+5.1	- 8	5600
R 7	0.2	+ .11	-.11	-30	-12	+ 6.9	+3.4	+5.3	- 8	5700
R 7	0.2	+ .11	-.11	-31	-12	+ 7.0	+3.5	+5.4	- 8	5800
R 7	0.2	+ .12	-.12	-31	-12	+ 7.1	+3.5	+5.6	- 9	5900
R 7	0.2	+ .12	-.12	-31	-13	+ 7.2	+3.6	+5.7	- 9	6000
R 7	0.2	+ .13	-.13	-31	-13	+ 7.3	+3.6	+5.8	- 9	6100
R 7	0.2	+ .13	-.13	-32	-13	+ 7.4	+3.7	+6.0	-10	6200
R 8	0.2	+ .14	-.14	-32	-13	+ 7.5	+3.7	+6.1	-10	6300
R 8	0.2	+ .14	-.14	-32	-13	+ 7.6	+3.8	+6.3	-10	6400
R 8	0.2	+ .15	-.15	-32	-13	+ 7.7	+3.8	+6.4	-10	6500
R 8	0.2	+ .16	-.16	-33	-13	+ 7.8	+3.8	+6.5	-11	6600
R 9	0.2	+ .16	-.16	-33	-13	+ 7.9	+3.9	+6.6	-11	6700
R 9	0.2	+ .17	-.17	-33	-13	+ 8.0	+3.9	+6.8	-11	6800
R 9	0.2	+ .18	-.18	-33	-13	+ 8.1	+4.0	+6.9	-11	6900
R 9	0.2	+ .19	-.19	-33	-13	+ 8.2	+4.0	+7.0	-12	7000
R 10	0.2	+ .20	-.20	-34	-13	+ 8.3	+4.0	+7.1	-12	7100
R 10	0.2	+ .21	-.21	-34	-13	+ 8.5	+4.1	+7.2	-12	7200
R 10	0.2	+ .22	-.22	-34	-13	+ 8.6	+4.1	+7.4	-12	7300
R 10	0.2	+ .24	-.24	-34	-13	+ 8.7	+4.2	+7.5	-13	7400
R 10	0.2	+ .26	-.25	-34	-12	+ 8.8	+4.2	+7.6	-13	7500
R 11	0.2	+ .28	-.27	-35	-12	+ 8.9	+4.2	+7.7	-13	7600
R 11	0.2	+ .30	-.28	-35	-12	+ 9.0	+4.2	+7.8	-14	7700
R 11	0.3	+ .32	-.30	-35	-12	+ 9.1	+4.3	+7.9	-14	7800
R 12	0.3	+ .34	-.32	-36	-12	+ 9.2	+4.3	+8.0	-14	7900
R 12	0.3	+ .37	-.34	-36	-12	+ 9.4	+4.3	+8.1	-15	8000
R 12	0.3	+ .41	-.36	-36	-12	+ 9.5	+4.3	+8.2	-15	8100
R 13	0.3	+ .45	-.39	-37	-12	+ 9.6	+4.3	+8.3	-15	8200
R 13	0.3	+ .49	-.42	-37	-12	+ 9.8	+4.4	+8.3	-16	8300
R 14	0.3	+ .53	-.46	-37	-12	+ 9.9	+4.4	+8.4	-16	8400
R 14	0.3	+ .58	-.51	-37	-12	+10.0	+4.4	+8.5	-17	8500
R 15	0.3	+ .63	-.57	-38	-12	+10.2	+4.4	+8.6	-17	8600
R 15	0.3	+ .69	-.64	-38	-12	+10.3	+4.4	+8.7	-18	8700
R 16	0.3	+ .78	-.72	-38	-12	+10.5	+4.4	+8.7	-18	8800
R 16	0.3	+ .96	-.81	-39	-12	+10.7	+4.4	+8.8	-19	8900
R 17	0.3	+1.39	-.94	-39	-12	+10.8	+4.4	+8.8	-19	9000

*Drift includes side jump.

**These values are to be used with propelling charge M1A1 (Green Bag).

***These values are to be used with propelling charge M2 (White Bag).

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 5
M.V. 1083 f/s

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, F.....	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second. Propelling charge M1A1 (Green Bag).....	-17	-15	-13	-10	- 8	- 5	- 3	0	+ 2	+ 5	+ 8
Change in velocity, feet per second. Propelling charge M2 (White Bag).....	-27	-23	-20	-16	-12	- 8	- 4	0	+ 4	+ 8	+12

TABLE C

Cant of carriage axle, deflection effect in mils due to.

Range, yards	1000	2000	3000	4000	5000	6000	7000
Cant of 10 mils, effect.....	0.4	0.9	1.4	2.0	2.6	3.3	4.2

Range, yards	7000	8000	8500	9000	9100	9200	9295
Cant of 10 mils, effect.....	4.2	5.3	6.1	7.3	7.6	8.1	9.4

Note: Right wheel above left causes left deflection effect.

The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be so cross leveled.

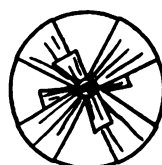
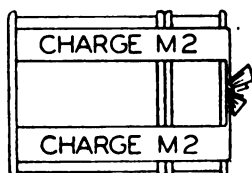
CHARGE 6
155-mm. HOWITZER
HE Shell, Mark I, Narrow Band;
Fuze M46 and M47

Powder Charge: Base Section and Increment Number 6.

Charge consists of base section and one increment, numbered 5 and 6 respectively. (Some M2 charges in storage are in 7 sections.) The cartridge cloth is white. The igniter pad is on the rear end of the base section.

Maximum Range: 11,460 yards.

(Data for ranges 7000—11,000 included herein.)



155-mm HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 6
M.V. 1359 f/s

TABLE A

1	2	3	4	5	6	Probable Error		9	10
Range	Elevation	Fork	Change in elevation for 100-yd. change in range	Change in range for 1-mil change in elevation	Time of flight	Range	Deflection	Slope of fall	Line number of metro message
R	El	F	c	1 mil	Time	e _{pr}	e _{pd}	Slope	Line
yds.	m	m	m	yds.	sec.	yds.	yds.	1/-	No.
7000	290.0	9	6.0	16	20.9	35	5	2.7	3
7100	296.0	10	6.2	16	21.3	35	5	2.6	3
7200	302.2	10	6.2	16	21.7	36	5	2.5	3
7300	308.4	10	6.2	16	22.1	36	5	2.5	3
7400	314.8	10	6.4	16	22.5	37	5	2.4	3
7500	321.2	10	6.4	15	22.9	37	5	2.4	3
7600	327.6	11	6.4	15	23.2	37	5	2.3	3
7700	334.2	11	6.6	15	23.6	38	5	2.3	3
7800	340.8	11	6.6	15	24.0	38	5	2.2	3
7900	347.4	11	6.6	15	24.4	39	5	2.2	3
8000	354.2	11	6.8	15	24.9	39	5	2.1	3
8100	361.0	12	6.8	14	25.3	39	5	2.1	3
8200	368.0	12	7.0	14	25.7	40	5	2.0	3
8300	375.0	12	7.0	14	26.2	40	5	1.99	3
8400	382.0	12	7.0	14	26.6	41	5	1.95	3
8500	389.2	12	7.2	14	27.0	41	5	1.91	3
8600	396.4	13	7.2	14	27.5	41	5	1.87	4
8700	403.8	13	7.4	13	27.9	42	5	1.83	4
8800	411.4	13	7.6	13	28.4	42	5	1.79	4
8900	419.0	13	7.6	13	28.8	43	6	1.75	4
9000	426.6	14	7.8	13	29.3	43	6	1.72	4
9100	434.4	14	7.8	13	29.8	43	6	1.68	4
9200	442.4	15	8.0	12	30.3	44	6	1.64	4
9300	450.4	15	8.2	12	30.7	44	6	1.61	4
9400	458.6	16	8.2	12	31.2	45	6	1.58	4
9500	467.0	16	8.4	12	31.7	45	6	1.54	4
9600	475.6	17	8.6	11	32.2	45	6	1.51	4
9700	484.4	17	8.8	11	32.7	46	6	1.48	4
9800	493.4	18	9.0	11	33.3	46	6	1.44	5
9900	502.6	18	9.2	11	33.8	47	6	1.41	5
10000	512.0	19	9.6	10	34.3	47	6	1.38	5
10100	521.6	19	10.0	10	34.9	47	7	1.35	5
10200	531.6	20	10.4	10	35.5	48	7	1.32	5
10300	542.2	21	10.8	9	36.1	48	7	1.29	5
10400	553.4	22	11.4	9	36.7	49	7	1.26	5
10500	565.0	24	12.0	8	37.3	49	7	1.22	5
10600	577.2	25	12.8	8	38.0	49	7	1.19	5
10700	590.2	26	13.6	7	38.7	50	7	1.16	6
10800	604.2	28	14.6	7	39.5	50	7	1.12	6
10900	619.2	31	15.8	6	40.3	50	8	1.09	6
11000	635.6	35	17.4	6	41.2	50	8	1.05	6

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 6
M.V. 1359 f/s

TABLE A

11	12	13	14	15	16	17	18	19	1
Deflection effect		Complementary angle of site for each		Range Effect of Increase of—					
Drift Dft.*	Lateral wind of 1 mile per hour (+) W-D	+1 mil of site	-1 mil of site	One Sq. in wt. of proj. Four Sq. is standard Wt.	One foot per sec. in M.V. VE	Air temperature 1 degree. Standard is 59° F. Temp.	Rear wind 1 mile per hour W-R	One percent in air density Den.	Range R
m	m	m	m	yds.	yds.	yds.	yds.	yds.	yds.
R 6	0.3	+ .10	-.10	-6	+5.6	+3.6	+ 7.4	-16	7000
R 6	0.3	+ .10	-.10	-6	+5.6	+3.7	+ 7.6	-17	7100
R 7	0.3	+ .10	-.10	-6	+5.7	+3.8	+ 7.8	-17	7200
R 7	0.3	+ .11	-.11	-5	+5.7	+3.9	+ 7.9	-18	7300
R 7	0.3	+ .11	-.11	-5	+5.8	+4.0	+ 8.1	-18	7400
R 7	0.4	+ .11	-.11	-5	+5.8	+4.1	+ 8.3	-18	7500
R 7	0.4	+ .12	-.12	-5	+5.8	+4.2	+ 8.5	-19	7600
R 8	0.4	+ .12	-.12	-4	+5.9	+4.3	+ 8.6	-19	7700
R 8	0.4	+ .13	-.13	-4	+5.9	+4.3	+ 8.8	-19	7800
R 8	0.4	+ .13	-.13	-4	+6.0	+4.4	+ 9.0	-20	7900
R 8	0.4	+ .13	-.13	-3	+6.0	+4.5	+ 9.2	-20	8000
R 8	0.4	+ .13	-.13	-3	+6.0	+4.6	+ 9.4	-21	8100
R 9	0.4	+ .14	-.14	-3	+6.1	+4.7	+ 9.6	-21	8200
R 9	0.4	+ .14	-.14	-3	+6.1	+4.7	+ 9.7	-21	8300
R 9	0.4	+ .15	-.15	-2	+6.2	+4.8	+ 9.9	-22	8400
R 9	0.4	+ .15	-.15	-2	+6.2	+4.9	+10.1	-22	8500
R 9	0.4	+ .16	-.16	-2	+6.2	+5.0	+10.3	-23	8600
R 10	0.4	+ .16	-.16	-1	+6.2	+5.0	+10.5	-23	8700
R 10	0.4	+ .17	-.17	-1	+6.3	+5.1	+10.7	-24	8800
R 10	0.4	+ .17	-.17	0	+6.3	+5.1	+10.9	-24	8900
R 10	0.4	+ .18	-.18	0	+6.3	+5.2	+11.1	-24	9000
R 10	0.4	+ .19	-.19	0	+6.3	+5.3	+11.3	-25	9100
R 11	0.4	+ .20	-.20	0	+6.4	+5.3	+11.5	-25	9200
R 11	0.4	+ .21	-.20	+1	+6.4	+5.4	+11.7	-26	9300
R 11	0.4	+ .22	-.21	+1	+6.5	+5.4	+11.9	-26	9400
R 11	0.4	+ .23	-.22	+1	+6.5	+5.5	+12.1	-27	9500
R 12	0.4	+ .25	-.23	+1	+6.5	+5.6	+12.3	-27	9600
R 12	0.4	+ .27	-.25	+2	+6.6	+5.6	+12.5	-28	9700
R 12	0.4	+ .29	-.27	+2	+6.6	+5.7	+12.7	-28	9800
R 13	0.4	+ .31	-.28	+3	+6.7	+5.7	+12.9	-29	9900
R 13	0.5	+ .33	-.30	+3	+6.7	+5.8	+13.1	-29	10000
R 13	0.5	+ .36	-.32	+3	+6.8	+5.9	+13.3	-30	10100
R 14	0.5	+ .40	-.35	+4	+6.8	+5.9	+13.5	-30	10200
R 14	0.5	+ .44	-.38	+4	+6.9	+6.0	+13.8	-31	10300
R 15	0.5	+ .49	-.41	+5	+6.9	+6.0	+14.0	-31	10400
R 15	0.5	+ .56	-.45	+5	+7.0	+6.1	+14.2	-32	10500
R 16	0.5	+ .64	-.50	+5	+7.0	+6.2	+14.4	-32	10600
R 16	0.5	+ .73	-.56	+6	+7.1	+6.2	+14.6	-33	10700
R 16	0.5	+ .84	-.63	+6	+7.1	+6.3	+14.9	-33	10800
R 17	0.5	+1.00	-.72	+7	+7.2	+6.3	+15.1	-34	10900
R 17	0.5	+1.24	-.83	+7	+7.2	+6.4	+15.3	-35	11000

*Drift includes side jump.

155-mm. HOW.
SHELL, H.E., MK I
FUZE M46 AND M47
CHARGE 6
M.V. 1359 f/s

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, F.-----	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second-----	-37	-32	-27	-21	-16	-11	- 5	0	+ 5	+11	+16

TABLE C

Cant of carriage axle, deflection effect in mils due to.

Range, yards	1000	2000	3000	4000	5000	6000	7000	8000	9000
Cant of 10 mils, effect.	0.3	0.6	1.0	1.4	1.9	2.4	3.0	3.7	4.5

Range, yards	9000	10000	10500	11000	11100	11200	11300	11400	11460
Cant of 10 mils, effect.	4.5	5.6	6.3	7.3	7.6	7.9	8.3	8.9	10.0

Note: Right wheel above left causes left deflection effect.

The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be so cross leveled.

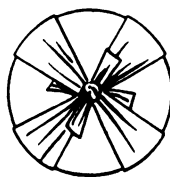
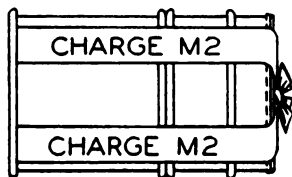
CHARGE 7
155-mm. HOWITZER
HE Shell, Mark I, Narrow Band;
Fuze M46 and M47

Powder Charge: Base Section and Increment Number 6 and 7.

Charge consists of base section and two increments, numbered 5, 6, and 7 respectively. (Some M2 charges in storage are in 7 sections.) The cartridge cloth is white. The igniter pad is on the rear end of the base section.

Maximum Range: 12,295 yards.

(Data for ranges 8000-12000 included herein.)



155-mm. HOW.
SHELL, H. E., MK. I
FUZE, M46 AND M47
CHARGE 7
M. V. 1478 f/s

TABLE A

1	2	3	4	5	6	7 8		9	10
Range R	Elevation El	Fork F	Change in elevation for 100-yd change in range c	Change in range for 1 mil change in elevation 1 mil	Time of flight Time	Probable Error		Slope of fall Slope	Line number of metro message Line
						Range e _{pr}	Deflection e _{pd}		
yds.	m	m	m	yds.	sec	yds.	yds.	1/-	No.
8000	308.6	9	6.0	17	23.3	39	5	2.4	3
8100	314.6	10	6.0	17	23.7	40	5	2.3	3
8200	320.6	10	6.0	17	24.1	40	5	2.3	3
8300	326.8	10	6.2	16	24.5	40	5	2.2	3
8400	333.0	10	6.2	16	24.9	41	5	2.2	3
8500	339.2	10	6.4	16	25.3	41	5	2.1	3
8600	345.4	11	6.4	16	25.7	42	6	2.1	3
8700	351.8	11	6.4	16	26.1	42	6	2.1	3
8800	358.2	11	6.6	15	26.6	42	6	2.0	3
8900	364.8	11	6.6	15	27.0	43	6	1.97	4
9000	371.6	12	6.8	15	27.4	43	6	1.93	4
9100	378.4	12	6.8	15	27.8	44	6	1.89	4
9200	385.2	12	7.0	15	28.3	44	6	1.86	4
9300	392.2	12	7.0	14	28.7	44	6	1.82	4
9400	399.2	13	7.2	14	29.2	45	6	1.79	4
9500	406.2	13	7.2	14	29.6	45	6	1.75	4
9600	413.4	13	7.4	14	30.1	46	6	1.72	4
9700	420.8	14	7.4	14	30.5	46	6	1.68	4
9800	428.2	14	7.6	13	31.0	46	6	1.65	4
9900	435.8	14	7.6	13	31.4	47	6	1.61	4
10000	443.6	15	7.8	13	31.9	47	6	1.58	4
10100	451.4	15	7.8	13	32.4	48	6	1.55	4
10200	459.4	16	8.0	12	32.9	48	7	1.52	4
10300	467.6	16	8.2	12	33.3	48	7	1.49	5
10400	476.0	17	8.4	12	33.8	49	7	1.46	5
10500	484.4	17	8.6	12	34.3	49	7	1.43	5
10600	493.0	18	8.8	11	34.8	50	7	1.40	5
10700	501.8	18	9.0	11	35.3	50	7	1.37	5
10800	511.0	19	9.2	11	35.8	50	7	1.34	5
10900	520.4	19	9.4	11	36.4	51	7	1.31	5
11000	530.0	20	9.8	10	37.0	51	7	1.28	5
11100	540.0	21	10.2	10	37.6	52	7	1.25	5
11200	550.2	22	10.6	10	38.2	52	7	1.23	5
11300	561.0	23	11.2	9	38.8	52	8	1.20	6
11400	572.4	24	11.8	9	39.5	53	8	1.17	6
11500	584.4	26	12.4	8	40.2	53	8	1.14	6
11600	597.2	28	13.2	8	41.0	53	8	1.11	6
11700	610.8	31	14.2	7	41.8	54	8	1.08	6
11800	625.4	34	15.4	7	42.6	54	8	1.05	6
11900	641.4	37	16.8	6	43.5	54	9	1.01	6
12000	659.4	41	19.2	5	44.5	54	9	0.97	6

155-mm. HOW.
SHELL, H. E., MK I
FUZE, M46 AND M47
CHARGE 7
M. V. 1478 f/s

TABLE A

11	12	13	14	15	16	17	18	19	1
Deflection effect		Complementary angle of site for each		Range effect of increase of					Range R
Drift* Dft.	Lateral wind of 1 MPH (+) W-D	+ 1 mil of site	- 1 mil of site	One sq. in. wt of proj. Four sq. "standard"	One foot per sec in M. V.	Air temperature 1 deg. Standard is 59° F.	Rear wind 1 MPH	One percent in air density	
				W. L.	VE	Temp.	W-R	Den.	
m	m	m	m	yds.	yds.	yds.	yds.	yds.	yds.
R 7	0.4	+ .10	- .10	- 6	+ 6.1	+ 3.5	+ 8.1	- 22	8000
R 8	0.4	+ .11	- .11	- 6	+ 6.1	+ 3.6	+ 8.3	- 23	8100
R 8	0.4	+ .11	- .11	- 6	+ 6.2	+ 3.7	+ 8.5	- 23	8200
R 8	0.4	+ .11	- .11	- 5	+ 6.2	+ 3.8	+ 8.6	- 24	8300
R 8	0.4	+ .12	- .12	- 5	+ 6.3	+ 3.9	+ 8.8	- 24	8400
R 8	0.4	+ .12	- .12	- 5	+ 6.3	+ 4.0	+ 9.0	- 24	8500
R 9	0.4	+ .13	- .12	- 4	+ 6.3	+ 4.1	+ 9.2	- 25	8600
R 9	0.4	+ .13	- .13	- 4	+ 6.4	+ 4.2	+ 9.4	- 25	8700
R 9	0.4	+ .14	- .13	- 4	+ 6.4	+ 4.3	+ 9.6	- 26	8800
R 9	0.4	+ .14	- .14	- 3	+ 6.4	+ 4.4	+ 9.8	- 26	8900
R 9	0.4	+ .15	- .14	- 3	+ 6.5	+ 4.5	+ 10.0	- 26	9000
R 10	0.4	+ .16	- .15	- 3	+ 6.5	+ 4.6	+ 10.2	- 27	9100
R 10	0.4	+ .16	- .15	- 2	+ 6.5	+ 4.7	+ 10.4	- 27	9200
R 10	0.4	+ .17	- .16	- 2	+ 6.5	+ 4.7	+ 10.7	- 28	9300
R 10	0.5	+ .17	- .16	- 1	+ 6.6	+ 4.8	+ 10.9	- 28	9400
R 10	0.5	+ .18	- .17	- 1	+ 6.6	+ 4.9	+ 11.1	- 29	9500
R 11	0.5	+ .19	- .17	0	+ 6.6	+ 5.0	+ 11.3	- 29	9600
R 11	0.5	+ .20	- .18	0	+ 6.6	+ 5.1	+ 11.6	- 30	9700
R 11	0.5	+ .20	- .19	+ 1	+ 6.7	+ 5.1	+ 11.8	- 30	9800
R 11	0.5	+ .21	- .20	+ 1	+ 6.7	+ 5.2	+ 12.1	- 31	9900
R 11	0.5	+ .22	- .21	+ 2	+ 6.7	+ 5.3	+ 12.3	- 31	10000
R 12	0.5	+ .23	- .22	+ 2	+ 6.7	+ 5.4	+ 12.5	- 32	10100
R 12	0.5	+ .24	- .23	+ 3	+ 6.7	+ 5.4	+ 12.7	- 32	10200
R 12	0.5	+ .25	- .24	+ 3	+ 6.8	+ 5.5	+ 13.0	- 33	10300
R 12	0.5	+ .26	- .25	+ 4	+ 6.8	+ 5.6	+ 13.2	- 33	10400
R 12	0.5	+ .28	- .26	+ 4	+ 6.8	+ 5.7	+ 13.4	- 34	10500
R 13	0.5	+ .30	- .27	+ 5	+ 6.8	+ 5.7	+ 13.6	- 34	10600
R 13	0.5	+ .32	- .29	+ 5	+ 6.8	+ 5.8	+ 13.8	- 35	10700
R 13	0.5	+ .34	- .31	+ 6	+ 6.9	+ 5.8	+ 14.0	- 35	10800
R 14	0.5	+ .36	- .33	+ 6	+ 6.9	+ 5.9	+ 14.2	- 36	10900
R 14	0.5	+ .39	- .35	+ 7	+ 6.9	+ 6.0	+ 14.4	- 36	11000
R 15	0.5	+ .43	- .38	+ 7	+ 6.9	+ 6.0	+ 14.6	- 37	11100
R 15	0.5	+ .48	- .41	+ 8	+ 7.0	+ 6.1	+ 14.8	- 37	11200
R 16	0.5	+ .53	- .45	+ 8	+ 7.0	+ 6.1	+ 15.0	- 38	11300
R 16	0.5	+ .59	- .49	+ 9	+ 7.0	+ 6.2	+ 15.2	- 38	11400
R 16	0.5	+ .67	- .54	+ 9	+ 7.1	+ 6.2	+ 15.4	- 39	11500
R 17	0.5	+ .77	- .60	+ 10	+ 7.1	+ 6.3	+ 15.6	- 40	11600
R 17	0.6	+ .89	- .67	+ 10	+ 7.2	+ 6.3	+ 15.8	- 40	11700
R 18	0.6	+ 1.06	- .75	+ 11	+ 7.2	+ 6.4	+ 15.9	- 41	11800
R 18	0.6	+ 1.35	- .85	+ 11	+ 7.3	+ 6.4	+ 16.1	- 42	11900
R 19	0.6	+ 1.88	- .99	+ 12	+ 7.3	+ 6.5	+ 16.3	- 42	12000

*Drift includes side jump.

SHELL, H. E., MK. I
FUZE M46 AND M47
CHARGE 7
M. V. 1478 f/s

TABLE B

Change in velocity due to change in temperature of powder.

Temperature of powder, deg. F-----	0	10	20	30	40	50	60	70	80	90	100
Change in velocity, feet per second-----	-48	-41	-34	-27	-20	-14	-7	0	+ 7	+14	+20

TABLE C

Cant of carriage axle, deflection effect in mils due to

Range, yards-----	1000	2000	3000	4000	5000	6000	7000	8000	9000
Cant of 10 mils, effect-----	0.2	0.5	0.8	1.2	1.6	2.1	2.6	3.2	3.9

Range, yards-----	9000	10000	11000	11500	12000	12100	12200	12295	-----
Cant of 10 mils, effect-----	3.9	4.7	5.8	6.6	7.7	8.0	8.4	9.6	-----

NOTE: Right wheel above left causes left deflection effect. The deflection due to cant is automatically compensated for by cross leveling the sight, and this table is to be used only in the event the sight cannot be used.

SECTION VI
APPENDIXES



APPENDIX I
CONDENSED TABLE OF MAXIMUM ORDINATES (feet)

Range yds.	75-MM GUN		105-MM. HOWITZER			155-MM. HOWITZER				
	Shrapnel, 21 sec., nor- mal charge	Shell Mk I, fuze M46 & M47, normal charge	Shell, Mk I, fuze M48 and M54		Charge 7	Shell Mk I, fuze M46 and M47		Charge 6	Charge 5	Charge 4
			Charge 3	Charge 5		Charge 3	Charge 4			
1000	16	16	58	34	16	54	42	20	46	19
2000	75	79	269	151	74	229	178	95	153	81
3000	213	229	670	365	195	551	423	236	339	202
4000	451	501	1338	695	396	1079	806	450	624	399
5000	814	930	2702	1173	692	1970	1375	782	1032	688
6000	1347	1569		1851	1096		2231	1218	1597	1082
7000	2129	2507		2858	1622		3769	1795	2378	1597
8000	3257	3943		4719	2291			2545	3498	2257
9000	5055	7010			3144			3518	5430	3099
10,000					4264			4822		4185

APPENDIX II

EQUIVALENT EROSION EFFECTS

75-MM. GUN		
Projectile	Number of rounds equivalent in erosion effect to 1 round full charge	Equivalent erosion effects in decimals
Shrapnel	1	1
Shell Mk I normal charge	2	.5
Shell Mk I reduced charge	30	.033
105-MM. HOWITZER		
Charge	Number of rounds equivalent in erosion effect to 1 full charge	Equivalent erosion effects in decimals
1	120	.00833
2	70	.0143
3	40	.0250
4	20	.050
5	10	.10
6	3	.33
7	1	1.00
155-MM. HOWITZER		
Charge	Number of rounds equivalent in erosion effect to 1 full charge	Equivalent erosion effects in decimals
1	120	.0083
2	70	.014
3	40	.025
4	20	.050
5	10	.10
6	3	.3
7	1	1.0

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